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abstractHtml: Edge functions at an edge location of a content delivery network (CDN) may use APIs of a datastore engine in order to read/write or create/delete local tables at the edge location. Data may be accumulated in the local tables and the new data may be used to enhance decision at the edge. Some of the local tables may be initially populated from a back-end database. This allows the functions to modify the data from the back-end database, without affecting the actual source data at the back-end database (modifications to local tables remain local to the edge location).

descriptionHtml: BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a logical block diagram illustrating a system for low latency query processing and data retrieval at the edge, according to some embodiments.
- (2) FIG. 2 is a logical block diagram illustrating low latency query processing and data retrieval at the edge, according to some embodiments.
- (3) FIG. 3 is a logical block diagram illustrating a system for low latency datastore access for event-triggered functions at the edge, according to some embodiments.
- (4) FIG. 4 illustrates an interface for specifying a function for low latency datastore access at the edge, according to some embodiments.
- (5) FIG. 5 illustrates processing an event-triggered function at the edge, according to some embodiments.
- (6) FIG. 6 illustrates inserting database data into a local edge datastore for low latency datastore access at the edge, according to some embodiments.
- (7) FIG. 7A illustrates obtaining database data at a local edge datastore, according to some embodiments.
- (8) FIG. 7B illustrates inserting a key-value object into a local edge datastore for low latency datastore access at the edge, according to some embodiments.
- (9) FIG. 8 is a logical block diagram illustrating a system for low latency writes to local tables by event-triggered functions at the edge, according to some embodiments.
- (10) FIG. 9 is a logical block diagram illustrating low latency datastore access for event-triggered functions at the edge, according to some embodiments.
- (11) FIG. 10 illustrates an interface for specifying a function for low latency writes to local tables at the edge, according to some embodiments.
- (12) FIG. 11 illustrates a flow diagram for low latency query processing and data retrieval at the edge, according to some embodiments.
- (13) FIG. 12 illustrates a flow diagram for specifying a function for low latency datastore access at the edge, according to some embodiments.
- (14) FIG. 13 illustrates a flow diagram for low latency datastore access for an event-triggered function at the edge, according to some embodiments.
- (15) FIG. 14 illustrates a flow diagram for specifying a function for low latency writes to local tables at the edge, according to some embodiments.
- (16) FIG. 15 illustrates a flow diagram for low latency writes to local tables by an event-triggered function at the edge,

according to some embodiments.

(17) FIG. 16 is a block diagram illustrating an example computing system, according to some embodiments.

(18) While embodiments are described herein by way of example for several embodiments and illustrative drawings, those skilled in the art will recognize that the embodiments are not limited to the embodiments or drawings described. It should be understood, that the drawings and detailed description thereto are not intended to limit embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include”, “including”, and “includes” mean including, but not limited to.

DETAILED DESCRIPTION

(19) The systems and methods described herein may be employed in various combinations and in various embodiments to implement low latency query processing and data retrieval at the edge, according to some embodiments. The systems and methods described herein may be employed in various combinations and in various embodiments to implement low latency datastore access for event-triggered functions at the edge, according to some embodiments. The systems and methods described herein may be employed in various combinations and in various embodiments to implement low latency writes to local tables by event-triggered functions at the edge, according to some embodiments.

(20) In embodiments, the various techniques described herein may reduce the amount of time to process a request from a client (e.g., a request to retrieve data from a database and/or a retrieve a result based on retrieved data), compared to traditional techniques for processing requests. In various embodiments, the time to retrieve data and/or process retrieved data by event-triggered functions may be reduced, while still maintaining security for function processing. For example, a function at an edge location may execute in an isolated execution environment and/or not make network calls, while still having access to data from database tables). Moreover, the function may insert data into a table that is based on data from a database table, wherein the inserted data remains local to the edge location.

(21) In various embodiments, the components illustrated in the figures may be implemented directly within computer hardware, as instructions directly or indirectly executable by computer hardware (e.g., a microprocessor or computer system), or using a combination of these techniques. For example, the components of the figures may be implemented by a system that includes one or more computing nodes, in one embodiment, each of which may be similar to the computer system embodiment illustrated in FIG. 16 and described below.

(22) This specification begins includes descriptions of a system for low latency query processing and data retrieval at the edge, a system for low latency datastore access for event-triggered functions at the edge, and a system for low latency writes to local tables by event-triggered functions at the edge. A number of different methods and techniques to implement the above techniques are discussed, some of which are illustrated in accompanying flowcharts. Finally, a description of an example computing system upon which the various components, modules, systems, and/or techniques described herein may be implemented is provided. Various examples are provided throughout the specification.

(23) FIG. 1 is a logical block diagram illustrating a system for low latency query processing and data retrieval at the edge, according to some embodiments.

(24) As shown, a CDN 102 may include any number of edge locations 104a-104n, each of which may obtain (e.g., fetch) data from any number of databases that are hosted by any number of origin servers 106. As depicted, the databases may be different types of databases that have different schemas. For example, database A may be a key-value database that uses a particular schema, database N may be a relational database that uses a different schema, any another of the databases may be a different type relational database that uses a different schema. In embodiments, the different types of databases process queries and provide responses that are each in different formats. For example, commands, syntax, and/or metadata used for creating or interpreting queries/responses for a particular backend

database at an origin server(s) may be different than commands, syntax, and/or metadata used for creating or interpreting queries/responses for any of the other backend databases at the origin server(s).

(25) In embodiments, an edge location 104 may include any number of edge servers that may perform any of the techniques described herein. In some embodiments, the edge location may be a data center of the CDN (e.g., a POP). In embodiments, one or more remote origin servers 106 of the CDN (e.g., at a remote data center) may provide data/objects to the edge location (e.g., in response to a client request for the data). In embodiments, any number of different clients 108 of the CDN may request data from the CDN (e.g., by sending a request from a remote network of the client) and the CDN may return the data to the requesting client. As shown, a given client may communicate with the CDN via a wide area network 110 (e.g., the internet).

(26) In the depicted embodiment, an edge location (e.g., edge location 104a) includes a datastore engine 112 that implements a query/response translator and an edge datastore 114 that stores data (e.g., as a local datastore object) obtained from any number of the databases on behalf of the client. In embodiments, the edge datastore may be an in-memory datastore of the edge server (e.g., random access memory), or any other type of volatile or non-volatile memory. As shown, the CDN may also include any number of configuration servers 116, which may be used by a client to configure/enable edge locations to perform any number of the techniques described herein on behalf of the client (e.g., by receiving input from a client via a management interface). For example, the a configuration server may receive, from the client 108a, an indication to enable storing of data of one or more of the databases into edge datastores at edge locations (and/or use event-driven functions described herein) and based on the indication, the CDN may configure the edge location 104a (and/or any other number of edge locations) to store data for the one or more databases in the edge datastore on behalf of the client (and/or use event-driven functions).

(27) As shown, the datastore engine may receive, from a client (e.g., client 108a), a query for data in database A. The query may be in a format for database A (different than query formats of the other databases). The datastore engine 112 may identify, based on the query, databases A as the target database for the query. For example, the datastore engine may determine that identifying data in the query indicates the query is for database A.

(28) The query/response translator may translate the query, based on the identified database A, from a format for database A into a native format for the edge datastore 114. The datastore engine may then determine, based on the translated query, whether the data exists at the edge datastore. For example, the datastore engine may apply the translated query to the edge datastore and determine whether the data is returned from the query. In embodiments, the native format for queries/responses used for the edge datastore is different than the various formats for queries/responses used for the databases at the origin server(s) 106 (e.g., databases A-N). Therefore, the query/response translator may have the capability to translate a responses/queries from different formats of the different back-end databases into the native format of the edge datastore, and vice-versa.

(29) If the datastore engine determines that the data does not exist at the edge datastore, then the datastore engine may forward the query that was received from the client to database A. In some embodiments, the datastore engine may provide security credentials associated with the client to authorize the request and forward the request/credentials to the origin server/target database. The datastore engine may receive the response with the requested data, in the format for database A. The datastore engine may parse the response, based on the format for database A, to obtain the data. The datastore engine may then store the data into the edge datastore (e.g., in a key-value format and/or any other suitable format). The datastore engine may also send the response to the client. In embodiments, the datastore engine may be capable of translating/parsing queries and responses in the same way for any of the databases of the CDN.

(30) In embodiments, if the datastore engine determines that the data exists at the edge datastore, then the datastore engine may generate a response in the format for the database (e.g., by translating from the native response format of the datastore engine or directly creating the response in the format for the database based on knowledge of the request/response format of the database), wherein the generated response includes the data. The datastore engine may then send the generated response to the client.

(31) In embodiments, the datastore engine may subsequently receive another query from the client for a portion of the data from database A (e.g., less than the entire amount of data indicated in the query above). The datastore engine may identify, based on the other query, database A as the target database for the other query. The datastore engine may translate the query, based on the identified database A, from a format for database A into a native format for the edge datastore 114. The datastore engine may then determine, based on the translated query, whether the data exists at the edge datastore.

(32) In response to determining that the portion of the data exists at the datastore, the datastore engine may generate a response in the format for database A, wherein the other response includes the portion of the data. The datastore engine may then send the other response to the client. By using the above technique, the datastore engine may quickly provide a response to a subsequent query for only a portion of the data that was previously added to the local edge datastore (due to the previous query), without the need to fetch data from the target database. This reduces the time for clients to retrieve data, compared to traditional techniques that use caching.

(33) In various embodiments, data that is stored into the edge datastore due to multiple queries may be provided in response to a subsequent query from a client. For example, a first query may result in portion A of data from a particular table being stored into the edge datastore and another query may result in portion B of data from the particular table being stored into the edge datastore. An additional query for at least some of portion A data and at least some of portion B data may be provided in a response by the edge datastore. By using multiple queries over time to re-construct data from the backend database tables at the edge, the datastore engine may provide low-latency access to data from the backend database tables (avoiding the need to retrieve data from the back-end tables to serve subsequent queries).

(34) In embodiments, the datastore engine may determine whether the requested data found in the edge datastore is stale. If so, then the datastore engine will retrieve the data from the source database, as described above. If not, then the datastore engine will generate a response for the requested data.

(35) In some embodiments, the datastore engine may not be able to parse data from certain databases (e.g., due to not having the necessary translator software installed). In such cases, the datastore engine may instead store the response with an associated cache key. For example, the datastore engine may determine that data requested in a query is not in the edge datastore, send a query to database X, receive a response from database X, and store the response and an associated cache key into the edge datastore. The datastore engine may also send the response to the client. The datastore engine may send the same cached response to the client for any subsequent queries for the same data. Subsequent queries for the same data may result in a match with the cache key and therefore the database engine can return the associated response.

(36) At some point, the datastore engine may receive an update (e.g., from an administrator or update service at the CDN) and install the update, which allows the datastore engine to translate queries/response for database X. Once this occurs, the datastore engine may begin to translate queries/response for database X and store data into the edge datastore as described above (avoiding the need to store the entire response with cache key).

(37) In embodiments, the datastore engine may associate a time to live (TTL) with data when it stores data into the edge datastore. Therefore, when the datastore engine determines that data exists in the edge datastore (in response to a query), the datastore engine may determine, based the TTL associated with the data at the edge datastore, whether the requested data is stale (e.g., greater than a time or an amount of time indicated by the TTL). In response to determining that the data is not stale, the datastore engine may generate a response that includes the data and send the response to the client.

(38) In embodiments, the database may require authorization of the requesting entity in order to response to queries. Therefore, when sending a query to a database, the datastore engine may providing, to the database, security credentials (e.g., along with the request or as part of the request). The security credentials may be assigned to the client and may be used by the database to authorize the query submitted to the database on behalf of the client.

(39) FIG. 2 is a logical block diagram illustrating low latency query processing and data retrieval at the edge, according

to some embodiments.

(40) In the example embodiment, a client may enable edge caching (e.g., enable use of the edge datastore for the client as described above). For example, a client user/administrator may select an option at a datastore/database to enable the use of edge caching. As shown, APIs at the CDN may obtain security credentials associated with the client (e.g., API_KEY) from the datastore/database application and provide, a request to enable edge caching for the client along with the security credentials. The security credentials may be stored at the edge location in association with the client's CDN distribution at the edge (e.g., by the CDN application and/or the datastore engine). The datastore engine and/or other application may then enable edge caching for the client (based at least on authorization using the security credentials).

(41) In embodiments, the datastore engine may use these security credentials (and/or other credentials associated with the client) to authorize a request from the client and/or to forward the request to the datastore/database at the origin server. As described herein, the response may then be received from the datastore/database and the returned data may be stored at the edge datastore along with a TTL. In embodiments, the response/data is also returned to the client (e.g., returned to the requesting layer of the CDN application and then sent to the client).

(42) As shown, when the client submits a POST query for books from a particular author, the datastore engine determines whether the data is in the edge datastore (e.g., as described above). In the depicted embodiment, the data is not there (e.g., a "miss"), so the request is forwarded to one of the databases. A response is provided to the datastore engine, where the data is stored into the edge datastore and the response is sent to the client.

(43) FIG. 3 is a logical block diagram illustrating a system for low latency datastore access for event-triggered functions at the edge, according to some embodiments.

(44) As in FIG. 1, the depicted CDN 102 includes a datastore engine 112, edge datastore 114, origin server(s) 106, and configuration server(s) 116. In the example embodiment, a given edge location (e.g., edge location 104a) also includes a scripts engine 302 capable of executing any number of registered functions 304 in its own execution environment (e.g., within an isolated environment/container). In some embodiments, any number of functions may execute concurrently, each in its own isolated execution environment/container of the scripts engine. In some embodiments, after execution of a given function, the function and/or the isolated execution environment/container terminates so that resources (memory, computing hardware/software, etc.) that were used by the function and/or the isolated execution environment/container are made available to initialize and/or run another function and/or the isolated execution environment/container. The edge location also includes an event handler 306, which detects different events and determines whether to trigger one of the functions in response to detecting the event (based on client-specified configurations of the registered functions). In some embodiments, the scripts engine, functions, and/or any other components of FIG. 3 may perform in the same or similar way as the scripts engine, functions, and/or any other components of other figures.

(45) As shown, the edge location (e.g., edge server) may receive, from a client (e.g., client 108a), a request (e.g., query) for data in database A (in a format for database A). One of the registered functions may be registered to execute in response to a particular event occurring during request processing (e.g., encountering a request for data from one of the tables that are associated with the function during configuration of the function by a user). During processing of the request, the event handler detects the particular event has occurred (e.g., the edge server encounters the event during the request processing). In response to encountering the event (e.g., in response to detection of the event), the event handler causes one of the registered functions to execute in the isolated execution environment of the scripts engine 302.

(46) As described in more detail below, the function accesses the requested data at one or more of the corresponding read-only local tables at the edge location, and the accessed data is based on data obtained at the edge location from the a particular table(s) of one of the databases (e.g., database A). In embodiments, the data of a given read-only local table is accessed within the isolated execution environment without making network calls. For example, the data may be

passed to the function as part of the function call. The data and/or result(s) of processing of the accessed data may then be sent to the client in a response.

(47) In some embodiments, a given function may use any data of the local table(s) to perform any number of operations/calculations to generate one or more results, which are returned to the client as part of a response. For embodiments in which functions have write access to a local table (e.g., using APIs as discussed for FIG. 8), the function may also (or instead) write the one or more results to the local table, at least some of which may be used as input for later operations/calculations performed by the function and/or other functions.

(48) In embodiments, the data in the read-only local table may first be obtained from the target database (using the same or similar techniques described above for FIG. 1) and then provided to the function for access during execution (e.g., in a portion of memory mapped to the function). For example, before execution of the function, the datastore engine may identify, based on the request for data, a table of a database, determine that the requested data and/or corresponding local table does not exist in the edge datastore, and in response, obtain the data from the table of database A (applying any filter(s) to reduce the amount of data obtained from the table, if any were specified during configuration of the function).

(49) The datastore engine may then create a read-only local table using the obtained data and store the read-only local table at the edge datastore (in some embodiments, if the read-only local table for the table of database A already exists in the edge datastore, then the obtained data may simply be added to the local table). The datastore engine may then send the read-only local table (or at least the requested data) to the function and/or to a destination associated with the function (e.g., to the event handler or other component that may initiate execution of the function or cause the function to be executed in the isolated execution environment). In embodiments, the datastore engine may convert the local table/data from a format (e.g., key-value format) into a different format (e.g., a serialized format) before sending the local table/data to the destination/function.

(50) The read-only local table may be exposed to the function during execution (e.g., stored in memory mapped to the function or otherwise provided to the function as a function parameter). Therefore, the function may have access to data that was obtained from a table at database A, even though the function is unable to make network calls outside of the execution environment and/or the edge location. In embodiments, any number of local tables may be sent to the function using the above techniques.

(51) In embodiments, prior to execution of the function, the event handler (or other component of the edge server/proxy server) may: identify, based on a pre-defined event (e.g., request for data from a particular database table), the function associated with the event and the one or more corresponding read-only local tables associated with the function (based on the function configuration); obtain, from a location (e.g., a datastore at the edge location or another location), code for the function; and obtain, from the datastore engine, one or more corresponding read-only local tables. Any number of those local tables may already exist in the edge datastore (e.g., due to previous use by other functions). However, if any of the local tables are not present or not usable (e.g., stale), then they may first need to be created (or updated) by obtaining the data from the source database table, as described herein.

(52) In various embodiments, providing a low-latency datastore further enhances the capability of edge functions. Without access to data (e.g., data from back-end origin databases), edge functions may have limited functionality, and allowing such edge functions to make network calls to access data (e.g., from origin databases) would add a high amount of latency in order to obtain the data. By allowing low latency datastore access to event-triggered edge functions as described herein, edge functions can access data from origin databases without incurring the high latency associated with network calls (by allowing access to the datastore that has been populated from the origin database).

(53) FIG. 4 illustrates an interface for specifying a function for low latency datastore access at the edge, according to some embodiments.

(54) In the example embodiment, a function specification window allows a client to specify a function configuration. The client may specify a triggering event that when encountered, will cause the function to execute. As shown, the client may

indicate whether the event is a request or response event. The client may also specify a function identifier (e.g., a unique name for the function). In embodiments, the client may not be required to specify a triggering event for the function. For example, the function may, by default, be invoked/executed in response to incoming requests for data (or invoked for certain types of requests for data, such as request directed to a particular target database and/or table).

(55) The client may also specify one or more tables of one or more of the databases that are to provide data for one or more corresponding read-only local tables at a given edge server. The client may specify a name for the read-only local table, a unique identifier for the source table and/or database, and optionally a filter to apply to the source table. This may reduce the amount of data that is pulled by the datastore engine when obtaining data based on a client query for data.

(56) FIG. 5 illustrates processing an event-triggered function at the edge, according to some embodiments.

(57) As shown, when a CDN proxy/edge server of the edge location detects a triggering event associated with a registered function (request from a client or data from a table), the CDN proxy fetches the function from a location that stores the function code (code origin) and also creates a datastore (read-only local table) in the datastore engine. The datastore engine fetches the data from the target database, creates a key-value object store of the data, serializes the data, and sends the serialized data to the CDN proxy.

(58) The CDN proxy then sends the function and serialized data to the scripts engine, where the function is executed and accesses the data. As shown, the scripts engine de-serializes the data and extracts the data into a key-value store, where it is embedded into memory and is accessible to the function. The function may return a result to the CDN proxy, which may then send a response to the client that includes the result.

(59) FIG. 6 illustrates inserting database data into a local edge datastore for low latency datastore access at the edge, according to some embodiments.

(60) As depicted, the CDN proxy sends a request for data to the datastore engine and in response to receiving the request, the datastore engine sends a request for the data to the target database in a format for the target database. The datastore engine receives a response in a format for the target database (e.g., JSON file). The datastore engine parses the response to obtain a list of "item" objects, which may then be stored into a key-value object store (or another type of table). The list may be serialized and stored in the edge datastore and/or returned to the CDN proxy/edge server as an in-memory DB, to be used by a function.

(61) Although the depicted example shows the datastore engine receives data from the target database in a JSON format and converts the data into a tabular key-value format for storage in the datastore engine, the datastore engine may receive data in any type of format and convert the data into any other type of format (e.g., for storage in the edge datastore). Furthermore, in various embodiments the datastore engine may convert the data into multiple different types of formats, each of which may be stored in the edge datastore.

(62) FIG. 7A illustrates obtaining database data at a local edge datastore, according to some embodiments.

(63) As shown, a query 702 is sent to a target database of the CDN. The response 704 is received in a format for the target database. For example, the format may be in JSON or any other format specific to the target database. As shown, the response includes two items that matched the query (data describing two different dogs).

(64) FIG. 7B illustrates inserting a key-value object into a local edge datastore for low latency datastore access at the edge, according to some embodiments.

(65) As depicted, the response of FIG. 7A is parsed and converted by the datastore engine into a key-value object 706 Pets. The Pets object may then be serialized and inserted into the edge datastore 708 associate with the key "D345.PETS," which may also include any number of other serialized objects. The serialized data may be passed to a destination (e.g., scripts engine) for use by a function.

(66) FIG. 8 is a logical block diagram illustrating a system for low latency writes to local tables by event-triggered functions at the edge, according to some embodiments.

(67) As in FIG. 3, the depicted CDN 102 includes components such as a datastore engine 112, edge datastore 114,

origin server(s) 106, configuration server(s) 116, and a scripts engine. As shown, the edge location (e.g., edge server) may receive, from a client (e.g., client 108a), a request (e.g., query for data in database A, in a format for database A).

(68) One of the registered functions may be registered to execute in response to a particular event occurring during request processing. During processing of the request, the event handler detects the particular event has occurred (e.g., the edge server encounters the event during the request processing). In response to encountering the event (e.g., in response to detection of the event), the event handler causes one of the registered functions to execute in the isolated execution environment of the scripts engine 302.

(69) As described in more detail below, the function may provide a request to the datastore engine (e.g., via an API of the datastore engine). For example, the function may provide, to the datastore engine, a request to create a local table or to insert data into a local table of the edge datastore. Note that in embodiments, the datastore engine and the edge datastore are external to the isolated execution environment and are hosted by the at least one edge server. In response to receiving the request (e.g., via the API), the datastore engine may create the local table or insert the data into a local table. In embodiments, the inserted data remains local to the edge location (e.g., changes do not propagate outside of the edge location to source tables at the origin server(s)).

(70) In embodiments, when a local read/write table is created, it may be initially populated/sourced by a target table of a database (the source of truth) as described herein. In embodiments, a function may make changes to data from the target database, but the changes remain local to the edge location. In embodiments, the function may also delete a local table via a datastore engine API. Note that in some embodiments, the datastore engine may obtain source data for a local table in the same or similar way as described above (e.g., for FIGS. 1 and/or 3).

(71) In some embodiments, any type of table operations may be available for a function to perform via APIs on local tables in the edge datastore (e.g., create table, insert data, fetch data, delete entry, delete table, etc.). For example, a function may provide to the datastore engine (e.g., via a delete API) a request to delete an entry from a local table (the request may indicate a particular key). In response to receiving the indication of the table and the particular key, the datastore engine may delete an entry from the local table that has the particular key. In embodiments, the deletion of the entry remains local to the edge location (e.g., does not propagate to backend database tables).

(72) If requested data (e.g., using the fetch API) is not present in a local table at the edge datastore (or the local table does not exist), then the datastore engine may obtain the data from the source table (or create the table) as described herein. For example, a function may submit (via the fetch API) a request for data from a local table (e.g., an item/row of the table) that corresponds to a particular key.

(73) In response to determining that the local table includes the particular key, the datastore engine may provide, to the function, the data that corresponds to the particular key. In response to determining that the local table does not include the particular key, the datastore engine may obtain, from the table assigned as the source of truth for the local table, the data that corresponds to the particular key. The datastore engine may then insert that data into the local table and send the data to the function. In embodiments, the data in the local table is accessible to other copies/instances of the same function that are each running in an isolated execution environment and/or other functions that are each running in an isolated execution environment.

(74) In various embodiments, the datastore engine APIs may perform any locking mechanisms for the local tables in order to protect data that is accessed by any number of functions. For example, if two different functions are attempting to write to the same data at the same local table, then the APIs will only allow one function to access and/or modify the data at a time.

(75) In embodiments, the use of APIs for local tables of the edge datastore may allow functions to accumulate data in the local tables. That accumulated data may be used to enhance decisions made at the edge. This may provide flexibility for a client that wishes to perform various operations on data and modifications to data to determine various possible results/outcomes, without affecting data at the backend source tables.

(76) FIG. 9 is a logical block diagram illustrating low latency datastore access for event-triggered functions at the edge,

according to some embodiments.

(77) As shown, a client may enable use of event-triggered functions for edge caching (e.g., enable use of the functions for the client as described above). APIs at the CDN may obtain security credentials associated with the client (e.g., API_KEY) from the datastore/database application and provide, to the edge location, a request to enable edge caching for the client along with the security credentials. The security credentials may be stored at the edge location in association with the client's CDN distribution at the edge (e.g., by the CDN application and/or the datastore engine). The datastore engine and/or other application may then enable edge caching for the client (based at least on authorization using the security credentials).

(78) In some embodiments, the datastore engine may use these security credentials (and/or other credentials associated with the client) to authorize a request from the client and/or to forward the request to the datastore/database at the origin server. As described herein, the response may then be received from the datastore/database and the returned data may be stored at the edge datastore along with a TTL. In embodiments, the response/data is also returned to the client (e.g., returned to the requesting layer of the CDN application and then sent to the client).

(79) In the example embodiment, when the client submits a GET query for books from a particular author, a function is triggered to access the data. In the depicted embodiment, the function sends, to the datastore engine via an API of the datastore engine, a request for the data. In embodiments, the request may indicate a particular key associated with the requested data.

(80) If the datastore engine determines that the requested data is not in the edge datastore (e.g., there is no local table that is sourced from the target table or a local table is sourced from the target table but does not include the particular key associated with the data), then the request is forwarded to the target database. A response is provided from the target table to the datastore engine, where the data is stored into the edge datastore (e.g., the book title "Intelligent Investor"). The data is stored into a read/write local table or a read-only local table, depending on how the function was configured by a user. The data is then provided, via the database engine API, to the function. The function may then return the data to the client and/or may process the data and return a result to the client based at least on the processing of the data. As described herein, other functions may access/modify data in the local table (e.g., via the datastore engine APIs).

(81) FIG. 10 illustrates an interface for specifying a function for low latency writes to local tables at the edge, according to some embodiments.

(82) As shown, a function specification window allows a client to specify a function configuration. The client may specify a triggering event that when encountered, will cause the function to execute. As shown, the client may indicate whether the event is a request or response event. The client may also specify a function identifier (e.g., a unique name for the function).

(83) The client may also specify one or more local tables that the function is to be assigned read and write access to (or read only access to). The client may specify a name for the read-only local table and a type (e.g., read/write, read only, sourced from a datastore, and/or not sourced from a datastore). The client may also specify a maximum size of the local table. If the local table is sourced from a database, then the client specifies a unique identifier for the source table and/or database, and optionally a filter to apply to the source table. In embodiments, this may reduce the amount of data that is pulled by the datastore engine when obtaining data based on a client query for data.

(84) FIG. 11 illustrates a flow diagram for low latency query processing and data retrieval at the edge, according to some embodiments. These techniques, as well as the techniques discussed with regard to FIGS. 12-15, may be implemented using components or systems as described above with regard to FIGS. 1-10, as well as other types of components or systems, and thus the following discussion is not intended to be limiting as to the other types of systems that may implement the described techniques.

(85) At block 1102, the edge server receives a query for data. At block 1104, the edge server identifies the target database for the query. At block 1106, the edge server translates the query from the target database format to the edge

datastore format. At block 1108, the edge server determines whether the requested data exists at the edge datastore. If so, then at block 1110, the edge server generates a response in the target database format and at block 1112, the edge server sends the response to the client.

(86) Returning to block 1108, if the edge server determines that the requested data does not at the edge datastore, then at block 1114, the edge server forward the received query to the target database. At block 1116, the edge server receives a response in the target database format. At block 1118, the edge server parses the response to obtain the data. At block 1120, the edge server stores the data into the edge datastore. At block 1122, the edge server sends the response to the client.

(87) FIG. 12 illustrates a flow diagram for a user specifying a function for low latency datastore access at the edge, according to some embodiments.

(88) In embodiments, the interface of FIG. 4 may be used to receive inputs from a user to specify functions. At block 1202, the configuration server receives, from an end user of the client, an indication of an event that, when encountered, will cause a function to execute. At block 1204, the configuration server receives, from the user, an indication of a database table that is to provide data for a corresponding read-only local table. At block 1206, the configuration server receives, from the user, an indication of a filter to use for selecting data from the database table. At block 1208, the configuration server determines whether the user specifies another read-only local table. If so, then the process returns to block 1204. If not, then at block 1210, the configuration server stores the function configuration and/or registers the function at one or more edge locations.

(89) FIG. 13 illustrates a flow diagram for low latency datastore access for event-triggered function at the edge, according to some embodiments.

(90) At block 1302, the edge server receives, at the edge location, a request for data. At block 1304, the edge server determines whether a client-specified event has been encountered during the request processing (any processing/action that occurs at the edge location as a result of the request, which may include responses). If none of the specified events were encountered, then the process ends at block 1306.

(91) Returning to block 1304, if the edge server determines that a client-specified event has been encountered during the request processing, then at block 1308, the edge server obtains function code from the code origin and obtains data from one or more read-only local tables. At block 1310, the edge server executes the function in an isolated execution environment. At block 1312, the function accesses the data of the one or more read-only local tables without making network calls. The accessed data is based on data obtained from one or more tables at one or more origin servers. At block 1314, the function sends a response to the client based at least on the accessing of the data.

(92) FIG. 14 illustrates a flow diagram for specifying a function for low latency writes to local tables at the edge, according to some embodiments.

(93) In embodiments, the interface of FIG. 10 may be used to receive inputs from a user to specify functions. At block 1402, the configuration server receives, from an end user of the client, an indication of an event that, when encountered, will cause a function to execute. At block 1404, the configuration server receives, from the user, an indication of a local table that the function is to be assigned read/write access to (or read only access).

(94) At block 1406, the configuration server receives, from the user, an indication of a maximum size for the local table. At block 1408, the configuration server determines whether the user specifies a source of truth for the local table. If not, then at block 1410, the configuration server determines whether the user specifies another local table. If so, the process returns to block 1402. If not, then the configuration server stores the function configuration and/or registers the function at one or more edge locations.

(95) Returning to block 1408, if the configuration server determines that the user specifies a source of truth for the local table, then at block 1414, then the configuration server receives, from the user, an indication a database table that is to provide data to the local table. At block 1416, the configuration server receives, from the user, an indication of a filter to use for selecting data from the database table. As discussed above, at block 1410, the configuration server determines

whether the user specifies another local table. If so, the process returns to block 1402. If not, then the configuration server stores the function configuration and/or registers the function at one or more edge locations.

(96) FIG. 15 illustrates a flow diagram for low latency writes to local tables by an event-triggered function at the edge, according to some embodiments.

(97) At block 1502, the edge server receives a request from a client (e.g., a request for data). At block 1504, the edge server determines whether a client-specified event has been encountered during the request processing (any processing/action that occurs at the edge location as a result of the request, which may include responses). If none of the specified events were encountered, then the process ends at block 1506.

(98) Returning to block 1504, if the edge server determines that a client-specified event has been encountered during the request processing, then at block 1508, the edge server executes the function in an isolated execution environment. At block 1510, the edge server reads and/or writes data to one or more local tables of an edge datastore using an API of a datastore engine, where the inserted data remains local to the edge location.

(99) The methods described herein may in various embodiments be implemented by any combination of hardware and software. For example, in one embodiment, the methods may be implemented by a computer system (e.g., a computer system as in FIG. 8) that includes one or more processors executing program instructions stored on a computer-readable storage medium coupled to the processors. The program instructions may implement the functionality described herein (e.g., the functionality of the datastore engine, scripts engine, functions, and other components that implement the techniques described herein). The various methods as illustrated in the figures and described herein represent example embodiments of methods. The order of any method may be changed, and various elements may be added, reordered, combined, omitted, modified, etc.

(100) Embodiments to implement low latency query processing and data retrieval at the edge, low latency datastore access for event-triggered functions at the edge, and low latency writes to local tables by event-triggered functions at the edge as described herein may be executed on one or more computer systems, which may interact with various other systems or devices. One such computer system is illustrated by FIG. 16. In different embodiments, computer system 1600 may be any of various types of devices, including, but not limited to, a personal computer system, desktop computer, laptop, notebook, or netbook computer, mainframe computer system, handheld computer, workstation, network computer, a camera, a set top box, a mobile device, a consumer device, video game console, handheld video game device, application server, storage device, a peripheral device such as a switch, modem, router, or in general any type of computing node or compute node, computing device, compute device, or electronic device.

(101) In the illustrated embodiment, computer system 1600 includes one or more processors 1610 coupled to a system memory 1620 via an input/output (I/O) interface 1630. Computer system 1600 further includes a network interface 1640 coupled to I/O interface 1630, and one or more input/output devices 1650, such as cursor control device 1660, keyboard 1670, and display(s) 1680. Display(s) may include standard computer monitor(s) and/or other display systems, technologies or devices, in one embodiment. In some embodiments, it is contemplated that embodiments may be implemented using a single instance of computer system 1600, while in other embodiments multiple such systems, or multiple nodes making up computer system 1600, may host different portions or instances of embodiments. For example, in one embodiment some elements may be implemented via one or more nodes of computer system 1600 that are distinct from those nodes implementing other elements.

(102) In various embodiments, computer system 1600 may be a uniprocessor system including one processor 1610, or a multiprocessor system including several processors 1610 (e.g., two, four, eight, or another suitable number). Processors 1610 may be any suitable processor capable of executing instructions, in one embodiment. For example, in various embodiments, processors 1610 may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors 1610 may commonly, but not necessarily, implement the same ISA.

(103) In some embodiments, at least one processor 1610 may be a graphics processing unit. A graphics processing unit

or GPU may be considered a dedicated graphics-rendering device for a personal computer, workstation, game console or other computing or electronic device, in one embodiment. Modern GPUs may be very efficient at manipulating and displaying computer graphics, and their highly parallel structure may make them more effective than typical CPUs for a range of complex graphical algorithms. For example, a graphics processor may implement a number of graphics primitive operations in a way that makes executing them much faster than drawing directly to the screen with a host central processing unit (CPU). In various embodiments, graphics rendering may, at least in part, be implemented by program instructions for execution on one of, or parallel execution on two or more of, such GPUs. The GPU(s) may implement one or more application programmer interfaces (APIs) that permit programmers to invoke the functionality of the GPU(s), in one embodiment.

(104) System memory 1620 may store program instructions 1625 and/or data accessible by processor 1610, in one embodiment. In various embodiments, system memory 1620 may be implemented using any suitable memory technology, such as static random access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. In the illustrated embodiment, program instructions and data implementing desired functions, such as those described above (e.g., the datastore engine, scripts engine, functions, etc.) are shown stored within system memory 1620 as program instructions 1625 and data storage 1635, respectively. In other embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media or on similar media separate from system memory 1620 or computer system 1600. A computer-accessible medium may include non-transitory storage media or memory media such as magnetic or optical media, e.g., disk or CD/DVD-ROM coupled to computer system 1600 via I/O interface 1630. Program instructions and data stored via a computer-accessible medium may be transmitted by transmission media or signals such as electrical, electromagnetic, or digital signals, which may be conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface 1640, in one embodiment.

(105) In one embodiment, I/O interface 1630 may be coordinate I/O traffic between processor 1610, system memory 1620, and any peripheral devices in the device, including network interface 1640 or other peripheral interfaces, such as input/output devices 1650. In some embodiments, I/O interface 1630 may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory 1620) into a format suitable for use by another component (e.g., processor 1610). In some embodiments, I/O interface 1630 may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface 1630 may be split into two or more separate components, such as a north bridge and a south bridge, for example. In addition, in some embodiments some or all of the functionality of I/O interface 1630, such as an interface to system memory 1620, may be incorporated directly into processor 1610.

(106) Network interface 1640 may allow data to be exchanged between computer system 1600 and other devices attached to a network, such as other computer systems, or between nodes of computer system 1600, in one embodiment. In various embodiments, network interface 1640 may support communication via wired or wireless general data networks, such as any suitable type of Ethernet network, for example; via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks; via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

(107) Input/output devices 1650 may, in some embodiments, include one or more display terminals, keyboards, keypads, touchpads, scanning devices, voice or optical recognition devices, or any other devices suitable for entering or retrieving data by one or more computer system 1600, in one embodiment. Multiple input/output devices 1650 may be present in computer system 1600 or may be distributed on various nodes of computer system 1600, in one embodiment. In some embodiments, similar input/output devices may be separate from computer system 1600 and may interact with one or more nodes of computer system 1600 through a wired or wireless connection, such as over network interface 1640.

(108) As shown in FIG. 16, memory 1620 may include program instructions 1625, that implement the various embodiments of the systems as described herein, and data store 1635, comprising various data accessible by program instructions 1625, in one embodiment. In one embodiment, program instructions 1625 may include software elements of embodiments as described herein and as illustrated in the Figures. Data storage 1635 may include data that may be used in embodiments (e.g., data, tables, code, etc.). In other embodiments, other or different software elements and data may be included.

(109) Those skilled in the art will appreciate that computer system 1600 is merely illustrative and is not intended to limit the scope of the embodiments as described herein. In particular, the computer system and devices may include any combination of hardware or software that can perform the indicated functions, including a computer, personal computer system, desktop computer, laptop, notebook, or netbook computer, mainframe computer system, handheld computer, workstation, network computer, a camera, a set top box, a mobile device, network device, internet appliance, PDA, wireless phones, pagers, a consumer device, video game console, handheld video game device, application server, storage device, a peripheral device such as a switch, modem, router, or in general any type of computing or electronic device. Computer system 1600 may also be connected to other devices that are not illustrated, or instead may operate as a stand-alone system. In addition, the functionality provided by the illustrated components may in some embodiments be combined in fewer components or distributed in additional components. Similarly, in some embodiments, the functionality of some of the illustrated components may not be provided and/or other additional functionality may be available.

(110) Those skilled in the art will also appreciate that, while various items are illustrated as being stored in memory or on storage while being used, these items or portions of them may be transferred between memory and other storage devices for purposes of memory management and data integrity. Alternatively, in other embodiments some or all of the software components may execute in memory on another device and communicate with the illustrated computer system via inter-computer communication. Some or all of the system components or data structures may also be stored (e.g., as instructions or structured data) on a computer-accessible medium or a portable article to be read by an appropriate drive, various examples of which are described above. In some embodiments, instructions stored on a computer-readable medium separate from computer system 1600 may be transmitted to computer system 1600 via transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link. This computer readable storage medium may be non-transitory. Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Accordingly, the present invention may be practiced with other computer system configurations.

(111) Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD-ROM, non-volatile media such as RAM (e.g. SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc., as well as transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

(112) The various methods as illustrated in the Figures and described herein represent example embodiments of methods. The methods may be implemented in software, hardware, or a combination thereof. The order of method may be changed, and various elements may be added, reordered, combined, omitted, modified, etc.

(113) Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. It is intended that the invention embrace all such modifications and changes and, accordingly, the above description to be regarded in an illustrative rather than a restrictive sense.

claimsHtml: 1. A system for implementing writes to local tables at an edge location, comprising: at least one origin server of a content delivery network, wherein the at least one origin server is configured to implement a plurality of databases

with different respective schemas, and wherein the plurality of databases are accessible to a plurality of clients of the content delivery network; a configuration server of the content delivery network, wherein the configuration server is configured to: receive user input, via a management interface, a specification for one or more functions, wherein for a function of the one or more functions, the specification indicates at least: an event that, when encountered at the edge location, will cause the function specified by the user input to execute within an isolated execution environment; and a local table of an edge datastore at the edge location that the function specified by the user input is to be assigned read and write access to via a datastore engine external to the isolated execution environment of the function; and an edge server at the edge location of the content delivery network, wherein the edge server is configured to: receive, from a remote client of the content delivery network, a query to insert data into a source table of a database of the plurality of databases of the origin server; encounter the event during processing of the query; and in response to the encountering of the event, execute the function in the isolated execution environment, wherein the function is configured to: provide, to the datastore engine, a request to insert the data into the local table of the edge datastore, wherein the local table includes data sourced from the source table of the database of the origin server, wherein the datastore engine and the edge datastore are external to the isolated execution environment and are hosted by the same edge server that executes the function in response to encountering the event during processing of the query received from the remote client of the content delivery network; and in response to reception, by the datastore engine from the function specified by the user input, of the request to insert the data into the local table of the edge datastore, insert the data into the local table of the edge datastore, wherein, based on identification of the database of the origin server as a target database of the query to insert the data into the source table of the database, the query to insert data into the source table of the origin server that was received from the remote client by the edge server of the content delivery network is translated from an origin format of the database of the origin server of the content delivery network into a native format of the edge datastore of the edge server of the same content delivery network and the data is inserted according to the translated query into the local table of the edge location instead of being inserted into the source table of the database, and wherein the data inserted into the local table of the edge location remains local to the edge location of the content delivery network without insertion of the data into the corresponding source table of the origin server of the content delivery network.

2. The system as recited in claim 1, wherein to provide, to the datastore engine, the request to insert the data into the local table of the edge datastore, the function is further configured to: provide, to the datastore engine via an application programming interface (API) of the datastore engine, an identifier of the table, a key to be inserted into the local table, and a value to be inserted into the local table.

3. The system as recited in claim 1, wherein the function is further configured to: prior to the providing of the request to insert the data into the local table of the edge datastore, provide, to the datastore engine, a request to create the local table; and wherein the edge server is further configured to: in response to reception, by the datastore engine, of the request to create the local table, create the local table of the edge datastore.

4. The system as recited in claim 3, wherein the specification for the function further indicates the source table of one of the plurality of databases to be the source for the local table, and wherein to create the local table of the edge datastore, the edge server is further configured to: assign the source table as the source for the local table.

5. The system as recited in claim 4, wherein the edge server is further configured to: receive, from the function at the datastore engine, a request for data from the local table that corresponds to a particular key; in response to determining that the local table does not include the particular key, obtain, by the datastore engine from the table assigned as the source, the data that corresponds to the particular key; insert into the local table, by the datastore engine, the data and the particular key; and send the data from the datastore engine to the function.

6. A method, comprising: performing, by one or more computing devices: receiving at an edge server of an edge location of a content delivery network, a query from a remote client of the content delivery network to insert data into a source table of a database, wherein the content delivery network comprises an origin server that hosts the database;

encountering an event during processing of the query, wherein a function and a local table of an edge datastore that the function is to be assigned read and write access to are associated with the event based on a specification previously received at the content delivery network from user input via a management interface, the specification indicating the event that, when encountered at the edge location, will cause the function specified by the user input to execute within an isolated execution environment and a local table at the edge location that the function specified by the user input is to be assigned read and write access to via a datastore engine external to the isolated execution environment of the function; in response to encountering the event: executing the function in the isolated execution environment at the edge location; providing, by the function to the datastore engine, a request to insert the data into the local table of the edge datastore, wherein the local table includes data sourced from the source table of the database of the origin server, wherein the datastore engine and the edge datastore are external to the isolated execution environment and are hosted at the same edge location that executes the function in response to encountering the event during processing of the query received from the remote client of the content delivery network; and in response to receiving, by the datastore engine from the function specified by the user input, the request to insert the data into the local table of the edge datastore, inserting the data into the local table of the edge datastore, wherein, based on identification of the database of the origin server as a target database of the query to insert the data into the source table of the database, the query to insert data into the source table of the origin server that was received from the remote client by the edge server of the content delivery network is translated from an origin format of the database of the origin server of the content delivery network into a native format of the edge datastore of the edge server of the same content delivery network and the data is inserted according to the translated query into the local table of the edge location instead of being inserted into the source table of the database, and wherein the data inserted into the local table of the edge location remains local to the edge location of the content delivery network without insertion of the data into the corresponding source table of the origin server of the content delivery network.

7. The method of claim 6, wherein the providing, by the function to a datastore engine, a request to insert data into the local table of the edge datastore comprises: providing, to the datastore engine via an application programming interface (API) of the datastore engine, an identifier of the table, a key to be inserted into the local table, and a value to be inserted into the local table.

8. The method of claim 6, further comprising: prior to providing the request to insert the data into the local table of the edge datastore, providing, to the datastore engine, a request to create the local table; in response to receiving, by the datastore engine, the request to create the local table, creating the local table of the edge datastore.

9. The method of claim 8, wherein the providing, to the datastore engine, a request to create the local table comprises: providing, to the datastore engine via an application programming interface (API) of the datastore engine, an identifier of the local table and a source for the local table.

10. The method of claim 9, wherein the source for the local table comprises one or more of the at least one database or the source table of the origin server.

11. The method of claim 6, further comprising: receiving, from the function at the datastore engine, a request for data from the local table that corresponds to a particular key; in response to determining, by the datastore engine, that the local table does not include the particular key, obtaining by the datastore engine, from the source table that is assigned as the source for the local table, the data that corresponds to the particular key; inserting, into the local table of the edge datastore by the datastore engine, the data and the particular key; and sending the data from the datastore engine to the function.

12. The method of claim 11, further comprising: receiving, from the function at the datastore engine, another request for the data from the local table that corresponds to the particular key; in response to determining that the local table includes the particular key, providing, from the datastore engine to the function, the data that corresponds to the particular key.

13. The method of claim 11, further comprising: receiving, from another function at the datastore engine, another

request for the data from the local table that corresponds to the particular key; in response to determining that the local table includes the particular key, providing, from the datastore engine to the other function, the data that corresponds to the particular key.

14. The method of claim 6, further comprising providing, by the function to the datastore engine, a request to delete an entry from the local table, wherein the request indicates the local table and a particular key and wherein: in response to receiving, at the datastore engine, the indication of the table and the particular key, delete an entry from the local table that has the particular key, wherein the deletion of the entry remains local to the edge location.

15. One or more non-transitory computer-accessible storage media storing program instructions that when executed on or across one or more processors cause the one or more processors to: receive at an edge location of a content delivery network, a query from a remote client to insert data into a source table of a database, wherein the content delivery network comprises an origin server that hosts the database; encounter an event during processing of the query, wherein a function is associated with the event based on a specification previously received at the content delivery network from user input via a management interface, the specification indicating the event that, when encountered at the edge location, will cause the function specified by the user input to execute within an isolated execution environment and a local table of an edge datastore at the edge location that the function specified by the user input is to be assigned read and write access to via a datastore engine external to the isolated execution environment of the function; in response to the encountered event: execute the function in the isolated execution environment at the edge location; and provide, by the function to the datastore engine, a request to insert the data into a local table of the edge datastore, wherein the local table includes data sourced from the source table of the database of the origin server, wherein the datastore engine and the edge datastore are external to the isolated execution environment and are hosted at the same edge location that executes the function in response to encountering the event during processing of the query received from the remote client of the content delivery network; and in response to reception, by the datastore engine from the function specified by the user input, of the request to insert the data into the local table of the edge datastore, insert the data into the local table of the edge datastore, wherein, based on identification of the database of the origin server as a target database of the query to insert the data into the source table of the database, the query to insert data into the source table of the origin server that was received from the remote client by the edge server of the content delivery network is translated from an origin format of the database of the origin server of the content delivery network into a native format of the edge datastore of the edge server of the same content delivery network and the data is inserted according to the translated query into the local table of the edge location instead of being inserted into the source table of the database, and wherein the data inserted into the local table of the edge location remains local to the edge location of the content delivery network without insertion of the data into the corresponding source table of the origin server of the content delivery network.

16. The one or more non-transitory computer-accessible storage media as recited in claim 15, wherein to provide, by the function to a datastore engine, a request to insert data into a local table of the edge datastore, the program instructions when executed on or across the one or more processors cause the one or more processors to: provide, to the datastore engine via an application programming interface (API) of the datastore engine, an identifier of the table, a key to be inserted into the local table, and a value to be inserted into the local table.

17. The one or more non-transitory computer-accessible storage media as recited in claim 15, further comprising program instructions that when executed on or across the one or more processors further cause the one or more processors to: prior to the providing of the request to insert the data into the local table of the edge datastore, provide, to the datastore engine, a request to create the local table; and in response to reception, by the datastore engine, of the request to create the local table, create the local table of the edge datastore.

18. The one or more non-transitory computer-accessible storage media as recited in claim 17, wherein to provide, to the datastore engine, a request to create the local table, the program instructions when executed on or across the one or more processors cause the one or more processors to: provide, from the function to the datastore engine via an

application programming interface (API) of the datastore engine, an identifier of the local table and a source for the local table.

19. The one or more non-transitory computer-accessible storage media as recited in claim 15, further comprising program instructions that when executed on or across the one or more processors further cause the one or more processors to: receive, from the function by the datastore engine, a request for data from the local table that corresponds to a particular key; in response to determining that the local table does not include the particular key, obtain, at the datastore engine from the source table that is assigned as the source for the local table, the data that corresponds to the particular key; insert, into the local table of the edge datastore by the datastore engine, the data and the particular key; and send the data from at the datastore engine to the function.

20. The one or more storage non-transitory computer-accessible media as recited in claim 15, further comprising program instructions that when executed on or across the one or more processors further cause the one or more processors to: receive, from another function by the datastore engine, a request for the same data from the local table, wherein the other function is in another isolated execution environment at the edge location; and in response to a determination that the local table includes the requested data, provide, from at the datastore engine to the other function, the requested data.

briefHtml: BACKGROUND
(1) Content delivery networks (CDNs) often use data centers at different geographic locations (referred to herein as “points of presence” (POPs) or “edge locations”) to deliver content to clients. A CDN may set up edge locations in different geographical locations that are physically close to clients in order to provide faster responses for content that is requested by the clients. When a client requests data from the CDN (e.g., via a web browser or other application), the requested data may be retrieved from an origin database, provided to the client, and stored at the edge location for quick data retrieval if the same request is made for the same data. However, different client requests for data require additional data retrieval from the origin database, which introduces network transmission delays to obtain data from the origin and to return the data to the client. A client may also have the ability to use scripts to handle processing of a request at the edge location. However, these scripts have limited capability. For various security reasons, they are unable to make network calls. For example, they are unable to access databases or make changes to databases at the CDN.

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20210900

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abstractHtml: An aqua-ozone hygienization system enables hygienization of items through selectable treatment modes: atomized mist for non-wetting deodorization, spray for wetting disinfection, and immersion for purifying, deodorizing, and disinfecting. The system includes a tank with a platform configured to support items and form a water reservoir below. An electrochemical generator produces ozonated concentrate liquid, while a circulating pump delivers the liquid to a switchable nozzle, enabling misting or spraying. The platform may be stationary or retractable, facilitating immersion by accumulating liquid above the platform or lowering items into the reservoir. Egress ports manage liquid flow, ensuring proper drainage post-treatment. Integrated sensors monitor liquid levels and ozone concentration, while a control system automates operations, offering precise treatment customization. This versatile system is ideal for environments requiring efficient, safe, and adaptable cleaning, such as fitness centers, laundromats, and homes, promoting hygiene without harmful byproducts.

descriptionHtml: BRIEF DESCRIPTION OF THE FIGURES

(1) The subject matter which is regarded as the invention is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other objects, features, and advantages of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIGS. 1-2 illustrate examples of an aqua-ozone hygienization system;

(3) FIG. 3 illustrates one example of a system block diagram for an aqua-ozone hygienization system;

(4) FIG. 4 illustrates one example of a control system for an aqua-ozone hygienization system;

(5) FIG. 5 illustrates one example of transitioning an aqua-ozone hygienization system between a deodorizing atomized mist, a disinfection spray, a portable spray, and a purifying, deodorizing, and disinfecting immersion;

(6) FIG. 6 illustrates examples of operating an aqua-ozone hygienization system;

(7) FIG. 7 illustrates one example of an aqueous ozone generator;

(8) FIG. 8 illustrates one example of a system and network diagram;

(9) FIG. 9 illustrates one example of an ozone disinfection database structure;

(10) FIG. 10 illustrates one example of a technician's use of a software application;

(11) FIG. 11 illustrates one example of a floor plan to monitor geofenced or geolocate room spaces that have installed an aqua-ozone hygienization system;

(12) FIGS. 12-15 illustrates examples of methods of transitioning an aqua-ozone hygienization system between a deodorizing atomized mist, a disinfection spray, and a purifying, deodorizing, and disinfecting immersion; and

(13) FIGS. 16-17 illustrate exemplary embodiments that can be used interchangeably with the methods of the present invention.

(14) The detailed description explains the preferred embodiments of the invention, together with advantages and features, by way of example with reference to the drawings.

DETAILED DESCRIPTION OF THE INVENTION

(15) The present invention relates to an aqua-ozone hygienization system and methods of use. In an exemplary embodiment, the present invention cleaning system is scientifically advanced using water infused with ozone, a powerful natural oxidizer, to achieve high levels of cleanliness and hygiene. The system works by delivering ozonated water, referred to as ozonated concentrate liquid, through different methods atomized misting (in a non-wetting manner), spraying (in a wetting manner), or immersion (in a soaking manner). Each method is tailored to specific purifying, deodorizing, and disinfection needs. Atomized Misting: Provides a fine spray of ozonated concentrate liquid that can deodorize and freshen surfaces or fabrics without making them too wet. In this regard, such surfaces and fabrics after treatment can be immediately used or worn. Spraying: Applies a more substantial amount of ozonated concentrate liquid

to thoroughly disinfect and sanitize surfaces or materials, ensuring harmful bacteria and germs are eliminated. Immersion: Involves purifying items in ozonated concentrate liquid, allowing for deep cleaning and disinfection that removes odors, sweat, and other contaminants.

(16) An advantage of the present invention is that it leverages the natural disinfecting power of ozone-infused water, making it an effective and versatile solution for maintaining cleanliness and hygiene in various environments. Furthermore, the ozone is produced from water by way of an electrochemical generator which increases the purity level of the ozone produced (in excess of 20%) and eliminates harmful nitrogen species compounds that are produced from corona discharge-created ozone gas.

(17) In the present invention, the term “hygienization” is intended to mean a comprehensive process aimed at improving the cleanliness and safety of a surface, a textile item, other items, or environment. This process typically involves: Sanitizing: Reducing the number of harmful microorganisms to a safe level; Deodorizing: Eliminating or neutralizing unpleasant odors caused by bacteria, mold, or other sources; and Disinfecting: Killing or inactivating pathogens, including bacteria, viruses, and fungi, to prevent the spread of illness.

(18) When aqueous ozone is used for hygienization, it acts as a powerful agent that not only reduces the microbial load but also removes odors and destroys pathogens, resulting in thoroughly cleaned items and safe environments. The term hygienization emphasizes both cleanliness and health, ensuring that the treated items and areas are hygienic, fresh, and free from harmful contaminants.

(19) In the present invention, the term “aqua-ozone hygienization” and its design as a system and methods of use are intended to mean a multifunctional cleaning system that employs aqueous ozone-created ozonated water, referred to as ozonated concentrate liquid, to achieve comprehensive hygiene through various application methods. The system allows for precise control of ozonated concentrate liquid delivery, enabling it to be atomized, misted, sprayed, or used in immersion, depending on the specific hygiene requirement.

(20) Turning now to the drawings in greater detail, it will be seen that in FIGS. 1 and 2 there is illustrated one example of an aqua-ozone hygienization system 100. In an exemplary embodiment, for example, and not a limitation, a fitness center use case is used to illustrate the different methods of treatment in the present invention. The aqua-ozone hygienization system 100 can be used in other exemplary embodiments, areas, and environments without limitation. Such other use cases can be factories, warehouses, auto shops, laundromats, retail stores, home use, locker rooms, and numerous other uses, locations, and environments, as may be required and/or desired in a particular embodiment.

(21) In an exemplary embodiment, of use in a fitness center 210: Towel 202A Treatment: Towel 202A can be placed 402 in the aqua-ozone hygienization system 100. The towel 202A can then be atomized and misted with ozonated concentrate liquid 118, allowing them to remain mostly dry and immediately useable while neutralizing odors. Alternatively, towels 202 can be sprayed to become damp, effectively disinfecting them by killing bacteria and other pathogens. As an example, user 304 may atomize mist the towel 202A before workout and use it so the towel is fresh and deodorized. Then after the workout and use of the towel wiping sweat from user 304 or exercise equipment 210, user 304 can spray the towel for a more comprehensive disinfection. Workout Clothes 202C-D Treatment: Workout clothes 202C-D can be placed 406 in the aqua-ozone hygienization system 100. The Sweaty workout clothes 202C-D can then be immersed in ozonated concentrate liquid 118, allowing them to be purified, rinsed, deodorized, and disinfected, leaving them fresh, clean, and odor-free. Post-Workout Clothing 202B Treatment: After a workout, a shirt or blouse 202B can be placed 404 in the aqua-ozone hygienization system 100. Such clothing can then be misted with ozonated concentrate liquid 118 in a non-wetting manner. This treatment keeps the fabric mostly dry while deodorizing it, ensuring that the clothing doesn't smell like the gym and is ready for immediate wear, whether to the office, school, home, or other occasions. Fitness Equipment Treatment: A spray bottle 144 can be filled with ozonated concentrate liquid 118 and then sprayed, in a sanitizing manner, onto the fitness equipment and other surfaces. Such fitness equipment 210 can be weights, treadmills, yoga mats, and other fitness equipment after each use, ensuring a clean and safe environment for all users 304.

(22) For disclosure purposes, towels 202A, shirts or clothing 202B, workout clothing 202C-D, and other items can all be referred to as item 202. As such item 202 can be placed in the aqua-ozone hygienization system 100 for treatment.

(23) An advantage of the present invention is that it is designed to adapt to various cleaning needs within the fitness center and numerous other environments, providing a versatile, ozone-powered solution for maintaining high standards of hygiene and cleanliness. In this regard, a multi-step process aimed at achieving a high standard of environmental cleanliness by utilizing chemical and physical methods to reduce, inactivate, or eliminate harmful microorganisms, pathogens, and organic compounds. In an aqua-ozone hygienization system 100, this process relies on the properties of ozonated water (water infused with ozone (O.sub.3) created by an electrochemical generator), to achieve its effects.

(24) Scientific Basis:

(25) Ozone as an Oxidizing Agent: Ozone is a triatomic molecule consisting of three oxygen atoms. It is a powerful oxidizing agent, meaning it can readily react with and break down the cell walls of bacteria, viruses, and other pathogens. When ozone comes into contact with organic matter or microorganisms, it oxidizes its cellular components, leading to the rupture of cell walls and the subsequent inactivation or death of the organism. Ozonated Water: Also referred to as ozonated concentrate liquid 118, in the aqua-ozone hygienization system 100, ozone is generated, by way of an electrochemical generator 516, in water, creating a potent cleaning solution. The ozonated concentrate liquid 118 retains the oxidizing properties of ozone while being safer and easier to handle than ozone gas and absent dangerous nitrogen species of compounds produced by corona discharge methods. The aqueous ozone acts upon contact, breaking down contaminants, neutralizing odors, and disinfecting surfaces.

Application Methods: Misting: Delivers a fine spray of ozonated water onto item 202 surface, allowing for broad surface coverage without significant moisture buildup. This is effective for deodorizing and lightly sanitizing surfaces where maintaining dryness is important. Spraying: Involves a more concentrated application of ozonated water, which wets the surface of item 202 more thoroughly, allowing for deeper penetration into contaminants. This is particularly useful for disinfecting and removing more stubborn microorganisms. Immersion: Items 202 are fully submerged in ozonated water, ensuring comprehensive contact with the solution. This method is ideal for deeply embedded contaminants or for items that require a more intensive cleaning process.

Chemical Reactions: When ozone (O.sub.3) encounters organic materials or pathogens, it oxidizes them, typically resulting in the formation of oxygen (O.sub.2) and harmless byproducts such as carbon dioxide (CO.sub.2) or water (H.sub.2O). The breakdown of these materials leads to effective deodorization, sanitization, and disinfection, as the underlying causes of odors and contamination are chemically neutralized.

Benefits: Broad-Spectrum Efficacy: Ozonated concentrate liquid 118 is effective against a wide range of microorganisms, including bacteria, viruses, fungi, and protozoa. Eco-Friendly: The use of ozone leaves no harmful residues, as it naturally decomposes back into oxygen after reacting, making it an environmentally friendly option. Versatility: The ability to mist, spray, or immerse items in ozonated water allows for tailored hygienization based on specific needs, ensuring optimal results across different scenarios.

(26) With reference to at least FIGS. 1 and 2, in an exemplary embodiment, an aqua-ozone hygienization system 100 can comprise a tank 104 that is configured to hold item 202 for hygienization. Hygienization is effectuated by way of user 304 selecting between treating item 202 with an atomized mist in a non-wetting deodorizing manner, a spray in a wetting disinfecting manner, or an immersion in a purifying, deodorizing, and disinfecting manner.

(27) The aqua-ozone hygienization system 100 can further comprise a platform positioned 142 within tank 104. Platform 142 forms a water reservoir 140 located beneath platform 142. Platform 142 has a top surface for placing item 202 to be treated.

(28) The aqua-ozone hygienization system 100 can further comprise an electrochemical generator 516 that is configured to generate an ozonated concentrate liquid 118 from water. A circulating pump is configured to draw water from the water reservoir 140 and supply the water to the electrochemical generator 516. A nozzle 102 that is in fluid communication with the electrochemical generator 516. The nozzle 102 can be switched between dispensing the

ozonated concentrate liquid 118 onto item 202 as the atomized mist or as the spray. An egress port 162 is configured to drain the ozonated concentrate liquid 118 from the top surface of platform 142.

(29) In operation, during treatments involving the atomized mist or the spray of the ozonated concentrate liquid 118 onto item 202, the egress port 162 can be configured to abate accumulation of the ozonated concentrate liquid 118 on the top surface of the platform 142. And, during treatments involving the immersion of item 202 in the ozonated concentrate liquid 118, nozzle 102 can be configured as the spray and the egress port 162 can be configured to allow the ozonated concentrate liquid 118 to accumulate on the top of the platform 142 immersing the item 202 in the ozonated concentrate liquid 118. Upon completion of the immersion, the egress port 162 can be configured to discharge the ozonated concentrate liquid 118 from the top surface of the platform 142.

(30) Referring to FIG. 2, there are illustrated examples of an aqua-ozone hygienization system 100. In an exemplary embodiment, in reference 'A' there is illustrated an aqua-ozone hygienization system 100 comprising a control panel 106 and an external nozzle 122.

(31) Reference 'B' better illustrates the control panel 106. In an exemplary embodiment, a user interface also referred to as a control panel 106 for the aqua-ozone hygienization system 100 can comprise at least one of the following a display 506, a display 506 with touchscreen, a communication interface 508 configured to data communicate with a remote data processing resource 702 such as a server 702 and/or a computing device 732.

(32) The control panel 106 for the aqua-ozone hygienization system 100 can further comprise a plurality of button input capabilities by way of the GPIO 510, or other user interfaces. The user interface is operationally related to the microcontroller 502

(33) In an exemplary embodiment, one such button can be a deodorize/disinfect button 110. Such a button 110 can be used to initiate treatment cycles for items placed on the platform 142.

(34) Another such button can be the ozonated concentrate liquid release button 124. Such ozonated concentrate liquid release button 124 can initiate the dispensing of ozonated concentrate liquid from the external nozzle 122. In this regard, an external nozzle 122 can be positioned external to tank 104 and can be configured to fill an external vessel 144.

(35) Additionally, a light bar/display 112, a battery charge indicator 116 (for battery operation), and a power port 126 for recharging batteries can be included. Such light bar/display 112 can be used to indicate states of operation through color change, flashing, text display, or other suitable methods. Such features can be part of display 506. General purpose inputs/outputs (GPIO) 510, power supply 526, or other suitable integrations with control system 500, as may be required and/or desired in a particular embodiment.

(36) In another exemplary embodiment, and with reference to reference 'C', an external nozzle 122 can be positioned within a concave portion 120 and configured to fill an external vessel 144 with the ozonated concentrate liquid 118. Tank 104 comprises the concave portion 120 along the exterior surface. The external vessel 114 fits and is storable, in a removable manner, within the concave portion 120. An overhang roof 154 can be part of the concave portion 120 to aid in positioning the external nozzle 122 to efficiently fill external vessel 114 with ozonate concentrate liquid 118.

(37) Referring to FIG. 3, there is illustrated one example of a system block diagram for an aqua-ozone hygienization system 100. In an exemplary embodiment, an aqua-ozone hygienization system 100 can comprise a tank 104 that is configured to hold item 202 for hygienization. Hygienization can be effectuated by way of user 304 selecting between treating the item with an atomized mist in a non-wetting deodorizing manner, a spray in a wetting disinfecting manner, or an immersion in a purifying, deodorizing, and disinfecting manner.

(38) Platform 142 can be positioned within tank 104. Platform 142 can form a water reservoir 140 located beneath platform 142. Platform 142 can have a top surface for placing item 202 to be treated. An electrochemical generator 516 can be configured to generate an ozonated concentrate liquid 118 from water 114. A circulating pump 128B can be configured to draw water 114 from the water reservoir 140 and supply the water 114 to the electrochemical generator 516.

(39) Nozzle 102 can be in fluid communication with the electrochemical generator 516. Nozzle 102 can be switchable

132 between dispensing the ozonated concentrate liquid 118 onto the item as the atomized mist or as the spray. In this regard, to switch between the atomized mist and spray, nozzle 102 can be mechanically changed (manually by user 304 or automatically by control system 500), the ozonated concentrate liquid 118 can be routed to different apertures in nozzle 102, the nozzle 102 can comprise more than one sub-nozzles—some atomized mist and some spray, valves can be controlled by the control system 500/valve control 524 to switch flow routes or direct flow to different sub nozzles, or other suitable methods can be implemented, as may be required and/or desired in a particular embodiment.

(40) An ingress port 164 can be configured to allow water 114 to fill the water reservoir 140. In an exemplary embodiment is the water source 114 is under sufficient pressure a pump 128A may not be needed. In such a configuration, a bypass 146 can be used. Otherwise, in low and no water pressure situations a pump 128A can utilized. Additionally, in some exemplary embodiment, a portion of the water supply 114 can be routed 148 to the input of the circulating pump 128B or alternatively routed 146 directly to the electrochemical generator 516. Valves 130C-D can be used to manage the flow of water 114 through out the system, as required and/or desired in a particular embodiment.

(41) In an exemplary embodiment, in operation, during treatments involving the atomized mist or the spray of the ozonated concentrate liquid 118 onto the item the ozonated concentrate liquid can be absent accumulation on the top surface of the platform 142. And, during treatments involving the immersion of item 202, water 114 can be added by way of the ingress port 164 to the water reservoir 140, raising the water level 410 to an immersion level 412 above platform 142, immersing item 202 that is placed on platform 142. The circulating pump 128B and the electrochemical generator 516 can be operated 416 (drawing water 114 from the water reservoir 140, producing ozonated concentrate liquid 118, and dispensing the ozonated concentrate liquid 118 into tank 104) to produce and dispense, by way of the nozzle 102, the ozonated concentrate liquid 118 into tank 104 mixing with water 114 creating a mixture 114/118 until a desired ozone concentration 420 of the mixture 114/118 is achieved. With reference to at least FIGS. 4 and 6, reference 'D', an ozone concentration sensor 522 can be configured to determine the ozone concentration level 420 of the ozonate concentrate liquid 118.

(42) In an exemplary embodiment, the electrochemical generator 516 can further comprise an ion exchange material 534. Additionally, a consumables cartridge 158 can be inserted, in a removable manner, into the aqua-ozone hygienization system 100. The consumables cartridge can comprise the electrochemical generator 516 as well as the ion exchange material 534 and other consumables. In this regard, as the consumables are consumed over time during the use of the aqua-ozone hygienization system 100, it is easy for a technician 302 to easily unscrew 134 or otherwise unfastened, remove, and replace the consumables cartridge 158 to keep the aqua-ozone hygienization system 100 operating efficiently.

(43) In an exemplary embodiment, an egress port 162 can be positioned to discharge 136 the ozonated concentrate liquid 118, water 114, of the mixture 114/118 at the platform level 162A and/or the water reservoir level 162B. Valves 130E-F can be used to control the discharge 136, as may be required and/or desired in a particular embodiment. Additionally, valves 130A-B can be used to control the flow of the ozonated concentrate liquid 118, as may be required and/or desired in a particular embodiment.

(44) In an exemplary embodiment, a control system 500 can comprise a microcontroller 502, a memory 504, and a liquid level sensor 518. The microcontroller 502 can be operationally related to the memory 504 and the liquid level sensor 518. The memory 504 can be encoded with instructions that when executed by the microcontroller 502 perform the steps of closing the egress port 130E-F/162A-B. Placing nozzle 102 in the spray mode of operation, operating the circulating pump 128B and the electrochemical generator 516, causing the ozonated concentrate liquid 118 to be produced and dispensed into tank 104. Stopping the circulating pump 128B and the electrochemical generator 516 when the liquid level sensor 518 determines that an immersion level 412 within tank 104 of the ozonated concentrate liquid 118 has been achieved. Delaying an immersion time period with item 202 immersed in the ozonated concentrate liquid to effectuate purification, deodorization, and disinfection of item 202. And, opening the egress port 130E-F/162A-B after the immersion time period elapses, allowing the ozonated concentrate liquid 118 to be discharged 136 from tank

104. Such immersion time period can be in the range of 5 to 60 seconds, or other desired range as may be required and/or desired in a particular embodiment.

(45) Referring to FIG. 4, there is illustrated one example of a control system 500 for the aqua-ozone hygienization system 100. In an exemplary embodiment, control system 500 can be integrated into and be responsive to the action of an aqua-ozone hygienization system 100. In addition, control system 500 can be a web-enabled control system.

(46) The term “web-enabled” or “web-enabled control system” or “web-enabled control system 500” in the present invention is intended to mean an Internet-of-things device. In this regard, a device that is capable of connecting a physical device such as an aqua-ozone hygienization system 100 to the digital world. Stated differently, web-enabling is equipping a device with the necessary electronics to be monitored, and controlled, and data communicate locally and remotely with other data-communicating devices. Such other data-communicating devices can be smartphones, tablets, laptops, mobile communication devices, other web-enabled devices, remote data processing resources, servers, and similar devices.

(47) In addition, and with reference to at least FIG. 8, such data communicating devices 732 can data communicate with remote data processing resources 702 and store and retrieve data from databases 706A-C, and other data processing resources, as may be required and/or desired in a particular embodiment. Laptops, smartphones, smartwatches, tablets, desktop computers, servers, mobile communication devices, and other types and kinds of data communication devices can all be data communicating devices 732 also referred to as computing devices 732.

(48) In operation, a technician 302, an administrator 304, or other authorized people can use computing device 732 to interact with the aqueous ozone generator 530 or aqua-ozone hygienization system. The aqueous ozone generator 530 can comprise the electrochemical generator 516 and ion exchange material 534.

(49) In this regard, a technician 302 can be a person who operates, maintains, cleans, configures, repairs and performs other functions on or with the aqueous ozone generator 530 or aqua-ozone hygienization system 100. An administrator 304 can be a person who administers, provides remote service or technical support, or be other types or kinds of authorized user, as may be required and/or desired in a particular embodiment.

(50) In operation the control system 500, by way of the communication interface 508 can data communicate with remote data processing resources 702. Such remote data processing resources 702 can be servers or other types or kinds of data processing resources. Furthermore, data communicating devices 732, remote data processing resources 702, data storage resources 706A-C, and other types and kinds of data communicating devices can data communicate over a global network 700. The Internet is a global network 700.

(51) In an exemplary embodiment and with reference to at least FIG. 4, the aqua-ozone hygienization system 100 can be equipped with a web-enabled control system 500. Such a web-enabled control system 500 can comprise a microcontroller 502 which is operationally related to a memory 504, a display 506, a plurality of communication interfaces 508, general purpose input and outputs (GPIO) 510, a plurality of sensors 512, a global position system (GPS) 514, an electrochemical generator 516, a liquid level sensor 518, a pump controller 520 that is operationally related to a plurality of pumps/valves 128, a plurality of ozone sensors 522, a valve controller 524, a power supply 526, current sensor 528, an aqueous ozone generator 530, and a water heater 536.

(52) The microcontroller 502 can be INTEL, ZILOG, MICROCHIP, AMD, ARM, and/or other types or kinds of microcontrollers.

(53) The memory 504 can be a combination of random access memory (RAM), read-only memory (ROM), flash, hard drives, solid-state drives, USB flash drives, and/or other types and kinds of memory.

(54) The display 506 can be a liquid crystal display (LCD), organic light emitting diode (OLED), or light emitting diode (LED), as well as have touch input capabilities and/or other types and kinds of displays and user inputs as may be required and/or desired in a particular embodiment.

(55) The communication interface 508 can be LAN, WAN, USB, Ethernet, RS232, RS485, serial, WiFi, 802.11abgn and similar, second-generation (2G), third-generation (3G), fourth-generation (4G), or fifth-generation (5G) compatible,

Bluetooth, TCP, UDP, Mesh Network, Zigbee, Pico Network, LORAN, and/or other types and kinds of communication interfaces and protocols.

(56) In an exemplary embodiment, the communication interface 508 is operationally related to the microcontroller 502. The control system 500, by way of the communication interface 508, data communicates with the remote data processing resource 702, data communication devices 732, and other data processing resources in a local area network environment or a wide area network environment across a global network 700 in a wired or wireless manner as may be required and/or desired in a particular embodiment. The Internet is a global network 700.

(57) In an exemplary embodiment, the control system 500 can comprise a microcontroller 502, a memory 504, and a communication interface 508. The microcontroller 502 can be operationally related to the memory 504 and the communication interface 508. The memory 504 can be encoded with instructions that when executed by the microcontroller 502 perform the step of sending an operational status message by way of the communication interface 508 to a remote data processing resource 702. Such an operational status message can be related to the operational performance of the electrochemical generator 516. Additionally, the control system 500 further comprises a global position system (GPS) 514 that can be operationally related to the microcontroller 502. In operation, the operational status message can be related to the GPS location of the aqua-ozone hygienization system 100. In a plurality of other exemplary embodiments, the operational status message can be related to other operational, performance, and other data or conditions as may be required and/or desired in a particular embodiment.

(58) The GPIO 510 can be transistor-to-transistor (TTL), complementary metal-oxide-semiconductor (CMOS), transistors, buffers, relays, pushbuttons, switches, and/or other types and kinds of GPIO circuits.

(59) The sensors 512 and/motion sensor 518 can be passive infrared (PIR) motion sensors, infrared, thermal, Doppler radar, ultrasonic, capacitance, touch-type, optical, Hall effect, switch, fingerprint, and other types of biometric sensors, and/or other types and kinds of sensors. Additionally, sensor 512 can be ambient condition sensors such as temperature, moisture, humidity, total dissolved solids (TDS), sunlight, and/or other types and kinds of sensors.

(60) In an exemplary embodiment, analog-type sensor determinations can be converted to digital values so that the microcontroller 502 can process the data. Alternatively, the microcontroller 502 can perform analog-to-digital conversions if equipped to perform such functions.

(61) The electrochemical generator 516 can be an electrolysis-based device that utilizes ion exchange material 534 and other devices and processes to produce chemical compounds from water such as ozone O.sub.3.

(62) The liquid level sensor 518 can be utilized to determine the water 114 level in the water reservoir 140, the ozonated concentrate liquid 118 accumulated in tank 104, mixture 114/118 of water 114 and the ozonated concentrate liquid 118 in tank 104, or other suitable liquid levels, as may be desired and/or desired in a particular embodiment.

(63) The pumps and/or valves 520, in addition to performing their fluid handling tasks, pumps and/or valves 520 can be actuated and/or controlled by way of a relay, metal-oxide-semiconductor field-effect transistor (MOSFET), or other types and kinds of controlling devices. In addition, valve controller 524 and operationally related to valves/actuators 130 (better illustrated in at least FIG. 3 as valves 130A-F) such as solenoids, and other fluid management devices can be integrated into the system, as may be required and/or desired in a particular embodiment. Such valves/actuators 130 can be opened and closed to control the flow of water 114, ozonated concentrate liquid 118, discharge 136, and used in other ways to control other fluids or gases as may be required and or desired in a particular embodiment.

(64) The ozone sensor 522 can be configured to monitor the ozone concentration supplied to the system or other sources of ozonated liquid, as may be required and/or desired in a particular embodiment.

(65) The power supply 526 can be AC, DC, battery including rechargeable, solar, and/or other types and kinds of power supplies.

(66) The current sensor 528 can be configured to measure the supplied electrical current to the electrochemical generator 516, the aqueous ozone generator 530, a combination 516/132 thereof, and/or other devices and systems, as may be required and/or desired in a particular embodiment.

(67) The aqueous ozone generator 530 receives water as an input and uses the electrochemical generator 516 which is integrated into the aqueous ozone generator 530 to produce high concentrations of aqueous ozone molecules. Such concentrations of aqueous ozone can range from 1 ppm to 10 ppm or other desired range, as may be required and/or desired in a particular embodiment.

(68) The water heater 538 can be a resistive type or other suit type of water heater. In an exemplary embodiment, the water heater 538 can be configured to warm the water 114 in the water reservoir 140. In this regard, as users 304 interact with and touch the water 114 and ozonated concentrate liquid 118, the water can be warmed to a comfort level making the experience more enjoyable for the user 304.

(69) In an exemplary embodiment, a user interface also referred to as a control panel 106 for the aqua-ozone hygienization system 100 can comprise at least one of the following a display 506, a display 506 with touchscreen, a communication interface 508 configured to data communicate with a remote data processing resource 702 such as a server 702 and/or a computing device 732.

(70) The control panel 106 for the aqua-ozone hygienization system 100 can further comprise a plurality of button input capabilities by way of the GPIO 510, or other user interfaces. The user interface is operationally related to the microcontroller 502.

(71) Referring to FIG. 5, there is illustrated one example of transitioning an aqua-ozone hygienization system between a deodorizing atomized mist, a disinfection spray, a portable spray, and a purifying, deodorizing, and disinfecting immersion.

(72) In an exemplary embodiment, in the application of, for example, and not a limitation, item 202 treatment (such as towels, clothing, other textiles, or other suitable non-textile items), the present invention utilizes ozonated concentrate liquid 118, delivered in several distinct ways, each tailored for specific treatment needs:

(73) Atomizing Mist: in reference 'A', the water level is low 410 typically confined to the water reservoir 140. The nozzle 102 is configured 132 for a finely atomized mist of the ozonated concentrate liquid 118. The ozonated concentrate liquid 118 is dispersed onto the surface of item 202, creating a non-wetting treatment. This method is ideal for deodorizing fabrics by neutralizing odors without significantly dampening the material, making it suitable for items that need to remain dry and ready for immediate use.

(74) Spray Application: In reference 'B', the water level is low 410 typically confined to the water reservoir 140. The nozzle 102 is configured 132 for spraying the ozonated concentrate liquid 118. In this regard, a more concentrated spray of ozonated liquid is applied, resulting in a wetting effect on item 202. This method is designed for disinfecting fabrics by thoroughly wetting the surface, allowing the ozonated concentrate liquid 118 to penetrate and eliminate pathogens, bacteria, and other harmful microorganisms.

(75) Immersion: In reference 'C', Textiles are fully submerged in the ozonated concentrate liquid 118, providing a comprehensive treatment that purifies, deodorizes, and disinfects in a wetting manner. This approach ensures deep cleaning by allowing the ozone to penetrate every fiber of the fabric, effectively removing contaminants, neutralizing odors, and eliminating harmful microorganisms. In an exemplary embodiment, item 202 on platform 142 is flooded to an immersion level 412. This can happen in a couple of different ways, three of which are illustrated below: (1) The nozzle 102 can be configured 132 to spray and the egress port 130E-F/162 is configured to allow the ozonated concentrate liquid 118 to accumulate on the top of the platform 142 immersing item 202 in the ozonated concentrate liquid 118. Upon completion of the immersion, the egress port 130E-F/162 can be configured to discharge 136 the ozonated concentrate liquid 118 from the top surface of the platform 142; (2) The nozzle 102 can be configured 132 to spray. Water 114 is added by way of the ingress port 164 to the water reservoir 140, raising the water 114 level to an immersion level 412 above platform 142, immersing item 202 on platform 142. The circulating pump 128B and the electrochemical generator 516 are operated to produce and dispense, by way of the nozzle 102, the ozonated concentrate liquid 118 into the tank 104 mixing with the water 114 creating a mixture 114/118 until a desired ozone concentration 420 of the mixture 114/118/422 is achieved; or (3) The nozzle 102 can be configured 132 to spray. Platform 142 with item 202 on top can

be transitioned from an extended position 156A (as better illustrated in at least FIG. 6, reference 'B' platform 142A) to a retracted position 156B (as better illustrated in at least FIG. 6, reference 'B' platform 142B) lowering item 202 into and immersing item 202 in the water reservoir 14. The circulating pump 128B and the electrochemical generator 516 are operated to produce and dispense, by way of nozzle 102, the ozonated concentrate liquid 118 into the tank 104 mixing with water 114 creating a mixture 114/118 until a desired ozone concentration 42 of the mixture 114/118/422 is achieved. Portable Spray: in reference 'D', a spray bottle 144 can be filled with ozonated concentrate liquid 118 and then sprayed, in a sanitizing manner, onto the surfaces and equipment, such a fitness equipment 210, and other equipment as an example, and not a limitation. Such fitness equipment 210 can be weights, treadmills, yoga mats, and other fitness equipment after each use, ensuring a clean and safe environment for all users 304.

(76) These methods provide flexibility in treating items 202 of all types including textiles and non-textiles, depending on the desired outcome, whether it's maintaining dryness while deodorizing, achieving deeper disinfection with wetting, or fully purifying through immersion.

(77) Referring to FIG. 6, there are illustrated examples of operating an aqua-ozone hygienization system. In an exemplary embodiment, in reference 'A', the aqua-ozone hygienization system 100 comprises tank 104, and platform 142 is positioned inside tank 104. Platform 142 forms a water reservoir 140 located beneath platform 142. Platform 142 has a top surface for placing item 202 such as a towel, shirt, other clothing, or other suitable items to be treated. Aqueous ozone generator 530 can comprise electrochemical generator 516. Electrochemical generator 516 can be configured to generate the ozonated concentrate liquid 118 from water 114. Circulating pump 128 can be configured 416 to draw water 114 from the water reservoir 140 and supply water 114 to the electrochemical generator 516. Nozzle 102 can be in fluid communication with the electrochemical generator 516. Nozzle 102 can be switchable 132 between dispensing the ozonated concentrate liquid 118 onto item 202 as the atomized mist or a spray. Egress port 162/130E can be configured to drain 136 the ozonated concentrate liquid 118 from the top surface of platform 142. In this regard, in an exemplary embodiment, platform 142 can form a liquid barrier and prevent mixing between the water reservoir 140 and ozonated concentrate liquid dispenses onto the top surface of platform 140.

(78) Treating item 202 with an immersion in a purifying, deodorizing, and disinfecting manner. During treatments involving immersion of item 202 in the ozonated concentrate liquid 118, nozzle 102 can be configured 132 as a spray, and the egress port 162/130E can be configured to allow the ozonated concentrate liquid 118 to accumulate on the top of platform 148 immersing item 202 in the ozonated concentrate liquid 118. Upon completion of the immersion, the egress port 162/130E can be configured to discharge 136 the ozonated concentrate liquid 118 from the top surface of platform 142.

(79) In an exemplary embodiment, in reference 'B', the aqua-ozone hygienization system 100 comprises tank 104, and platform 142 is positioned within tank 104. Platform 142 in an extended position 142A extended, motion 156A, forms a water reservoir 140 which comprises water located beneath platform 142, and in a retracted position 142B retracted, motion 156B, platform 142 is immersed in the water 114 within the water reservoir 140. Platform 142 has a top surface for placing item 202 such as a towel, shirt, other clothing, or other suitable items to be treated. Aqueous ozone generator 530 can comprise electrochemical generator 516. Electrochemical generator 516 can be configured to generate the ozonated concentrate liquid 118 from water 114. Circulating pump 128 can be configured 416 to draw water 114 from water reservoir 140 and supply water 114 to the electrochemical generator 516. Nozzle 102 can be in fluid communication with the electrochemical generator 516. Nozzle 102 can be switchable 132 between dispensing the ozonated concentrate liquid 118 onto item 202 as the atomized mist or a spray.

(80) Treating item 202 with an immersion in a purifying, deodorizing, and disinfecting manner. During treatments involving the immersion of item 202, platform 142 can be is the retracted position lowering item 202 into and immersing item 202 in the water in the water reservoir 140. Circulating pump 128 and the electrochemical generator 516 are operated 416 to dispense, by way of nozzle 102, the ozonated concentrate liquid 118 into tank 104 mixing with the water 114 creating a mixture 114/118 until a desired ozone concentration 420 of mixture 114/118/422 is achieved. Such ozone

concentration 422 of the mixture 114/118 can be determined by way of ozone sensor 522, or other suitable devices or methods.

(81) In an exemplary embodiment, in reference 'C', the aqua-ozone hygienization system 100 comprises tank 104, and platform 142 is positioned inside tank 104. Platform 142 forms a water reservoir 140 located beneath platform 142. Platform 142 has a top surface for placing item 202 such as a towel, shirt, other clothing, or other suitable items to be treated. Electrochemical generator 516 can be configured to generate the ozonated concentrate liquid 118 from water 114. Circulating pump 128 can be configured 416 to draw water 114 from the water reservoir 140 and supply water 114 to the electrochemical generator 516. Nozzle 102 can be in fluid communication with the electrochemical generator 516. Nozzle 102 can be switchable 132 between dispensing the ozonated concentrate liquid 118 onto item 202 as the atomized mist or a spray. Ingress port 164 can be configured to allow water 114 to fill the water reservoir 140.

(82) Treating item 202 with an immersion in a purifying, deodorizing, and disinfecting manner. During treatments involving immersion of item 202, water 114 can be added by way of the ingress port 164 to the water reservoir 140, raising the water level 410, permeating 152 platform 142, to an immersion level 412 above platform 142, immersing item 202 on platform 142. Circulating pump 128 and the electrochemical generator 516 are operated 416 to dispense, by way of the nozzle 102, the ozonated concentrate liquid 118 into tank 104 mixing with water 114 creating a mixture 114/118 until a desired ozone concentration 420 of mixture 114/118/422 is achieved. In this regard, in an exemplary embodiment, platform 142 can be permeable 152 allowing an overfilling of the water reservoir level 410 to rise through platform 142, allowing the water level to reach an immersion level 412. Such ozone concentration 422 of the mixture 114/118 can be determined by way of ozone sensor 522.

(83) In an exemplary embodiment, in FIG. 6, reference 'D' chart 418 illustrates the relationship between ozone concentration 422 of the mixture 114/118 of water 114 and ozonated concentrate liquid 118 in relationship to the system run time of the circulating pump 128 and the electrochemical generator 516. In this regard, as the system runs the ozonated concentrate liquid 118 is dispensed into tank 104. In immersion treatments that mix the water 144 and the ozonated concentrate liquid 118, in the tank, the ozone concentration 442 of the mixture 114/118/422 can be determined by way of ozone sensor 522, or other suitable devices or methods.

(84) In an exemplary embodiment, during operation, the circulating pump 128 and the electrochemical generator 516 can be operated to produce and dispense, by way of the nozzle 102, the ozonated concentrate liquid 118 into the tank mixing with the water 114 creating a mixture 114/118 until a desired ozone concentration 420 of the mixture 114/118 is achieved. Such ozone concentration 422 of the mixture 114/118 can be determined by way of ozone sensor 522.

(85) Referring to FIG. 7, there is illustrated one example of an aqueous ozone generator 530. In an exemplary embodiment, aqueous ozone production involves electrolysis and an ion exchange material 534 which consumes certain components and materials during the process. Thus, when the service life of the aqueous ozone generator 530 and/or electrochemical generator 516 is over the consumables 516/534 need to be replaced. To ease and speed consumable 516/534 exchange by technician 302 a plumbed housing 160 can be permanently plumbed or otherwise fastened in place and the aqueous ozone generator 530 comprising the consumables 516/534, screwed 132 on to or otherwise fastened 132, in a removable manner, to the plumbed housing 160. In this regard, the aqueous ozone generator 530 can be easily and quickly removed as a cartridge 158 and exchanged for a new aqueous ozone generator 530 and electrochemical generator 516/ion exchange material 534 when the consumables 516/534 need to be replaced.

(86) An advantage, in the present invention, is that once the plumbed housing 160 is installed and the liquid lines connected, the liquid lines don't need to be removed or disconnected to change the aqueous ozone generator 530. This saves technician 302 time, and cost, and reduces the chance of creating leaks in the system by having to disconnect/reconnect liquid carrying hoses.

(87) Another advantage, in the present invention, is how the outflow of the assembled unit 158/160 can be configured. In this regard, in reference 'A' the assembled unit 158/160 can be configured to produce an ozonated concentrate liquid 118, or in reference 'B' and as better illustrated in at least FIG. 3 configured to produce an outflow of an ozonated

concentrate liquid 118 stream and a water 114 stream. These streams 114/118 can be mixed at the point of dispense or mixed in the air as they are sprayed from one or more of the nozzles 102/122, as may be required and/or desired in a particular embodiment.

(88) In an exemplary embodiment, in operation, a plumbed housing 160 can be fastened in fluid communication pathways with the water source 114 and the ozonated concentrate liquid 118. An electrochemical generator 516 is integrated into the aqueous ozone generator 530. The electrochemical generator 516 comprises an ion exchange material 534. The aqueous ozone generator 530 is interchangeable and removably fastened to the plumbed housing 160 in a replaceable cartridge 158 manner.

(89) In an exemplary embodiment, a consumables cartridge 158 can be inserted, in a removable manner, into the aqua-ozone hygienization system 100. The consumables cartridge 158 can comprise the electrochemical generator 516 and other consumable components such as the ion exchange material 534.

(90) In an exemplary embodiment, an electrochemical generator 516 can be integrated into the aqueous ozone generator 530. The electrochemical generator comprises an ion exchange material 534. A computing device 732, operated by technician 302, can data communicate the test ozone concentration to a remote data processing resource 702, and receive from the remote data processing resource 702, by way of the computing device 732 a plurality of aqueous ozone generator service life data that corresponds to the remaining service life of the electrochemical generator 516.

(91) Referring to FIG. 8, there is illustrated one example of a system and network diagram. In an exemplary embodiment, users of the platform and network can include technicians 302, administrators 304, or other authorized persons.

(92) Each of the users uses computing devices 732A-C to data communicate over a global communication network 700 with one or more data processing resources 702. The computing devices 732A-C can be laptop computers, desktop computers, smartphones, tablets, or other types and kinds of computing devices, as may be required and/or desired in a particular embodiment. For disclosure purposes, computing devices 732A-C can be referred to as computing devices 732. Additionally, laptop and desktop types of computing devices 732 can be referred to as computing devices 712C, computing devices 732 such as smartphones can be referred to as computing devices 732B, and computing devices 732 such as tablets can be referred to as computing devices 732A. In operation, any of the users can use any of the types of computing devices 732A-C, without limitation to the type or kind of computing device 732, as may be required and/or desired in a particular embodiment. The global communication network 700 can be the Internet.

(93) The computing devices 732 can comprise a microprocessor 704B/704C, a database 706B/706C, memory 708B/708C, a communication interface 710B/710C, a display 712B/712C, and a plurality of general-purpose inputs and outputs (GPIO) 714B/714C.

(94) Additionally, mobile type of computing device 732A/732B (tablets, smartphones, and others) can comprise a global positioning system (GPS) 716, and a microphone and/or camera 718.

(95) In general, computing devices 232 can be configured with other functions and features, as may be required and/or desired in a particular embodiment.

(96) In an exemplary embodiment, the microprocessor 704B is operationally related to database 706B, memory 708B, communication interface 710B, display 712B, and GPIO 714B.

(97) In an exemplary embodiment, the microprocessor 704C is operationally related to database 706C, memory 708C, communication interface 710C, display 712C, GPIO 714C, and if equipped, with GPS 716, and microphone and/or camera 718. The computing devices 732 each rely on a suitable power source 720B/720C which can include a rechargeable battery, external power supply, or other types and/or kinds of power sources.

(98) Microprocessor 704B/704C can be INTEL, ZILOG, MICROCHIP, AMD, ARM, and/or other types or kinds of microprocessors.

(99) Database 706B/706C can be SQL, MYSQL, MARIADB, ORACLE, MS ACCESS, network-accessible storage, flat

files, a combination thereof, or other types and kinds of databases.

(100) Memory 708B/708C can be a combination of RAM, ROM, flash, hard drives, solid-state drives, USB flash drives, micro-SD cards, or other types of removable memory, and/or other types and kinds of memory.

(101) The communication interfaces 710B/710C can be local area network (LAN), wide area network (WAN), universal serial bus (USB), Ethernet, RS232, RS485, serial, Wi-Fi, 802.11abgn and similar, 2G 3G 4G 5G compatible, Bluetooth, transmission control protocol (TCP), user datagram protocol (UDP), Mesh Network, Zigbee, Pico Network, long-range navigation (LORAN), and/or other types and kinds of communication interfaces and protocols.

(102) Display 712B/712C can be a liquid crystal display (LCD), light emitting diode (LED), organic light emitting diode (OLED), or other types and kinds of displays.

(103) The general-purpose inputs and outputs (GPIO) 714B/714C can be TTL, CMOS, MOSFET, transistors, buffers, relays, pushbuttons, switches, and/or other types and kinds of GPIO circuits. In an exemplary embodiment, some of the GPIO 214 lines can be used to drive a touch screen input, biometric input devices, keyboards, and/or types and kinds of computing device input devices.

(104) Global positioning system (GPS) device 716 can be used to determine the geographic location of technician 302 and others who are carrying a computing device 732 equipped with a GPS 716. In this regard, such computing devices 732 are typically mobile computing devices such as tablets 732A, smartphones 732B, and other similar types and/or kinds of mobile computing devices 732.

(105) Microphone and/or camera 718 can be used to record audio, and video, and take pictures. In this regard, users 304/306 can use their computing devices equipped with a microphone and/or camera 718 to make digital media records that can be selectively shared as appropriate including on social media and other digital media outlet locations.

(106) With reference to FIG. 8, the data processing resource 702 can be a server, network storage device, or other types and kinds of data processing resources. Such data processing resources can be AMAZON WEB SERVICES (AWS), MICROSOFT AZURE, or other types and kinds of hosted data processing resource services. For disclosure purposes, a remote data processing resource 702 can also be referred to as server 702.

(107) The data processing resource 702 can comprise a microprocessor 704A, a database 706A, memory 708A, and a communication interface 710A. The microprocessor 704A is operationally related to database 706A, memory 708A, and communication interface 710A.

(108) The microprocessor 704A can be INTEL, ZILOG, MICROCHIP, AMD, ARM, and/or other types or kinds of microprocessors.

(109) The database 706A can be SQL, MYSQL, MARIADB, ORACLE, MS ACCESS, network accessible storage, flat files, a combination thereof, or other types and kinds of databases.

(110) The memory 708A can be a combination of RAM, ROM, flash, hard drives, solid-state drives, USB flash drives, micro-SD cards, or other types of removable memory, and/or other types and kinds of memory.

(111) The communication interfaces 710A can be LAN, WAN, USB, Ethernet, RS232, RS485, serial, Wi-Fi, 802.11abgn and similar, 2G 3G 4G 5G compatible, Bluetooth, TCP, UDP, Mesh Network, Zigbee, Pico Network, LORAN, and/or other types and kinds of communication interfaces and protocols.

(112) Referring to FIG. 9, there is illustrated one example of an ozone disinfection/deodorization database structure 800. In an exemplary embodiment, at least one database 706A/706B/706C can be implemented on at least one of the data processing resources 702 also referred to as server 702, or computing devices 732. In operation, one or more databases 706A/706B/706C can be accessed/created/managed/maintained as appropriate by more than one stakeholder. In this regard, in addition to system administrators and other authorized persons, other stakeholders can access/create/manage/maintain as appropriate.

(113) In an exemplary embodiment, such databases 706A/706B/706C can be SQL, MYSQL, MARIADB, ORACLE, MS ACCESS, network-accessible storage, flat files, a combination thereof, or other types and kinds of databases.

(114) In an exemplary embodiment, the ozone disinfection/deodorization database 900 can reside on a remote data

processing resource 702 in database 706A. In this regard, the ozone disinfection/deodorization database 800 can comprise a series of tables, records, fields, and accounts that include account record 802, location record 804, geofence record 806, equipment record 808, self-test results 810, technician record 812, notification record 814, ozonator record 816, and/or other types or kinds of records as may be required and/or desired in a particular embodiment. The database structure illustrated in FIG. 9 also illustrates some of the relationships between the various tables and fields.

(115) In an exemplary embodiment, the data structure is illustrative and can be expanded and modified without particular limitation as needed and as appropriate to support the functionality and methods of the present invention and to support future functionality and methods of the present invention as it grows and evolves over time.

(116) Referring to FIG. 10, a technician's use of a software application. In an exemplary embodiment, a software application or website can be used in combination with the computing device 732A/B to identify the aqua-ozone hygienization system 100, view operational statuses, record the test results and other results, and see other useful data by way of data communicating with a remote data processing resource 702. In some embodiments, certain air deodorizing system 100 may have the ability to data communicate 734 directly with a remote data processing resource 702, eliminating the need for computing device 732A/B to act as an intermediary device to record test results on the remote data processing resource 702.

(117) In an exemplary embodiment and with reference to at least FIG. 10, a computing device 732, operated by technician 302, data communicates with a remote data processing resource 702, and receives from the remote data processing resource 702, by way of the computing device 732 a plurality of aqua-ozone hygienization system 100 location 838 and service life data 840/842 that corresponds to the remaining service life the electrochemical generator 516 which can include the ion exchange material 534, and other service life information, as may be required and/or desired in a particular embodiment.

(118) Referring to FIG. 11, there is illustrated one example of a floor plan 844 to monitor geofenced or geolocate room spaces 'A' 846A, 'B' 846B, 'C' 846C, and 'D' 846D that have installed 208 an aqua-ozone hygienization system 100. In an exemplary embodiment, technician 302, by way of computing platform 732, can data communicate with server 732 or data communicate 206 with individual aqua-ozone hygienization system 100 to ascertain the operational status and location with a floor plan area 844.

(119) Referring to FIG. 12, there is illustrated one example of a method of transitioning an aqua-ozone hygienization system 100 between a deodorizing atomized mist, a disinfection spray, and a purifying, deodorizing, and disinfecting immersion. In an exemplary embodiment, the method begins in step 1002 by treating an item 202 with an atomized mist of the ozonated concentrate liquid 118, in a non-wetting deodorizing manner.

(120) As better illustrated in at least FIGS. 3 and 4, the aqua-ozone hygienization system 100 comprises tank 104, and platform 142 is positioned inside tank 104. Platform 142 forms a water reservoir 140 located beneath platform 142. Platform 142 has a top surface for placing item 202 such as a towel, shirt, other clothing, or other suitable items to be treated. Electrochemical generator 516 can be configured to generate the ozonated concentrate liquid 118 from water 114. Circulating pump 128A can be configured 416 to draw water 114 from the water reservoir 140 and supply water 114 to the electrochemical generator 516. Nozzle 102 can be in fluid communication with the electrochemical generator 516. Nozzle 102 can be switchable 132 between dispensing the ozonated concentrate liquid 118 onto item 202 as the atomized mist or a spray. Egress port 162/130E can be configured to drain 136 the ozonated concentrate liquid 118 from the top surface of platform 142. In this regard, in an exemplary embodiment, platform 142 can form a liquid barrier and prevent mixing between the water reservoir 140 and ozonated concentrate liquid dispenses onto the top surface of platform 140.

(121) The method continues in step 1004 by treating item 202 with a spray of the ozonated concentrate liquid in a wetting disinfecting manner. During treatments involving the atomized mist or the spray of the ozonated concentrate liquid 118 onto item 202, the egress port 162/130E can be configured to abate the accumulation of the ozonated concentrate liquid 118 on the top surface of the platform 142.

(122) The method continues in step 1006 by treating item 202 with an immersion in a purifying, deodorizing, and disinfecting manner. During treatments involving immersion of item 202 in the ozonated concentrate liquid 118, nozzle 102 can be configured 132 as a spray, and the egress port 162/130E can be configured to allow the ozonated concentrate liquid 118 to accumulate on the top of platform 148 immersing item 202 in the ozonated concentrate liquid 118. Upon completion of the immersion, the egress port 162/130E can be configured to discharge 136 the ozonated concentrate liquid 118 from the top surface of platform 142.

(123) Referring to FIG. 13, there is illustrated one example of a method of using the aqua-ozone hygienization system. In an exemplary embodiment, the method begins in step 1102 by transitioning, by user 304, the aqua-ozone hygienization system 100 between steps 1104, 1106, 1108, and 1110:

(124) In step 1104 a deodorizing mist, wherein nozzle 102 can be configured to dispense the atomized mist onto the item that is placed on platform 142 in a non-wetting deodorizing manner.

(125) In step 1106, a disinfecting spray, wherein nozzle 102 can be configured to dispense the spray onto the item that is placed on platform 142 in a wetting manner.

(126) In step 1108, an immersion, wherein nozzle 102 can be configured for the spray and the ozonated concentrate liquid 118 accumulates on platform 142 immersing the item that is placed on platform 142 in the ozonated concentrate liquid 118 in a purifying, deodorizing, and disinfecting manner.

(127) In step 1110, filling the external vessel 144 with the ozonated concentrate liquid 118, wherein the ozonated concentrate liquid 118 can be sprayed on items exterior to tank 104 and item 202 in a deodorizing and disinfecting manner.

(128) Referring to FIG. 14, there is illustrated one example of a method of transitioning an aqua-ozone hygienization system 100 between a deodorizing atomized mist, a disinfection spray, and a purifying, deodorizing, and disinfecting immersion. In an exemplary embodiment, the method begins in step 1202 by treating item 202 with an atomized mist of the ozonated concentrate liquid 118, in a non-wetting deodorizing manner.

(129) As better illustrated in at least FIGS. 3 and 4, the aqua-ozone hygienization system 100 comprises tank 104, and platform 142 is positioned inside tank 104. Platform 142 forms a water reservoir 140 located beneath platform 142. Platform 142 has a top surface for placing item 202 such as a towel, shirt, other clothing, or other suitable items to be treated. Electrochemical generator 516 can be configured to generate the ozonated concentrate liquid 118 from water 114. Circulating pump 128A can be configured 416 to draw water 114 from the water reservoir 140 and supply water 114 to the electrochemical generator 516. Nozzle 102 can be in fluid communication with the electrochemical generator 516. Nozzle 102 can be switchable 132 between dispensing the ozonated concentrate liquid 118 onto item 202 as the atomized mist or a spray. Ingress port 164 can be configured to allow water 114 to fill the water reservoir 140.

(130) The method continues in step 1204 by treating item 202 with a spray of the ozonated concentrate liquid in a wetting disinfecting manner. During treatments involving the atomized mist or the spray of the ozonated concentrate liquid 118 onto item 202, the egress port 162/130E can be configured to abate the accumulation of the ozonated concentrate liquid 118 on the top surface of the platform 142.

(131) The method continues in step 1206 by treating item 202 with an immersion in a purifying, deodorizing, and disinfecting manner. During treatments involving immersion of item 202, water 114 can be added by way of the ingress port 164 to the water reservoir 140, raising the water level 410, permeating 152 platform 142, to an immersion level 412 above platform 142, immersing item 202 on platform 142. Circulating pump 128A and the electrochemical generator 516 are operated 416 to dispense, by way of the nozzle 102, the ozonated concentrate liquid 118 into tank 104 mixing with water 114 creating a mixture 114/118 until a desired ozone concentration 420 of mixture 114/118/422 is achieved. In this regard, in an exemplary embodiment, platform 142 can be permeable 152 allowing an overfilling of the water reservoir level 410 to rise through platform 142, allowing the water level to reach an immersion level 412.

(132) Referring to FIG. 15, there is illustrated one example of a method of transitioning an aqua-ozone hygienization system 100 between a deodorizing atomized mist, a disinfection spray, and a purifying, deodorizing, and disinfecting

immersion In an exemplary embodiment, the method begins in step 1302 by treating an item 202 with an atomized mist of an ozonated concentrate liquid 118, in a non-wetting deodorizing manner.

(133) As better illustrated in at least FIGS. 3 and 4, the aqua-ozone hygienization system 100 comprises tank 104, and platform 142 positioned within tank 104. Platform 142 in an extended position (better illustrated in at least FIG. 6 as platform 142A extended, motion 156A) forms a water reservoir 140 which comprises water located beneath platform 142 and in a retracted position (better illustrated in at least FIG. 6 as platform 142B retracted, motion 156B) platform 142 is immersed in the water 114 within the water reservoir 140. Platform 142 has a top surface for placing item 202 such as a towel, shirt, other clothing, or other suitable items to be treated. Electrochemical generator 516 can be configured to generate the ozonated concentrate liquid 118 from water 114. Circulating pump 128A can be configured 416 to draw water 114 from water reservoir 140 and supply water 114 to the electrochemical generator 516. Nozzle 102 can be in fluid communication with the electrochemical generator 516. Nozzle 102 can be switchable 132 between dispensing the ozonated concentrate liquid 118 onto item 202 as the atomized mist or a spray.

(134) The method continues in step 1304 by treating item 202 with a spray of the ozonated concentrate liquid in a wetting disinfecting manner. During treatments involving the atomized mist or the spray of the ozonated concentrate liquid 118 onto item 202 platform 142 is in the extended position and the ozonated concentrate liquid 118 is absent accumulation on the top surface of the platform 142.

(135) The method continues in step 1306 by treating item 202 with an immersion in a purifying, deodorizing, and disinfecting manner. During treatments involving the immersion of item 202, platform 142 can be in the retracted position lowering item 202 into and immersing item 202 in the water in the water reservoir 140. Circulating pump 128A and the electrochemical generator 516 are operated 416 to dispense, by way of nozzle 102, the ozonated concentrate liquid 118 into tank 104 mixing with the water 114 creating a mixture 114/118 until a desired ozone concentration 420 of mixture 114/118/422 is achieved.

(136) Referring to FIG. 16, there are illustrated exemplary embodiments that can be used interchangeably with the methods of the present invention. In an exemplary embodiment, the method begins in step 1402 by closing the egress port 162, then in step 1404 by placing the nozzle 102 in the spray mode 132 of operation.

(137) The method continues in step 1406 by operating 416 (drawing water 114 from the water reservoir 140, producing ozonated concentrate liquid 118, and dispensing the ozonated concentrate liquid 118 into tank 104) the circulating pump 128B, and the electrochemical generator 516, causing the ozonated concentrate liquid 118 to be produced and dispensed into the tank 104.

(138) The method continues in step 1408 by stopping 416 the circulating pump 128B and the electrochemical generator when the liquid level sensor 518 determines that an immersion level 412 within tank 104 of the ozonated concentrate liquid 118 has been achieved.

(139) The method continues in step 1410 by delaying an immersion time period with item 202 immersed in the ozonated concentrate liquid 118. Such immersion time period can be in the range of 5 to 60 seconds, or other desired range as may be required and/or desired in a particular embodiment.

(140) The method continues in step 1412 by opening the egress port 130E-F/162 after the immersion time period elapses, allowing the ozonated concentrate liquid 118 to be discharged 136 from tank 104. The method is then exited.

(141) Referring to FIG. 17, there are illustrated exemplary embodiments that can be used interchangeably with the methods of the present invention.

(142) In step 1502, sending an operational status message by way of a control system 500 to a remote data processing resource 702.

(143) In an exemplary embodiment, the communication interface 508 can effectuate the actual data communication exchange with the remote data processing resource 702. In this regard, sending an operational status message by way of a communication interface 508 to a remote data processing resource 702. The control system comprises the communication interface 508. The control system can be integrated into the aqua-ozone hygienization system 100.

(144) In an exemplary embodiment, in operation, the operational status message can be related to the operational performance of the electrochemical generator 516.

(145) In an exemplary embodiment, in operation, the control system 500 can further comprise a global position system (GPS) 514. The operational status message can be related to the GPS location of the aqua-ozone hygienization system 100.

(146) In a plurality of other exemplary embodiments, the operational status message can be related to other operational, performance, and other data or conditions as may be required and/or desired in a particular embodiment.

(147) In step 1504, filling the water reservoir 140 by way of an ingress port 164. The aqua-ozone hygienization system 100 comprises the ingress port 164.

(148) In step 1506, positioning an external nozzle, within a concave portion 120, to fill an external vessel 144 with the ozonated concentrate liquid 118. Tank 104 comprises the concave portion 120 along the exterior surface. The external vessel 144 fits and is storable, in a removable manner, within the concave portion 120.

(149) In an exemplary embodiment, an external nozzle 122 can be positioned within a concave portion 120 and configured to fill an external vessel 144 with the ozonated concentrate liquid 118.

(150) In step 1508, replacing a consumables cartridge 158 that is inserted, in a removable manner, into the aqua-ozone hygienization system 100. The consumables cartridge 158 comprises the electrochemical generator 516. In an exemplary embodiment, the consumables cartridge 158 can also comprise the ion exchange material 534, and other consumer items, as may be required and/or desired in a particular embodiment.

(151) In step 1510, draining the ozonated concentrate liquid 118 from the top surface of the platform 142 and the water reservoir 140 by way of an egress port 162. The aqua-ozone hygienization system 100 comprises the egress port 162.

(152) In step 1512, draining the mixture (of water 114 and ozonated concentrate liquid 118) from the water reservoir 140 by way of an egress port 162. The aqua-ozone hygienization system comprises the egress port 162.

(153) The capabilities of the present invention can be implemented in software, firmware, hardware, or some combination thereof.

(154) As one example, one or more aspects of the present invention can be included in an article of manufacture (e.g., one or more computer program products) having, for instance, computer-usable media. The media has embodied therein, for instance, computer-readable program code means for providing and facilitating the capabilities of the present invention. The article of manufacture can be included as a part of a computer system or sold separately.

(155) Additionally, at least one program storage device readable by a machine, tangibly embodying at least one program of instructions executable by the machine to perform the capabilities of the present invention can be provided.

(156) The flow diagrams depicted herein are just examples. There may be many variations to these diagrams or the steps (or operations) described therein without departing from the spirit of the invention. For instance, the steps may be performed in a differing order, or steps may be added, deleted, or modified. All of these variations are considered a part of the claimed invention.

(157) While the preferred embodiment of the invention has been described, it will be understood that those skilled in the art, both now and in the future, may make various improvements and enhancements which fall within the scope of the claims which follow.

claimsHtml: 1. An aqua-ozone hygienization system comprising: a tank that is configured to hold an item for treatment with ozonated concentrated liquid; an electrochemical generator that is configured to generate an ozonated concentrate liquid from water; a circulating pump that is configured to draw water from a water reservoir and supply the water to the electrochemical generator; a nozzle that is in fluid communication with the electrochemical generator, the nozzle is switchable between treating the item by dispensing the ozonated concentrate liquid onto the item in the tank as an atomized mist or as a spray; and an egress port from the tank that is configured to abate the accumulation of the ozonated concentrate liquid in the tank, or to allow the accumulation of ozonated concentrated liquid in the tank, during treatment of the item.

2. The aqua-ozone hygienization system in accordance with claim 1, the electrochemical generator further comprises an ion exchange material.

3. The aqua-ozone hygienization system in accordance with claim 1, further comprising: an ingress port that is configured to allow water to fill the water reservoir.

4. The aqua-ozone hygienization system in accordance with claim 1, further comprising: an ozone concentration sensor that is configured to determine the ozone concentration level of the ozonate concentrate liquid.

5. The aqua-ozone hygienization system in accordance with claim 1, further comprising: an external nozzle that is positioned within a concave portion of the aqua-ozone hygienization system and configured to fill an external vessel with the ozonated concentrate liquid, the tank comprises the concave portion along the exterior surface, the external vessel fits and is storable, in a removable manner, within the concave portion.

6. The aqua-ozone hygienization system in accordance with claim 1, further comprising: an external nozzle positioned external to the tank that is configured to fill an external vessel.

7. The aqua-ozone hygienization system in accordance with claim 1, further comprising: a consumables cartridge that is inserted, in a removable manner, into the aqua-ozone hygienization system, the consumables cartridge comprises the electrochemical generator.

8. A method of using the aqua-ozone hygienization system of claim 1, the method comprising the steps of: the user of the aqua-ozone hygienization system selecting one of the following methods of treatment: a deodorizing mist, wherein the nozzle dispenses the atomized mist onto the item; a disinfecting spray, wherein the nozzle dispenses the spray onto the item and the egress port abates the accumulation of the ozonated concentrate liquid in the tank; and an immersion, wherein the nozzle dispenses the spray onto the item, and the egress port allows the accumulation of the ozonated concentrate liquid in the tank.

9. The aqua-ozone hygienization system in accordance with claim 1, further comprising: a control system comprising a microcontroller, a memory, and a communication interface, the microcontroller is operationally related to the memory and the communication interface, the memory is encoded with instructions that when executed by the microcontroller perform the step of: sending an operational status message by way of the communication interface control system to a remote data processing resource.

10. The aqua-ozone hygienization system in accordance with claim 9, wherein the operational status message is related to the operational performance of the electrochemical generator.

11. The aqua-ozone hygienization system in accordance with claim 1, the control system further comprises a global position system (GPS) that is operationally related to the microcontroller, wherein the operational status message is related to the GPS location of the aqua-ozone hygienization system.

12. The aqua-ozone hygienization system in accordance with claim 1, further comprising: a control system comprising a microcontroller, a memory, and a liquid level sensor, the microcontroller is operationally related to the memory and the liquid level sensor, the memory is encoded with instructions that when executed by the microcontroller perform the steps of: closing the egress port; placing the nozzle in the spray mode of operation; operating the circulating pump and the electrochemical generator, causing the ozonated concentrate liquid to be produced and dispensed into the tank; stopping the circulating pump and the electrochemical generator when the liquid level sensor determines that an immersion level within the tank of the ozonated concentrate liquid has been achieved; delaying an immersion time period with the item immersed in the ozonated concentrate liquid; and opening the egress port after the immersion time period elapses, allowing the ozonated concentrate liquid to be discharged from the tank.

13. The aqua-ozone hygienization system in accordance with claim 1, further comprising: a water line from the water reservoir to the tank that adds water from the water reservoir to the tank.

14. The aqua-ozone hygienization system in accordance with claim 13, further comprising: an ozone concentration sensor that is configured to determine the ozone concentration level of the mixture in the tank.

15. The aqua-ozone hygienization system in accordance with claim 13, further comprising: an egress port that is

configured to drain the ozonated concentrate liquid from the top surface of the platform and the water reservoir.

16. The aqua-ozone hygienization system in accordance with claim 13, further comprising: an external nozzle that is positioned within a concave portion of the aqua-ozone hygienization system and configured to fill an external vessel with the ozonated concentrate liquid, the tank comprises the concave portion along the exterior surface, the external vessel fits and is storable, in a removable manner, within the concave portion.

17. A method of using the aqua-ozone hygienization system of claim 13, the method comprising the steps of: transitioning, by the user, the aqua-ozone hygienization system between: deodorizing mist, wherein the nozzle is configured to dispense the atomized mist onto the item that is placed on the platform in a non-wetting deodorizing manner; disinfecting spray, wherein the nozzle is configured to dispense the spray onto the item that is placed on the platform in a wetting manner; immersion, wherein the water level is increased in the tank, the ozonated concentrate liquid mixes with the water forming the mixture that immerses the item on the platform in a purifying, deodorizing, and disinfecting manner; and filling the external vessel with the ozonated concentrate liquid, wherein the ozonated concentrate liquid is sprayed on items exterior to the tank in a deodorizing and disinfecting manner.

18. The aqua-ozone hygienization system in accordance with claim 13, further comprising: a control system comprising a microcontroller, a memory, and a communication interface, the microcontroller is operationally related to the memory and the communication interface, the memory is encoded with instructions that when executed by the microcontroller perform the step of: sending an operational status message by way of a control system to a remote data processing resource.

19. An aqua-ozone hygienization system comprising: a tank that is configured to hold an item for treatment with ozonated concentrated liquid; a platform that is positioned within the tank that can be in an extended position above a water reservoir or in a retracted position immersed in the water reservoir, the platform having a top surface for placing the item to be treated; an electrochemical generator that is configured to generate an ozonated concentrate liquid from water; a circulating pump that is configured to draw water from a water reservoir and supply the water to the electrochemical generator; a nozzle that is in fluid communication with the electrochemical generator, the nozzle is switchable between treating the item by dispensing the ozonated concentrate liquid onto the item in the tank as an atomized mist or as a spray; and an egress port from the tank that is configured to abate the accumulation of the ozonated concentrate liquid in the tank, or to allow the accumulation of ozonated concentrated liquid in the tank, during treatment of the item.

20. The aqua-ozone hygienization system in accordance with claim 19, further comprising: an ozone concentration sensor that is configured to determine the ozone concentration level of the mixture in the tank.

21. The aqua-ozone hygienization system in accordance with claim 19, further comprising: an egress port that is configured to drain the mixture from the water reservoir.

22. A method of using the aqua-ozone hygienization system of claim 19, the method comprising the steps of: the user of the system selecting one of the following methods of treatment: a deodorizing mist, wherein the nozzle dispenses the atomized mist onto the item; a disinfecting spray, wherein the nozzle dispenses the spray onto the item and the egress port abates the accumulation of the ozonated concentrate liquid in the tank; an immersion, wherein the nozzle dispenses the spray onto the item, and the egress port allows the accumulation of the ozonated concentrate liquid in the tank; and filling the external vessel with the ozonated concentrate liquid, wherein the ozonated concentrate liquid is sprayed on items exterior to the tank in a deodorizing and disinfecting manner.

briefHtml: TECHNICAL FIELD OF THE INVENTION
(1) The present invention relates to systems and methods for hygienization using ozonated aqueous solutions. Specifically, the invention pertains to an aqua-ozone hygienization system designed to sanitize, deodorize, and disinfect items through selectable treatment modes, including atomized misting, spraying, and immersion. The invention addresses the need for efficient and adaptable hygienization systems that minimize environmental and health risks by utilizing ozonated water generated through electrochemical processes. Applications of the invention span various environments, including fitness centers, homes, and commercial facilities,

where safe, effective, and versatile cleaning solutions are essential.
BACKGROUND OF THE INVENTION
(2)

Before our invention, washing systems face several significant shortcomings that limit their public use effectiveness, safety, and convenience, particularly in modern, fast-paced environments. One shortcoming is the use of ozone generation by way of corona discharge in many conventional wash systems. While this method produces ozone for use, it also inadvertently generates harmful nitrogen species compounds as byproducts. These nitrogen species compounds, when inhaled, can pose serious health risks, including respiratory issues and other adverse effects. As a result, corona discharge-generated ozone systems are unsafe for use in environments where humans may be exposed to the ozone gas (and undesirable nitrogen species), such as in wash areas or laundries, where users could come into contact with or breathe the gas. This safety concern significantly restricts the practical application of such systems in everyday environments and in particular in public environments.
(3) Another shortcoming of prior wash systems, such as those resembling kitchen sinks, is their reliance on soaking items in water. While soaking is effective for thorough cleaning, it completely wets the clothing or textile items, requiring a drying phase before the items can be used or worn again. This process is particularly inconvenient when users need the items to be immediately usable after washing, as is often the case in time-sensitive situations or on-the-go scenarios.
(4) Furthermore, existing wash systems are generally not designed to meet the needs of users in fast-paced, on-the-go environments like fitness centers, gyms, or other active venues. In these settings, users often need to quickly freshen up textile items such as towels and workout clothes before use, thoroughly clean them after use, and freshen up items like shirts or blouses to be worn immediately after a workout without residual odors like gym smells, sweat, or cigarette smoke. Traditional wash systems do not offer the flexibility to accommodate these varied needs, particularly in a way that allows for immediate use of the items without a drying period. This lack of adaptability makes them ill-suited for modern lifestyle demands, where efficiency, safety, and convenience are essential.
(5) The present invention addresses these and other shortcomings by providing a new solution and treatment methods that are safe, effective, and adaptable to a wide range of environments.

For these reasons and shortcomings as well as other reasons and shortcomings there is a long-felt need that gives rise to the present invention.
SUMMARY OF THE INVENTION
(6)

The shortcomings of the prior art are overcome and additional advantages are provided through the provision of a treatment system comprising a tank configured to hold an item for treatment with ozonated concentrate liquid. The system includes an electrochemical generator that produces ozonated concentrate liquid from water and a circulating pump that delivers water from a reservoir to the electrochemical generator. A nozzle, in fluid communication with the generator, is switchable to dispense the ozonated concentrate liquid onto the item in the tank as either an atomized mist for deodorization or a spray for disinfection. An egress port from the tank facilitates precise control of the liquid, either preventing accumulation for misting and spraying or allowing accumulation for immersion treatments. This system provides a versatile and efficient solution for hygienization, addressing safety, environmental, and operational challenges found in prior systems.
(7) Additional shortcomings of the prior art are overcome and additional advantages are provided through the provision of a method for using an aqua-ozone hygienization system. The method involves a user selecting a desired treatment mode: a deodorizing mist, where the nozzle dispenses atomized mist onto the item; a disinfecting spray, where the nozzle dispenses the spray onto the item while the egress port prevents liquid accumulation in the tank; and immersion, where the nozzle dispenses the spray onto the item and the egress port allows the accumulation of ozonated concentrate liquid in the tank. This method provides users with the flexibility to select and execute tailored treatment modes, ensuring optimal hygienization for various items and applications.
(8) Additional shortcomings of the prior art are overcome and additional advantages are provided through the provision of a treatment system comprising a tank configured to hold an item for treatment with ozonated concentrate liquid and a platform positioned within the tank. The platform can transition between an extended position above a water reservoir and a retracted position immersed within the reservoir, with a top surface for placing the item to be treated. The system includes an electrochemical generator that produces ozonated concentrate liquid from water and a circulating pump that supplies water to the generator. A switchable nozzle dispenses the liquid onto the item as an atomized mist for deodorization or as a spray for disinfection. An egress port

manages liquid flow, either preventing accumulation during misting and spraying or enabling accumulation for immersion treatments. This versatile system provides adaptable, efficient, and safe hygienization options.

(9) Additional shortcomings of the prior art are overcome and additional advantages are provided through the provision of a method for using an aqua-ozone hygienization system. The method includes transitioning the system between various treatment modes: a deodorizing mist, where the nozzle dispenses atomized mist onto an item on the platform in a non-wetting manner; a disinfecting spray, where the nozzle dispenses a wetting spray onto the item on the platform; and immersion, where the water level in the tank is increased to mix with the ozonated concentrate liquid, forming a purifying, deodorizing, and disinfecting mixture that immerses the item on the platform. Additionally, the method includes filling an external vessel with the ozonated concentrate liquid, which can be sprayed on items exterior to the tank for deodorization and disinfection. This method offers flexible, efficient, and user-controlled hygienization options.

(10) System and computer program products corresponding to the above-summarized methods are also described and claimed herein.

(11) Additional features and advantages are realized through the techniques of the present invention. Other embodiments and aspects of the invention are described in detail herein and are considered a part of the claimed invention. For a better understanding of the invention with advantages and features, refer to the description and the drawings.

backgroundTextHtml: CROSS-REFERENCE TO RELATED TREATMENTS

(1) This application contains subject matter which is related to the subject matter of the following co-pending application. The below-listed applications are hereby incorporated herein by reference in their entirety:

(2) This U.S. non-provisional application is a continuation in part of the following: U.S. non-provisional application Ser. No. 18/888,218, inventor Gavin Hsu et al., entitled "AQUA-OZONE HYGIENIZATION", filed Sep. 18, 2024; and U.S. non-provisional application Ser. No. 18/888,222, inventor Gavin Hsu et al., entitled "AQUA-OZONE HYGIENIZATION", filed Sep. 18, 2024; U.S. non-provisional application Ser. No. 18/782,055, inventor Darren Simmons et al., entitled "MICROFOAMING AQUEOUS OZONE DISINFECTION", filed Jul. 24, 2024; U.S. non-provisional application Ser. No. 18/966,217, inventor Gavin Hsu et al., entitled "DEODORIZING AIR USING AQUEOUS OZONE AS A CATALYST", filed Dec. 3, 2024; U.S. non-provisional application Ser. No. 18/919,065, inventor Darren Simmons et al., entitled "IMMERSION CONTAINER FOR TREATING PRODUCE WITH AN OZONATED CONCENTRATE LIQUID", filed Oct. 18, 2024; and U.S. non-provisional application Ser. No. 18/646,394, inventor Darren Simmons et al., entitled "AQUEOUS OZONE FLOOR DISINFECTION SYSTEM", filed Apr. 25, 2024.

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abstractHtml: A method and a system for detecting a semiconductor device are provided. The method comprises obtaining an image of the semiconductor device, evaluating a feature of the image, detecting a defect of the semiconductor device based on the feature, extracting a defect information for the defect, calculating a defect die ratio (DDR) in response to the defect and analyzing a relation between the DDR and the defect information.

descriptionHtml: BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the embodiments of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various structures are not drawn to scale. In fact, the dimensions of the various structures can be arbitrarily increased or

reduced for clarity of discussion.

(2) FIG. 1 is a schematic view of a system for measuring a semiconductor device, in accordance with some embodiments of the present disclosure.

(3) FIG. 2A is a schematic view of bonding two semiconductor wafers, in accordance with some embodiments of the present disclosure.

(4) FIG. 2B is a schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(5) FIG. 3 is a flowchart showing operations for detecting a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(6) FIG. 4A is a schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(7) FIG. 4B is another schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(8) FIG. 4C is another schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(9) FIG. 5A is a schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(10) FIG. 5B is another schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(11) FIG. 5C is another schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(12) FIG. 5D is another schematic view of a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(13) FIG. 6A is a schematic chart illustrating the defect die ratio and the defect size, in accordance with some embodiments of the present disclosure.

(14) FIG. 6B is another schematic chart illustrating the defect die ratio and the defect size, in accordance with some embodiments of the present disclosure.

(15) FIG. 7 is a block diagram of a system of detecting a semiconductor device, in accordance with some embodiments.

(16) FIG. 8 is a flowchart showing operations for detecting a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

(17) FIG. 9 is a flowchart showing other operations for detecting a defect in a semiconductor device, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(18) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features can be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(19) Further, spatially relative terms, such as "beneath," "below," "lower," "above," "over," "upper," "on" and the like, can be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the

device in use or operation in addition to the orientation depicted in the figures. The apparatus can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(20) As used herein, although terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may only be used to distinguish one element, component, region, layer or section from another. Terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.

(21) Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately” and “about” generally mean within a value or range that can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately” and “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

(22) FIG. 1 is a schematic view of a system 100 for detecting a semiconductor device, in accordance with some embodiments of the present disclosure. FIG. 1 shows a system 100 for testing or measuring a semiconductor wafer 10. FIG. 1 also shows a semiconductor wafer 10 on which a semiconductor die to be tested or measured can be included.

(23) The semiconductor wafer 10 can include an elementary semiconductor such as silicon, germanium, or diamond. The semiconductor wafer 10 may include one or more dies 12 formed thereon. A plurality of scribe lines 16 and 18 can be provided between adjacent dies 12 so that dies 12 can be separated or singulated in subsequent processing. In some embodiments, the dies 12 can be integrated circuits (ICs) or chips. The semiconductor wafer 10 may include a plurality of dies 12 and several process control monitoring (PCM) devices (not shown in FIG. 1A). A PCM device can be regarded as a benchmark device that can be utilized to evaluate the characteristics or performance of the dies 12.

(24) Referring to FIG. 1, the system 100 can constitute testing or measurement equipment. The system 100 may include hardware and software components that provide a suitable operational or functional environment in which the dies 12 can be tested. In some embodiments, the system 100 may include a processor 102, a signal generator 104, a monitor 106, a coupler 108, and optical sensors 110.

(25) Signals and commands can be transmitted between each of the processor 102, the signal generator 104, the monitor 106, and the coupler 108. In some embodiments, the signals transmitted within the system 100 can include power signals having adjustable voltage levels. In some embodiments, the signals transmitted within the system 100 can include optical signals having various frequencies. The system 100 can include a C-mode scanning acoustic microscope (C-SAM).

(26) In various embodiments, the processor 102 may include, but is not limited to, at least one hardware processor, including at least one microprocessor such as a CPU, a portion of at least one hardware processor, or any other suitable dedicated processor such as those developed based on Field Programmable Gate Array (FPGA) and Application

Specific Integrated Circuit (ASIC).

(27) The signal generator 104 may be configured to provide or receive test signals. All types of optical signals and electrical signals, such as acoustic signals, microwave signals, data signals, clock signals, or power signals, can be provided to a PCM device associated with the die 12. In some embodiments, optical signals constituting an image of the semiconductor wafer 10 can be provided to the signal generator 104.

(28) The monitor 106 may be configured to determine whether the semiconductor wafer 10 complies with testing criteria. The signals fed back from the semiconductor wafer 10 can be evaluated by the monitor 106, and a determination can be made whether the semiconductor wafer 10 complies with the testing criteria. The monitor 106 can provide information and/or instructions to the user. The monitor 106 can display information and/or instructions to the user. In some embodiments, the monitor 106 can display pop-up notifications. The monitor 106 can provide alarm messages to the user when the semiconductor wafer fails to comply with certain test criteria. In some embodiments, the coupler 108 may be configured to couple the processor 102 and the monitor 106 to the optical sensor 110 for measuring the semiconductor wafer 10.

(29) The optical sensor 110 can detect or receive optical signals on the semiconductor wafer 10 in order to measure characteristics and performance of the semiconductor wafer 10. In some embodiments, the optical sensor 110 may include APS, CMOS image sensor, CCD, infra-red sensor, optical-sensing transistor, or various optical cameras.

(30) FIG. 2A is a schematic view of bonding two semiconductor wafers 210 and 220, in accordance with some embodiments of the present disclosure. The semiconductor device 20 can include two semiconductor wafers 210 and 220. The semiconductor device 20 may include stacked semiconductor wafers. Bonding techniques, which may refer to bonding that involves two or more materials, can be used to form a semiconductor device 20.

(31) In some embodiments, hybrid bonding technique can be used to form a semiconductor device 20. The semiconductor device 20 can be formed by heat and compression. Hybrid bonding can refer to bonding that involves two or more materials (e.g. metal-to-metal bonding and dielectric-to-dielectric bonding). The semiconductor wafer 210 can be bonded to the semiconductor wafer 220.

(32) In some embodiments, the semiconductor wafer 220 can be bonded to the semiconductor wafer 210 by fusion. The semiconductor wafer 220 can be bonded to the semiconductor wafer 210 by an adhesive layer therebetween. The semiconductor wafer 220 can be located on the semiconductor wafer 210. An interface 230 can exist between the semiconductor wafers 220 and 210. The semiconductor wafer 220 may be in direct contact with the semiconductor wafer 210. The interface 230 can include or contact a lower surface of the semiconductor wafer 220. The interface 230 can include or contact an upper surface of the semiconductor wafer 210.

(33) FIG. 2B is a schematic view of a defect 240 of a semiconductor device 30, in accordance with some embodiments of the present disclosure. In some embodiments, a defect 240 may exist on the interface 230 between the semiconductor wafers 210 and 220. The defect 240 may occur during the process of bonding the semiconductor wafers 210 and 220. The defect 240 may occur after the process of bonding the semiconductor wafers 210 and 220. The defect 240 may occur due to a particle being introduced during the process of bonding the semiconductor wafers 210 and 220. The defect 240 may occur due to peeling off related to the semiconductor wafers 210 and 220.

(34) In some embodiments, the defect 240 can be a trap void on the interface 230. The trap void can include an air gap between the semiconductor wafers 210 and 220. A portion of the semiconductor wafer 210 may not directly contact the semiconductor wafer 220 due to the trap void. The defect 240 could result in failure of the semiconductor device 20.

(35) In some embodiments, a plurality of optical signals can be emitted from the system 100 for detecting the defects of semiconductor device 20. As shown in FIG. 2B, the optical signal L1 can transmit or penetrate the semiconductor wafer 220, the interface 230 and the semiconductor wafer 210. The optical signal L2 may be transmitted into the semiconductor wafer 220 and reflected by the defect 240 without being transmitted to the semiconductor wafer 210. The defect 240 can be detected or measured by evaluating the features of the optical signals L1 and L2. The defect 240 can be detected or measured by evaluating the features of the optical signal L2.

(36) FIG. 3 is a flowchart showing operations for detecting a defect in a semiconductor device, in accordance with some embodiments of the present disclosure. In operation 300, a semiconductor wafer is bonded to another semiconductor wafer of a semiconductor device. In operation 302, the semiconductor device is scanned to obtain an image. In operation 304, the defect of the semiconductor device and its features are identified. In operation 306, outgoing quality assurance (OQA) yield defect inspection is performed. More details for the operations 300, 302, 304, and 306 are provided in the embodiments of FIG. 4A to FIG. 4C.

(37) Note that the disclosed ordering of such acts or events is not intended to be interpreted in a limiting sense. For example, some operations may occur in different order and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(38) FIG. 4A is a schematic view of an image of a defect 440 of a semiconductor device 400, in accordance with some embodiments of the present disclosure. The embodiment of FIG. 4A may correspond to the operation 302 of FIG. 3. The image of FIG. 4A is a top view of the semiconductor wafer 40. The image can be obtained by scanning the semiconductor device 400. The semiconductor wafer 40 includes a plurality of dies, such as dies 41, 42, 43, 44, and 45. In addition, a defect 440 exists within the portion 46 near the edge of the semiconductor wafer 40.

(39) FIG. 4B is another schematic view of the portion 46 including the defect 440 in the semiconductor device 400, in accordance with some embodiments of the present disclosure. The embodiment of FIG. 4B may correspond to the operation 304 of FIG. 3. In some embodiments, the defect 440 of the semiconductor device 400 and its defect information are identified. The defect information can include but is not limited to at least one of a defect position, a defect perimeter, a defect area, a defect size and a total number of the defects.

(40) In some embodiments, the defect information can be used to calculate a defect die ratio (DDR) for evaluating the impact or damage caused by the defect on the semiconductor device. For example, as shown in FIG. 4B, the defect 440 is located between the edge of the semiconductor wafer 40 and the dies 44 and 45. Accordingly, the dies 44 and 45 may be affected and damaged since they are near or close to the defect 440.

(41) The defect information can be extracted or evaluated by executing machine-readable instructions, including source code or various appropriate programming languages. Some examples of programming languages that may be used to store the instruction set comprise, but are not limited to: Java, C, C++, Python, Objective-C, Visual Basic, or .NET programming. In some embodiments a compiler or interpreter is comprised to convert the instruction set into machine executable code for execution by the processor.

(42) FIG. 4C is another schematic view of a defect 441 of a semiconductor device 400, in accordance with some embodiments of the present disclosure. The embodiment of FIG. 4C may correspond to the operation 306 of FIG. 3. In some embodiments, the OQA yield defect inspection can be executed to determine the quality of the semiconductor device in the final stage of OQA. As shown in FIG. 4C, a defect 441 is detected, and thus the semiconductor device 400 may be scrapped or discarded.

(43) In some embodiments, a criteria such as determining a threshold defect size can be applied to evaluate whether to ship or scrap the semiconductor device 400 before the final stage of OQA. More specifically, when the defect size exceeds a threshold, the semiconductor device will be scrapped or discarded. When the defect size is smaller than or equal to the threshold, the semiconductor device will be shipped. As a result, the defect can be identified prior to the OQA stage, such that costs of the outlier wafer being released to the OQA stage can be reduced.

(44) FIG. 5A is a schematic view of an image of a defect 540 in a semiconductor device 500A, in accordance with some embodiments of the present disclosure. The image of FIG. 5A is similar to that of FIG. 4A except that further adjustments can be performed on the image for detecting and evaluating the defects.

(45) In some embodiments, wafer edge for length reference, such as the stripes in FIG. 5A, can be defined so that the image can be analyzed in detail with reference to pixels. Each die and defect can be precisely positioned by pixel

values. Therefore, defect information including brightness, position, perimeters, area, and size can be calculated or obtained accordingly.

(46) FIG. 5B is another schematic view of an image of a defect 540 of a semiconductor device 500B, in accordance with some embodiments of the present disclosure. The image of FIG. 5B is similar to that of FIG. 5A except that further adjustments can be performed on the image for detecting and evaluating the defects. In some embodiments, smoothing and blurring can be performed on the image in order to remove the outlier pixel. The image in FIG. 5A can be more uniform, and image contrast increased. As a result, the defect can be monitored and detected accurately.

(47) FIG. 5C is another schematic view of an image of a defect 540 in a semiconductor device 500C, in accordance with some embodiments of the present disclosure. The embodiment of FIG. 5C can be executed after that of FIG. 5B. In some embodiments, the brightness of the defect can exceed that of other portions of the image. Therefore, the brightness can be used to identify the defect 540.

(48) The brightness of the image of the semiconductor wafer 50 can be in a range of 0 (which indicates black) to 255 (which indicates white). A threshold brightness value can be determined to assess or judge the defect. The threshold brightness value may be determined by a user. For example, the threshold brightness value may be determined as 190. When the brightness is greater than the threshold brightness value, it can be identified as a defect.

(49) FIG. 5D is another schematic view of an image of a defect 540 of a semiconductor device 500D, in accordance with some embodiments of the present disclosure. The embodiment of FIG. 5D can be executed after that of FIG. 5C. In some embodiments, the defect 540 can be further contoured in order to extract multiple defect information.

(50) In some embodiments, each defect can be assigned an identification number in association with multiple defect information, such as a defect position, a defect perimeter, a defect area, and a defect size. The defect position can include pixel values corresponding to the location or position where the defect appears. The defect perimeter can indicate the size of the defect. The defect perimeter is generally proportional to the size of the defect. The defect area can indicate the region occupied by the defect. The defect area is generally proportional to the size of the defect.

(51) In some embodiments, a ratio can be obtained by dividing the defect area with the defect perimeter. The obtained ratio can be used to depict the shape of the defect. Accordingly, the defect type can be determined according to the obtained ratio related to the defect shape and the defect size. The proposed method of the present disclosure can be applied to identify or recognize the defect type. In some embodiments, the relation between the DDR and the defect information can be analyzed when the defect type is the trap void. Therefore, the defect can be inspected more precisely and efficiently.

(52) More specifically, the defect type can be identified as a trap void when the ratio is in a range of 0.8 to 2. The defect type can be identified as a trap void when the ratio is smaller than 2. The defect type can be identified as a trap void when the ratio is approximately equal to one. In some embodiments, the defect type can be identified as an edge overpolish when the ratio is in a range of 2 to 4. The defect type can be identified as an edge overpolish when the ratio is greater than 2.

(53) FIG. 6A is a schematic chart illustrating the DDR and the defect size, in accordance with some embodiments of the present disclosure. Each dot of FIG. 6A represents the DDR versus the defect size. In some embodiments, the DDR is substantially proportional to the defect size. In other words, DDR increases with defect size.

(54) In some embodiments, a threshold defect size DT1 can be determined to further analyze the trend of DDR. The threshold defect size DT1 can be determined to maximize a difference between an average DDR 610 below the threshold defect size DT1 and another average DDR 620 above the threshold defect size DT1. As shown in FIG. 6A, the threshold defect size DT1 is about 0.38 cm, the average DDR 610 is about 8.2%, and the average DDR 620 is about 17.9%. The difference between the average DDR 610 and the average DDT 620 is up to about 9.7%.

(55) Upon calculation, if the threshold defect size DT1 is greater than or smaller than 0.38 cm, the difference between the average DDR 610 and the average DDR 620 will be decreased from the maximum value of 9.7%. Therefore, the difference or gap can be determined so that the average DDR 620 is higher than the average DDR 610 as much as

possible. Afterwards, the defect can be scrapped when the defect size is greater the threshold defect size since it causes high DDR, and thus the quality of the semiconductor wafer can be improved.

(56) FIG. 6B is another schematic chart illustrating the DDR and the defect size, in accordance with some embodiments of the present disclosure. The embodiments of FIG. 6A can include three contributing factors which are related to defects and results in the DDR. The contributing factors can include the particle, the peeling and other kinds of defects caused during the bonding process. The embodiments of FIG. 6B mainly include one contributing factor including the peeling.

(57) As illustrated in FIG. 6B, the threshold defect size DT2 is about 0.38 cm, the average DDR 640 is about 0.8%, and the average DDR 650 is about 6.1%. The difference between the average DDR 640 and the average DDT 650 is up to about 5.3%. Although the difference between the average DDR 640 and the average DDR 650 in FIG. 6B is less than the difference between the average DDR 620 and the average DDR 610 in FIG. 6A, the peeling factor can be the dominant factor among all contributing factors due its higher contribution than other factors. As a result, the main cause of the defect can be analyzed and investigated by the proposed method of the present disclosure.

(58) FIG. 7 is a block diagram of a system 70 of detecting a semiconductor device, in accordance with some embodiments. FIG. 7 is a block diagram of a system 70 of detecting a semiconductor device, in accordance with some embodiments. The system 70 can include, for example, an electronic design automation (EDA) system.

(59) In some embodiments, the system 70 includes an automatic placement and routing (APR) system. Methods described herein of detecting defects in a semiconductor device, in accordance with one or more embodiments, are implementable, for example, using the system 70, in accordance with some embodiments.

(60) In some embodiments, the system 70 is a general purpose computing device including a hardware processor 702 and a memory 704. The memory 704 may be a computer-readable storage medium 704. The storage medium 704, amongst other things, is encoded with, computer program code 706 or a set of executable instructions. Execution of instructions 706 by hardware processor 702 represents (at least in part) an EDA tool which implements a portion or all of a method according to an embodiment, e.g., the methods described herein in accordance with one or more embodiments (hereinafter, the noted processes and/or methods).

(61) The processor 702 may be electrically coupled to the memory 704 (such as computer-readable storage medium) via the bus 708. The processor 702 may be electrically coupled to an I/O interface 710 by bus 708. A network interface 712 may be electrically connected to processor 702 via bus 708. Network interface 712 may be connected to a network 714, so that processor 702 and the memory 704 are capable of connecting to external elements via network 714. Processor 702 may be configured to execute computer program code 706 encoded in memory 704 in order to cause system 70 to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, the processor 702 is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(62) In one or more embodiments, the memory 704 may be an electronic, magnetic, optical, electromagnetic, infrared, and/or semiconductor system (or apparatus or device). For example, the memory 704 may include a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, memory 704 includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(63) In one or more embodiments, the memory 704 may store computer program code (instructions) 706 configured to cause system 70 (where such execution represents, at least in part, the EDA tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, the memory 704 may store information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, the memory 704 may store library 707 of standard cells including such standard cells as disclosed herein and one or more layout diagrams 705 such as are disclosed herein.

(64) The system 70 may include I/O interface 710. I/O interface 710 may be coupled to external circuitry. In one or more embodiments, I/O interface 710 may include a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to the processor 702.

(65) The system 70 may include network interface 712 coupled to processor 702. Network interface 712 may allow system 70 to communicate with network 714, to which one or more other computer systems are connected. Network interface 712 may include wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA. Network interface 712 may include wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more systems 70.

(66) The system 70 may be configured to receive information through I/O interface 710. The information received through I/O interface 710 may include one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor 702. The information may be transferred to processor 702 via the bus 708. System 70 may be configured to receive information related to a UI through I/O interface 710. The information may be stored in memory 704 as user interface (UI) 742.

(67) In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods are implemented as a software application running on the system 70. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

(68) In some embodiments, the processor 702 is realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium may include, but not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

(69) FIG. 8 is a flowchart showing operations for detecting a defect in a semiconductor device, in accordance with some embodiments of the present disclosure. In operation 800, an image of the semiconductor device can be obtained. In operation 802, a feature of each pixel of the image is calculated or evaluated, such as the brightness. In operation 804, a defect of the semiconductor device can be detected or inspected based on the feature of the pixel.

(70) Furthermore, in operation 806, a defect information for the defect can be extracted, such as a defect position, a defect perimeter, a defect area, a defect size and a total number of defects. In operation 808, a DDR in response to the defect can be calculated or simulated. In operation 810, a relationship between the DDR and the defect information can be analyzed or evaluated.

(71) It should be noted that some operations may occur in different order and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(72) FIG. 9 is a flowchart showing other operations for detecting a defect in a semiconductor device, in accordance with some embodiments of the present disclosure. In operation 900, a defect is detected or identified according to an image of the semiconductor device comprising a plurality of semiconductor dies. In operation 902, a correspondence between a defect size of the defect and a DDR can be developed or created. In operation 904, a threshold defect size can be determined so that a gap is maximized between a first average DDR below the threshold defect size and a second average DDR above the threshold defect size. In addition, the identified defect can be scrapped when the defect size

exceeds the threshold defect size.

(73) Some embodiments of the present disclosure provide a method for detecting a semiconductor device are provided. The method comprises utilizing an optical sensor to obtain an image of the semiconductor device; evaluating a feature of the image by a processor electrically coupling to the optical sensor; detecting a defect of the semiconductor device based on the feature by the processor; extracting a defect information for the defect by the processor; calculating a defect die ratio (DDR) in response to the defect by the processor; utilizing the processor to analyze a relation between the DDR and the defect information for filtering the semiconductor device; and discarding the semiconductor device when the defect information dissatisfies a threshold.

(74) Some embodiments of the present disclosure provide a method for detecting a semiconductor device are provided. The method comprises capturing an image of the semiconductor device by an optical sensor, wherein the semiconductor device comprises a plurality of dies; detecting a defect according to an the image of the semiconductor device comprising a plurality of dies by a processor electrically coupling to the optical sensor; utilizing the processor to developing a correspondence between a defect size of the defect and a defect die ratio (DDR) by the processor; and determining a threshold defect size by the processor so that a gap is maximized between a first average DDR below the threshold defect size and a second average DDR above the threshold defect size.

(75) Some embodiments of the present disclosure provide a system for detecting a semiconductor device. The system includes at least one processing unit and at least one memory including computer program code for one or more programs. The at least one memory, the computer program code and the at least one processing unit are configured to cause the system to perform: evaluate a feature of the image of the semiconductor device obtained from an optical sensor; detecting a defect of the semiconductor device based on the feature; extract a defect information for the defect; calculate a DDR in response to the defect; and analyze a relation between the DDR and the defect information; and scraping the semiconductor device when the defect information fails to meet a threshold.

(76) The foregoing outlines structures of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

claimsHtml: 1. A method for detecting a semiconductor device, comprising: utilizing an optical sensor to obtain an image of the semiconductor device; evaluating a feature of the image by a processor electrically coupling to the optical sensor; detecting a defect of the semiconductor device based on the feature by the processor; extracting a defect information for the defect by the processor; calculating a defect die ratio (DDR) in response to the defect by the processor; utilizing the processor to analyze a relation between the DDR and the defect information for filtering the semiconductor device; and discarding the semiconductor device when the defect information dissatisfies a threshold.

2. The method of claim 1, wherein the defect information comprises at least one of a defect position, a defect perimeter, a defect area, a defect size and a total number of the defect.

3. The method of claim 2, wherein the DDR is substantially proportional to the defect size.

4. The method of claim 3, further comprising: determining the threshold as a threshold defect size by the processor to maximize a difference between a first average DDR below the threshold defect size and a second average DDR above the threshold defect size.

5. The method of claim 1, wherein the feature comprises a brightness.

6. The method of claim 5, wherein the defect is detected when the brightness is greater than a threshold brightness value.

7. The method of claim 2, further comprising: utilizing the processor to obtain a ratio by dividing the defect area with the

defect perimeter; and determining a defect type according to the ratio by the processor.

8. The method of claim 7, further comprising: determining the defect type as a trap void by the processor when the ratio is approximately equal to one.

9. The method of claim 8, further comprising: analyzing the relation between the DDR and the defect information by the processor when the defect type is the trap void.

10. A method for detecting a semiconductor device, comprising: capturing an image of the semiconductor device by an optical sensor, wherein the semiconductor device comprises a plurality of dies; detecting a defect according to the image by a processor electrically coupling to the optical sensor; utilizing the processor to develop a correspondence between a defect size of the defect and a defect die ratio (DDR) by the processor; and determining a threshold defect size by the processor so that a gap is maximized between a first average DDR below the threshold defect size and a second average DDR above the threshold defect size.

11. The method of claim 10, further comprising: scraping the semiconductor device when the defect size is greater than the threshold defect size.

12. The method of claim 10, further comprising: utilizing the processor to calculate a defect area and a defect perimeter based on the image.

13. The method of claim 12, further comprising: determining that the defect is a trap void by the processor when the defect area is approximately equal to the defect perimeter.

14. The method of claim 13, wherein the trap void is generated during the process of bonding a first wafer of the semiconductor device to a second wafer of the semiconductor device.

15. The method of claim 10, wherein the DDR is linearly proportional to the defect size in the correspondence.

16. A system for detecting a semiconductor device, comprising: at least one processing unit; and at least one memory including computer program code for one or more programs; wherein the at least one memory, the computer program code and the at least one processing unit are configured to cause the system to perform: evaluate a feature of the image of the semiconductor device obtained from an optical sensor; detect a defect of the semiconductor device based on the feature; extract a defect information for the defect; calculate a defect die ratio (DDR) in response to the defect; analyze a relationship between the DDR and the defect information; and scraping the semiconductor device when the defect information fails to meet a threshold.

17. The system of claim 16, wherein the feature comprises a brightness, and the defect is detected when the brightness is greater than a threshold brightness value.

18. The system of claim 16, wherein the defect information comprises at least one of a defect position, a defect perimeter, a defect area, a defect size and a total number of the defect.

19. The system of claim 18, wherein the at least one memory, the computer program code and the at least one processing unit are further configured to cause the system to: determine the threshold as a threshold defect size to maximize a difference between a first average DDR below the threshold defect size and a second average DDR above the threshold defect size.

20. The system of claim 18, wherein the at least one memory, the computer program code and the at least one processing unit are further configured to cause the system to: determine that the defect is a trap void when the defect area is approximately equal to the defect perimeter.

briefHtml: BACKGROUND
(1) The present disclosure relates, in general, to methods and systems for detecting a semiconductor device. Specifically, the present disclosure relates to inspecting defects in a semiconductor device.
(2) Defect inspection is an important part of semiconductor device manufacture, with image processing widely used therein. However, the process can be time consuming and costly. Therefore, more efficient methods and systems for defect inspection of a semiconductor device are called for.

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curlIntlPatentClassificationPrimary: G07F17/32 20060101
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cpcOrigInventiveClassificationHighlights: G07F17/3267 20130101, G07F17/3211 20130101
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abstractHtml: Gaming systems and method that provide a banked bonus player selection feature including a symbol display such as a displayed wheel having a plurality of displayed player selectable sections for a play of a multi-player game.

descriptionHtml: BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 is a flow chart of an example process for operating one example embodiment of a gaming system of the present disclosure providing a banked bonus player selection feature.

(2) FIGS. 2A, 2B, 2C, and 2D are front views of certain displays of parts of one example play of a game including an example banked bonus player selection feature of one example embodiment of the gaming system of the present disclosure.

(3) FIG. 3 is a perspective view of an example gaming system of the present disclosure.

(4) FIG. 4 is a schematic block diagram of one embodiment of an electronic configuration of an example electronic gaming machine of a gaming system of the present disclosure.

(5) FIGS. 5A and 5B are perspective views of example alternative embodiments of electronic gaming machines of a gaming system of the present disclosure.

(6) FIG. 5C is a front view of an example personal gaming device of the gaming system of the present disclosure.

DETAILED DESCRIPTION

(7) Various embodiments of the present disclosure relate to a gaming system and method that provides a banked bonus selection feature in a gaming environment. In various embodiments, the banked bonus player selection feature includes a symbol display object (such as a wheel) including a plurality of displayed player selectable sections for a play of a multi-player game (such as for a play of a secondary multi-player game). In various embodiments, a quantity of the player selectable sections of the symbol display object are selectable by a plurality of qualified players for the play of the multi-player game before the gaming system displays the awards associated with those sections of the symbol display object. In various embodiments, the quantity of sections of the symbol display object that selectable by each of the qualified players is one, a plurality of, or all of the sections of the symbol display object. After the qualified players select the respective sections of the symbol display object, the gaming system displays the symbols (such as the award symbols) associated with the sections of the symbol display object. The gaming system randomly determines and displays a determination of one of the sections of the symbol display object. The gaming system determines and display any awards to the qualified players that selected the determined section based on the symbols (such as the award symbols) displayed in association with the determined section of the symbol display object.

(8) In various embodiments, the symbol display object is a physical wheel and the gaming system causes the wheel to spin and stop spinning for each play of the multi-player game. In various other embodiments, the wheel is in video wheel that is displayed by a video display device of the gaming system and the gaming system causes the display device to display the video wheel spinning and stopping for each play of the multi-player game. In various embodiments, the physical or video wheel can be replaced with a different type of symbol display object such as a physical or video reel. It should thus be appreciated that the use of the wheel as an example in the present disclosure is thus meant to be non-limiting. In various embodiments, the indicator for the wheel can be mechanical such as for a mechanical wheel or can be a video indicator such as for a video wheel. In various embodiments, more than one indicators can be associated with the wheel.

(9) In various embodiments, for each play of the multi-player game, the gaming system displays the symbol display object in the form of a video wheel including a plurality of individually player selectable sections. Each of the sections of

the wheel is associated with one of a plurality of different awards; however, the gaming system initially displays the wheel and each of the sections of the wheel without displaying one, a plurality, or any of the respective award symbols that indicate the awards associated with the respective sections of the wheel. The gaming system enables each of the qualified players to make one or more inputs to select their allowed respective quantity of the displayed sections of the wheel for that play of the multi-player game. In various embodiments, the qualified players can all have the same quantity of sections to select or two or more of the qualified players can have different quantities of sections to select. After each of the qualified player selects their respective quantity of sections of the wheel for that play of the multi-player game, the gaming system displays the award symbols associated with each of the respective player selected sections of the wheel for that play of the multi-player game. In various embodiments, the gaming system can additionally display the award symbols associated with one or more or each of the respective non-selected sections (if any) of the wheel (i.e., the sections of the wheel that none of the qualified players selected for that play of the multi-player game). For the play of the multi-player game, the gaming system displays the wheel spinning and randomly determines one of the sections of the wheel to indicate from the sections of the wheel. The gaming system displays the wheel stopped and an indication of the randomly determined section of the wheel. The gaming system determines, displays, and provides each qualified player that selected the randomly determined and indicated section of the wheel any determined award based on the award symbols associated with that randomly determined and indicated section of the wheel for the play of the multi-player game. In various embodiments, the award associated with the indicated section is split evenly amongst qualified players that picked that section. In various other embodiments, the award associated with the indicated section is provided in full to each of the qualified players that picked that indicated section. In various other embodiments, for the play of the multi-player game, the gaming system displays the wheel spinning and randomly determines more than one of the sections of the wheel to indicate from the sections of the wheel, and determines, displays, and provides any awards based on those indicated sections.

(10) In various embodiments, the quantity of sections of the wheel for the play of the multi-player game can be determined based on any one or more of different wheel section quantity factors. These wheel section quantity factors can include for example, but are not limited to one or more of the following: (1) a predetermined quantity for the play of the multi-player game; (2) a randomly determined quantity for the play of the multi-player game; and/or (3) the triggering event for the play of the multi-player game.

(11) In various embodiments, the quantity of player selectable sections of the wheel for each qualified player of the play of the multi-player game can be determined based on any one or more of different player quantity selection factors. These player quantity selection factors can include for example, but are not limited to one or more of the following: (1) a predetermined quantity for the play of the multi-player game; (2) a randomly determined quantity for the play of the multi-player game; (3) the respective qualified player's accumulated qualifications (as discussed below) associated with plays of a primary wagering game by that qualified player; and/or (4) the triggering event for the play of the multi-player game.

(12) In various embodiments, the awards associated with the respective sections of the wheel for the play of the multi-player game can be determined based on any one or more of different award factors. These award factors can include for example, but are not limited to one or more of the following: (1) predetermined awards of the play of the multi-player game; (2) randomly determined awards for the play of the multi-player game; (3) the qualified players' accumulated qualifications associated with plays of primary wagering games; and/or (4) the triggering event for the play of the multi-player game. In various embodiments, two or more or each of the awards associated with the sections of the wheel are different. In various embodiments, two or more of the awards associated with two or more of the sections of the wheel are the same.

(13) In various embodiments, the gaming system provides only one spin of the wheel for the play of the multi-player game. In various other embodiments, the gaming system provides more than one spin of the wheel for the play of the multi-player game. In various embodiments, the quantity spins of the wheel for the play of the multi-player game can be

determined based on any one or more of different activation factors. These activation factors can include for example, but are not limited to one or more of the following: (1) a predetermined quantity for the play of the multi-player game; (2) a randomly determined quantity for the play of the multi-player game; (3) the qualified players' accumulated qualifications associated with plays of primary wagering games; and/or (4) the triggering event for the play of the multi-player game.

(14) In various embodiments, the gaming system provides only one section indicator associated with the wheel and configured to each indicate only one section of the wheel for the play of the multi-player game. In various other embodiments, the gaming system provides more than one section indicator associated with the wheel and configured to each indicate more than one sections of the wheel for the play of the multi-player game. In various embodiments, the quantity section indicators for the wheel for the play of the game can be determined based on any one or more of different indicator factors. These indicator factors can include for example, but are not limited to one or more of the following: (1) a predetermined quantity for the play of the multi-player game; (2) a randomly determined quantity for the play of the multi-player game; (3) the qualified players' accumulated qualifications associated with plays of primary wagering games; and/or (4) the triggering event for the play of the multi-player game.

(15) In various embodiments, the gaming system can associate one or additional features with one or more of the sections of the wheel selected by one or more of the qualified players. For example, the gaming system can associate a modifier (such as a multiplier) or an additional spin or reactivation feature with one or more of the sections selected by one or more of the qualified players. In various embodiments, the additional feature(s) associated with the player selectable sections of the wheel for the play of the multi-player game can be determined based on any one or more of different additional feature factors. These additional feature factors can include for example, but are not limited to one or more of the following: (1) predetermined additional features for the play of the multi-player game; (2) randomly determined additional features for the play of the multi-player game; (3) the qualified players' accumulated qualifications associated with plays of primary wagering games; and/or (4) the triggering event for the play of the multi-player game.

(16) In various embodiments, the gaming system can associate various colors or other distinguishing indications with one or more the sections of the wheel including the sections of the wheel selected by the respective qualified players. In various embodiments, the gaming system can associate various colors or other distinguishing indications with the sections of the wheel not selected by any of the qualified players. In various embodiments, after the sections of the wheel are selected by the qualified players, the gaming system keeps displaying all of the sections of the wheel. In various alternative embodiments, after the sections of the wheel are selected by the qualified players, the gaming system can stop displaying the sections of the wheel not selected by any of the qualified players. In certain such embodiments, the gaming system can expand one or more or all of the sections of the wheel selected by the qualified players such that they encompass the entire wheel.

(17) In various embodiments, after the sections of the wheel are selected by the qualified players for the play of the multi-player game, the gaming system can cause the display of the awards and/or award symbols associated with those sections on the respective sections. In various embodiments, after the sections of the wheel are selected by the qualified players and after the wheel spins and stops spinning (rather than before the wheel is spun) for the play of the multi-player game, the gaming system can cause the display of the awards and/or award symbols associated with those sections on the respective sections.

(18) In various embodiments, after the sections of the wheel are selected by the qualified players for the play of the multi-player game, the gaming system can cause the awards and/or award symbols associated with the sections not selected by the any of the qualified players to be displayed. In various embodiments, after the sections of the wheel are selected by the players for the play of the multi-player game, the gaming system can continue to cause the awards and/or award symbols associated with the sections of the wheel not selected by any of the qualified players to not be displayed.

(19) As mentioned above, the present disclosure contemplates that the banked bonus player selection feature can be

employed in connection with play of a secondary multi-player game but can alternatively be employed in association with a play of a primary multi-player game (such as a primary wagering multi-player game).

(20) The present disclosure additionally contemplates that while the credit balances, the wagers, and various awards are displayed as amounts of monetary credits or currency in the various example embodiments described herein, one or more of such credit balances, such wagers, and/or such awards can be for non-monetary credits, promotional credits, player tracking points, or in other suitable forms.

(21) The present disclosure additionally contemplates that, in various example embodiments, the gaming system includes multiple electronic gaming machines ("EGM") that each include various input and display devices such as described below, and that enable multiple qualified players to play the game as provided herein. In various embodiments such as shown in FIG. 3, the gaming system includes a plurality of EGMs 2000a adjacently arranged (such as in a bank of EGMs) and a display device above the EGMs 2000a such as in the form of an overhead central display 5000 positioned above and associated with the EGMs 2000a. The overhead central display 5000 displays the symbol display object (such as the wheel) and all can display all associated actions of the wheel and indications regarding the wheel (or other symbol display object) as described herein. Each EGM 2000a is configured to provide one or more suitable wagering primary or secondary games such as described below in addition to the multi-player game described herein. The gaming system including the EGMs are also configured to enable the players of such EGMs to become qualified players for plays of the multi-player game such as discussed herein or in any other suitable manner.

(22) In various embodiments, the EGMs 2000a are networked together for triggering the plays of the multi-player game for the players of the EGMs. In various such embodiments, the central display 5000 can be controlled by a central display processor that controls the central display for the multi-player game (such as overhead display 5000). The central display processor can be connected to each of the EGMs for receiving various input related signal and displaying the multi-player game or other events or images.

(23) In various embodiments, one of the EGMs 2000a functions as a master EGM that controls the triggering of the multi-player game for all of the EGMs, wherein all the EGMs participate in the multi-player game at the same time if they are qualified. For example, if the master EGM determines that the multi-player game should occur, the master EGM can send signals to the other EGMs. Each of the EGMs can keep track of its own player's current eligibility for the multi-player game and therefore will make a decision whether it will enable the player of that EGM to participate in a triggered play of the multi-player game. Each EGM, upon receipt of a multi-player game signal determines if the player of that EGM is qualified for playing the multi-player game, and if so facilitates that player's participation in the play of the multi-player game. This can be done in any suitable manner.

(24) In various different embodiments, each of the players of the EGMs of the gaming system can become a qualified player for a play of the multi-player game in any one or more different manners in accordance with the present disclosure. In various embodiments, for each EGM, a multi-player game triggering event needs to occur on that EGM for the player of that EGM to become a qualified player for the next play of the multi-player game. In various other embodiments, only one multi-player game triggering event needs to occur on one of the EGMs for the players of all of the EGMs to become or potentially become qualified players for the next play of the multi-player game. In various other embodiments, the multi-player game triggering event occurs on a predetermined basis (such as every ten minutes). In various other embodiments, the multi-player game triggering event occurs on a random basis (not linked to specific random game play related determinations by any of the EGMs). In various other embodiments, the multi-player game triggering event occurs based on the qualifications accumulated by the players of the EGMs of the gaming system.

(25) Each EGM can keep track of banked or other accumulated qualifications for the player of such EGM (such as based on eligible time, event eligibility-based credits, wager activity, and/or accumulated multipliers). For each play of the multi-player game, if the player of an EGM has not banked or otherwise accumulated a minimum threshold level of qualifications, the player or EGM cannot participate in the multi-player game. If the EGM has accumulated at least the

minimum threshold qualifications, the EGM enables the player of that EGM to play the multi-player game. Thus, the qualification process for each player can be any suitable process. For example, the qualification process for a player can be based on one or more factors such as but not limited to: (a) whether the player was actively playing the EGM at the time of the trigger of the play of the multi-player game; (b) whether the player has any eligibility time remaining in a bank or meter at the time of the trigger of the play of the multi-player game; (c) whether the player has any eligibility units remaining in a bank or meter at the time of the trigger of the play of the multi-player game; or (d) whether the player has any multipliers remaining in a bank or meter at the time of the trigger of the play of the multi-player game. The EGMs can track each player's eligibility in any suitable manner. In the embodiments where the player has accumulated a multiplier for the bonus event or is entitled to a multiplier for the bonus event based on their eligibility, the gaming system can provide that multiplier for that player to any award earned in the multi-player game.

(26) In various embodiments, when the qualified players enter or qualify for the play of the multi-player game with one or more accumulated multipliers, the gaming system can provide the player a choice as to whether to use any of their accumulated multipliers for the play of the game. This can be done at any suitable time for the play of the multi-player game. For example, this can be done before the awards or award symbols are displayed in association with the sections of the wheel, or after the awards or award symbols are displayed in association with the sections of the wheel. In other embodiments, that can be done after the wheel is spun, stopped, and the qualified player that selected the indicated section knows what the award is that they can win.

(27) In various embodiments, when a qualified player enters a play of the multi-player game, the gaming system partly or fully uses or reduces the player's accumulated qualifications (such as their accumulated multipliers or other qualifications) regardless of whether the player wins any award in the play of the multi-player game (such as by selecting the correct section of the wheel that is subsequently indicated in the play of the multi-player game). In various other embodiments, when a qualified player enters a play of the multi-player game, the gaming system does not partly or fully use or reduce the player's accumulated qualifications if the player does not win any award in the play of the multi-player game (such as by selecting the correct section of the wheel that is subsequently indicated in the play of the multi-player game). In such embodiments, the gaming system partly or fully uses or reduces the player's accumulated qualifications if the player wins an award in the play of the multi-player game (such as by selecting the correct section of the wheel that is subsequently indicated in the play of the multi-player game). Thus, qualified players can participate in plays of the multi-player game without risking some or all of their accumulated qualifications (even if they do not win in the play of the multi-player game).

(28) FIG. 1 illustrates a flowchart of an example process or method 100 of operating the gaming system of one embodiment of the present disclosure. In various embodiments, process 100 is represented by a set of instructions stored in one or more memories and executed by one or more processors. Although process 100 is described with reference to the flowchart shown in FIG. 1, it should be appreciated that many other processes of performing the acts associated with this illustrated process may be employed. For example, the order of certain of the illustrated blocks may be changed, certain of the illustrated blocks may be optional, and/or certain of the illustrated blocks may not be employed.

(29) In this example embodiment, responsive to a triggering event for a play of the multi-player game, the gaming system displays a wheel including a plurality of player selectable sections without displaying any of the awards associated with those sections, as indicated by block 102. The gaming system can additionally display an indication to each of the qualified players of how many sections of the wheel that that qualified player can select. Any suitable manner of enabling each qualified player to select a section of the wheel can be employed in accordance with the present disclosure. It should also be appreciated that in other embodiments the gaming system can be configured to enable one or more of the qualified players to select more than one but not all of the sections of the wheel, such as based on the qualifications for the play of the multi-player game. It should also be appreciated that in other embodiments the gaming system can be configured to enable one or more of the qualified players to select all of the sections of the

wheel, such as based on the qualifications for the play of the multi-player game. For example, if a player has a maximum level of qualifications for the play of the multi-player game, the gaming system can be configured to enable that qualified players to select all of the sections of the wheel.

(30) Responsive to each player selection of one of the sections of the wheel, the gaming system displays an indication that that section is selected by that qualified player for the play of the multi-player game, as indicated by block 104. This step is repeated until each qualified player selects the total quantity of player selectable sections of the wheel of that qualified player. The gaming system can display a different player indication for each qualified player for each section picked by that player. For example, the player can have assigned numbers and after a qualified player picks a section of the wheel, the gaming system can display the number associated with the player on the section of the wheel picked by the player. Thus, the gaming system can display multiple numbers on a section picked by multiple qualified players. These player indications can be displayed as the qualified players pick the sections or after all of the qualified players pick all of the sections. In the first alternative, the gaming system enables the qualified players to see the sections previously picked by the other qualified players. The gaming system can also provide a display a limited time period for the qualified player to pick the sections of the wheel.

(31) Responsive to the qualified players selecting the total quantity of sections of the wheel that each qualified player is able to select for the play of the multi-player game, the gaming system displays an indication of the award symbols (and thus the awards) associated with each of the sections of the wheel, as indicated by block 106. The gaming system can display these award symbols directly on the sections of the wheel or in another suitable manner.

(32) After the qualified players select the total quantity of sections of the wheel that the qualified players are able to select for the play of the game, the gaming system also causes a display of the wheel spinning, makes a random determination of one of the sections of the wheel to indicate, and causes a display of the wheel stopped such that the indicator associated with the wheel indicates the randomly determined section, as indicated by block 108.

(33) The gaming system determines any award for that indicated section and causes a display of that award, as indicated by block 110. The gaming system also provides that award to the qualified player or qualified players that selected that section, as indicated by block 110.

(34) FIGS. 2A, 2B, 2C, and 2D illustrate certain screen shots displayed by the central display 5000 for an example play of a multi-player game provided by a gaming system of one example embodiment of the present disclosure. In this example, the symbol display object is a wheel 200 that includes a plurality of adjacently arranged sections, and specifically sections 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, and 232. It should be appreciated that the quantity of sections of the wheel can vary in accordance with the present disclosure. The central display 5000 of the gaming system also displays an indicator 160 associated with the wheel 200.

(35) More specifically, FIG. 2A illustrates a point in the play of the multi-player game after the play of the game has been triggered and the gaming system and particularly the EGMs 2000a have displayed to the qualified players thereof an indication that they have qualified for the play of the multi-player game. At this point, the central display 5000 of the gaming system does not display any awards or award symbols on or associated with these sections 210, 212, 214, 216, 218, 220, 222, 224, 226, 228, 230, and 232 of the wheel 200. In this example embodiment, each of the EGMs 2000a also display the play of the multi-player game as generally shown in FIG. 3. At this point in the play of the multi-player game, the central display 5000 also displays an indication to the qualified players of the quantity of sections of the wheel 200 that the qualified players can pick for this play of the multi-player game. In this example, that quantity is one for each qualified player. The gaming system and specifically each of the EGMs 2000a enables each qualified player to make an input to select one of the sections of the wheel 200 using an input device of the respective EGM 2000a that the qualified player is playing.

(36) FIG. 2B illustrates a subsequent point in the play of the multi-player game, where each qualified player has picked one of the sections of the wheel 200, and particularly in this example the qualified players have picked sections 212, 216, 224, and 232. At this point, the central display 5000 displays player indications on the picked sections 212, 216,

224, and 232 of the wheel 200 to indicate that the qualified players have picked by the respective sections as well as the award symbols that have previously been associated with these picked sections 212, 216, 224, and 232. In this example, the different awards and award symbols are displayed on the picked sections 212, 216, 224, and 232. In this example, the gaming system also displays the player indications of which player picked which of the selects using the P1, P2, P3, and P4 player indications displayed on the picked sections 212, 216, 224, and 232. At this point in the play of the multi-player game, the gaming system does not display any awards or award symbols on or associated with the non-picked sections of the wheel 200. In alternative embodiments, the gaming system can display one or more or all of those other awards and/or award symbols for the non-picked sections of the wheel 200. In various embodiments, the gaming system causes the central display 5000 to highlight, brighten, emphasize, or otherwise display an indication of each of the player selected sections of the wheel 200 relative to the other non-selected sections of the wheel 200 to provide an indication to the qualified player as to which sections of the wheel 200 the qualified players have chosen for this play of the multi-player game. For each play of the multi-player game, any suitable section indications can be used to indicate the set of sections of the wheel 200 that the players selected to employ for that play of the multi-player game. It should thus be appreciated that for each play of the multi-player game, each of the players can select the same or different sections. It should be appreciated that the player indications on the selected sections can be displayed adjacent to the award symbols on the selected sections.

(37) It should be appreciated that for each play of the multi-player game, the awards and award symbols associated with the respective sections of the wheel 200 can be different so that the qualified players do not learn which awards and/or award symbols are associated with each section during multiple plays of the multi-player game. In various embodiments, the awards and/or award symbols associated with the respective sections can be randomly or otherwise suitably determined.

(38) FIG. 2C illustrates a subsequent point in the play of the multi-player game after the four qualified players have picked the four sections 212, 216, 224, and 232 of the wheel 200, after the gaming system has displayed the player indications and the awards and/or award symbols associated with the player picked sections 212, 216, 224, and 232, and after the gaming system has caused the wheel 200 to start to spin.

(39) FIG. 2D illustrates a subsequent point in the play of the multi-player game after the gaming system has randomly selected one of the sections of the wheel (and particularly section 216 of the wheel 200 in this example), and after the gaming system has caused the wheel 200 to stop spinning such that the randomly determined section 216 of the wheel 200 is indicated by indicator 160. The gaming system thus determines the award of 5000 credits for this play of the multi-player game based on selected and indicated section 216 of the wheel 200. The gaming system displays the award such as in a win display (not shown) and updates the respective qualified player's credit balance (not shown) indicated by credit meter (not shown). In this example, since player 3 picked section 216, the EGM that the player 3 is playing provides the player with the award of 5000 credits. The gaming system can display any suitable message in message display indicating the award determination such as shown in FIG. 2D.

(40) In various different embodiments, one or more awards provided in association with the games of the present disclosure include one or more of: a quantity of monetary credits, a quantity of non-monetary credits, a quantity of promotional credits, a quantity of player tracking points, a progressive award, a modifier, such as a multiplier, a quantity of free plays of one or more games, a quantity of plays of one or more secondary or bonus games, a multiplier of a quantity of free plays of a game, one or more lottery based awards, such as lottery or drawing tickets, a wager match for one or more plays of one or more games, an increase in the average expected payback percentage for one or more plays of one or more games, one or more comps, such as a free dinner, a free night's stay at a hotel, a high value product such as a free car, or a low value product, one or more bonus credits usable for online play, a lump sum of player tracking points or credits, a multiplier for player tracking points or credits, an increase in a membership or player tracking level, one or more coupons or promotions usable within and/or outside of the gaming establishment (e.g., a 20% off coupon for use at a convenience store), virtual goods associated with the gaming system, virtual goods not

associated with the gaming system, an access code usable to unlock content on an internet.

(41) It should be appreciated from the above that in various embodiments, the gaming system, for a play of a multi-player game, the gaming system communicates data that results in a display, by a central device, of a wheel and a plurality of selectable sections of the wheel, wherein the sections are respectively associated with a plurality of different awards for the play of the multi-player game, and wherein the sections are displayed without displaying any of the awards in association with the sections. In various other embodiments, the gaming system communicates data that results in a display, by a display device, of a wheel and a plurality of selectable sections of the wheel, wherein the sections are respectively associated with a plurality of awards, and the awards are displayed separately from the sections and without displaying which awards are associated with which sections of the wheel. In various such embodiments, the gaming system communicates data that results in a display, by the display device, of a table that includes the awards displayed separately from the wheel and separately from the sections of the wheel. In various other embodiments, the gaming system communicates data that results in a display, by a display device, of a wheel and a plurality of selectable sections of the wheel, wherein the sections are respectively associated with a plurality of awards, and the awards are displayed in association with such sections so that the qualified players can pick sections partly based on or knowing the awards that are associated with which sections of the wheel.

(42) It should be appreciated that in different embodiments, one or more of the determinations of the present disclosure is/are predetermined, randomly determined, randomly determined based on one or more weighted percentages, determined based on a generated game outcome, determined independent of a generated game outcome, determined based on a random determination by the central controller, determined independent of a random determination by the central controller, determined based on a random determination at the gaming system, determined independent of a random determination at the gaming system, determined based on at least one play of at least one game, determined independent of at least one play of at least one game, determined based on a player's selection, determined independent of a player's selection, determined based on one or more side wagers placed, determined independent of one or more side wagers placed, determined based on the player's primary game wager, determined independent of the player's primary game wager, determined based on time (such as the time of day), determined independent of time (such as the time of day), determined based on an amount of coin-in accumulated in one or more pools, determined independent of an amount of coin-in accumulated in one or more pools, determined based on a status of the player (i.e., a player tracking status), determined independent of a status of the player (i.e., a player tracking status), determined based on one or more other determinations of the present disclosure, determined independent of any other determination of the present disclosure or determined based on any other suitable method or criteria.

(43) It should be appreciated that the present disclosure provides an improvement in gaming technology, in part, by enabling each qualified player to make a meaningful input in each play of the multi-player game that partially determines the award(s) that can be provided to the qualified player. The gaming system also enables the qualified players to see immediate results from the player's selections.

Gaming Systems

(44) The above-described embodiments of the present disclosure may be implemented in accordance with or in conjunction with one or more of a variety of different types of gaming systems, such as, but not limited to, those described below.

(45) The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. A "gaming system" as used herein refers to various configurations of: (a) one or more central servers, central controllers, or remote hosts; (b) one or more electronic gaming machines such as those located on a casino floor; and/or (c) one or more personal gaming devices, such as desktop computers, laptop computers, tablet computers or computing devices, personal digital assistants, mobile phones, and other mobile computing devices. Moreover, an EGM as used herein refers to any suitable electronic gaming machine which enables a player to play a game (including but not limited to a game of chance, a game of skill, and/or a game of partial skill) to

potentially win one or more awards and/or place zero, one or more wagers on one or more sporting events, wherein the EGM include, but is not limited to: a slot machine, a video poker machine, a video lottery terminal, a terminal associated with an electronic table game, a video keno machine, a video bingo machine located on a casino floor, or a sports betting terminal.

(46) In various embodiments, the gaming system of the present disclosure includes: (a) one or more electronic gaming machines in combination with one or more central servers, central controllers, or remote hosts; (b) one or more personal gaming devices in combination with one or more central servers, central controllers, or remote hosts; (c) one or more personal gaming devices in combination with one or more electronic gaming machines; (d) one or more personal gaming devices, one or more electronic gaming machines, and one or more central servers, central controllers, or remote hosts in combination with one another; (e) a single electronic gaming machine; (f) a plurality of electronic gaming machines in combination with one another; (g) a single personal gaming device; (h) a plurality of personal gaming devices in combination with one another; (i) a single central server, central controller, or remote host; and/or (j) a plurality of central servers, central controllers, or remote hosts in combination with one another.

(47) For brevity and clarity and unless specifically stated otherwise, "EGM" as used herein represents one EGM or a plurality of EGMs, "personal gaming device" as used herein represents one personal gaming device or a plurality of personal gaming devices, and "central server, central controller, or remote host" as used herein represents one central server, central controller, or remote host or a plurality of central servers, central controllers, or remote hosts.

(48) As noted above, in various embodiments, the gaming system includes an EGM (or personal gaming device) in combination with a central server, central controller, or remote host. In such embodiments, the EGM (or personal gaming device) is configured to communicate with the central server, central controller, or remote host through a data network or remote communication link. In certain such embodiments, the EGM (or personal gaming device) is configured to communicate with another EGM (or personal gaming device) through the same data network or remote communication link or through a different data network or remote communication link. For example, the gaming system includes a plurality of EGMs that are each configured to communicate with a central server, central controller, or remote host through a data network.

(49) In certain embodiments in which the gaming system includes an EGM (or personal gaming device) in combination with a central server, central controller, or remote host, the central server, central controller, or remote host is any suitable computing device (such as a server) that includes at least one processor and at least one memory device or data storage device. As further described herein, the EGM (or personal gaming device) includes at least one EGM (or personal gaming device) processor configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the EGM (or personal gaming device) and the central server, central controller, or remote host. The at least one processor of that EGM (or personal gaming device) is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the EGM (or personal gaming device). Moreover, the at least one processor of the central server, central controller, or remote host is configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the central server, central controller, or remote host and the EGM (or personal gaming device). The at least one processor of the central server, central controller, or remote host is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the central server, central controller, or remote host. One, more than one, or each of the functions of the central server, central controller, or remote host may be performed by the at least one processor of the EGM (or personal gaming device). Further, one, more than one, or each of the functions of the at least one processor of the EGM (or personal gaming device) may be performed by the at least one processor of the central server, central controller, or remote host.

(50) In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the EGM (or personal gaming device) are executed by the central server, central controller, or remote host. In such "thin client" embodiments, the central server, central controller,

or remote host remotely controls any games (or other suitable interfaces) displayed by the EGM (or personal gaming device), and the EGM (or personal gaming device) is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the EGM (or personal gaming device) are communicated from the central server, central controller, or remote host to the EGM (or personal gaming device) and are stored in at least one memory device of the EGM (or personal gaming device). In such "thick client" embodiments, the at least one processor of the EGM (or personal gaming device) executes the computerized instructions to control any games (or other suitable interfaces) displayed by the EGM (or personal gaming device).

(51) In various embodiments in which the gaming system includes a plurality of EGMs (or personal gaming devices), one or more of the EGMs (or personal gaming devices) are thin client EGMs (or personal gaming devices) and one or more of the EGMs (or personal gaming devices) are thick client EGMs (or personal gaming devices). In other embodiments in which the gaming system includes one or more EGMs (or personal gaming devices), certain functions of one or more of the EGMs (or personal gaming devices) are implemented in a thin client environment, and certain other functions of one or more of the EGMs (or personal gaming devices) are implemented in a thick client environment. In one such embodiment in which the gaming system includes an EGM (or personal gaming device) and a central server, central controller, or remote host, computerized instructions for controlling any primary or base games displayed by the EGM (or personal gaming device) are communicated from the central server, central controller, or remote host to the EGM (or personal gaming device) in a thick client configuration, and computerized instructions for controlling any secondary or bonus games or other functions displayed by the EGM (or personal gaming device) are executed by the central server, central controller, or remote host in a thin client configuration.

(52) In certain embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a local area network (LAN) in which the EGMs (or personal gaming devices) are located substantially proximate to one another and/or the central server, central controller, or remote host. In one example, the EGMs (or personal gaming devices) and the central server, central controller, or remote host are located in a gaming establishment or a portion of a gaming establishment.

(53) In other embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a wide area network (WAN) in which one or more of the EGMs (or personal gaming devices) are not necessarily located substantially proximate to another one of the EGMs (or personal gaming devices) and/or the central server, central controller, or remote host. For example, one or more of the EGMs (or personal gaming devices) are located: (a) in an area of a gaming establishment different from an area of the gaming establishment in which the central server, central controller, or remote host is located; or (b) in a gaming establishment different from the gaming establishment in which the central server, central controller, or remote host is located. In another example, the central server, central controller, or remote host is not located within a gaming establishment in which the EGMs (or personal gaming devices) are located. In certain embodiments in which the data network is a WAN, the gaming system includes a central server, central controller, or remote host and an EGM (or personal gaming device) each located in a different gaming establishment in a same geographic area, such as a same city or a same state. Gaming systems in which the data network is a WAN are substantially identical to gaming systems in which the data network is a LAN, though the quantity of EGMs (or personal gaming devices) in such gaming systems may vary relative to one another.

(54) In further embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data

network is an internet (such as the Internet) or an intranet. In certain such embodiments, an Internet browser of the EGM (or personal gaming device) is usable to access an Internet game page from any location where an Internet connection is available. In one such embodiment, after the EGM (or personal gaming device) accesses the Internet game page, the central server, central controller, or remote host identifies a player before enabling that player to place any wagers on any plays of any wagering games. In one example, the central server, central controller, or remote host identifies the player by requiring a player account of the player to be logged into via an input of a unique player name and password combination assigned to the player. The central server, central controller, or remote host may, however, identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identification number associated with the player by the central server, central controller, or remote host; or by identifying the EGM (or personal gaming device), such as by identifying the MAC address or the IP address of the Internet facilitator. In various embodiments, once the central server, central controller, or remote host identifies the player, the central server, central controller, or remote host enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games, and displays those plays via the Internet browser of the EGM (or personal gaming device). Examples of implementations of Internet-based gaming are further described in U.S. Pat. No. 8,764,566, entitled "Internet Remote Game Server," and U.S. Pat. No. 8,147,334, entitled "Universal Game Server".

(55) The central server, central controller, or remote host and the EGM (or personal gaming device) are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via: a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile Internet network), or any other suitable medium. The expansion in the quantity of computing devices and the quantity and speed of Internet connections in recent years increases opportunities for players to use a variety of EGMs (or personal gaming devices) to play games from an ever-increasing quantity of remote sites. Additionally, the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.

EGM Components

(56) FIG. 4 is a block diagram of an example EGM 1000 and FIGS. 5A and 5B include two different example EGMs 2000a and 2000b. The EGMs 1000, 2000a, and 2000b are merely example EGMs, and different EGMs may be implemented using different combinations of the components shown in the EGMs 1000, 2000a, and 2000b. Although the below refers to EGMs, in various embodiments personal gaming devices (such as personal gaming device 2000c of FIG. 5C) may include some or all of the below components.

(57) In these embodiments, the EGM 1000 includes a master gaming controller 1012 configured to communicate with and to operate with a plurality of peripheral devices 1022.

(58) The master gaming controller 1012 includes at least one processor 1010. The at least one processor 1010 is any suitable processing device or set of processing devices, such as a microprocessor, a microcontroller-based platform, a suitable integrated circuit, or one or more application-specific integrated circuits (ASICs), configured to execute software enabling various configuration and reconfiguration tasks, such as: (1) communicating with a remote source (such as a server that stores authentication information or game information) via a communication interface 1006 of the master gaming controller 1012; (2) converting signals read by an interface to a format corresponding to that used by software or memory of the EGM; (3) accessing memory to configure or reconfigure game parameters in the memory according to indicia read from the EGM; (4) communicating with interfaces and the peripheral devices 1022 (such as input/output devices); and/or (5) controlling the peripheral devices 1022. In certain embodiments, one or more components of the

master gaming controller 1012 (such as the at least one processor 1010) reside within a housing of the EGM (described below), while in other embodiments at least one component of the master gaming controller 1012 resides outside of the housing of the EGM.

(59) The master gaming controller 1012 also includes at least one memory device 1016, which includes: (1) volatile memory (e.g., RAM 1009, which can include non-volatile RAM, magnetic RAM, ferroelectric RAM, and any other suitable forms); (2) non-volatile memory 1019 (e.g., disk memory, FLASH memory, EPROMs, EEPROMs, memristor-based non-volatile solid-state memory, etc.); (3) unalterable memory (e.g., EPROMs 1008); (4) read-only memory; and/or (5) a secondary memory storage device 1015, such as a non-volatile memory device, configured to store gaming software related information (the gaming software related information and the memory may be used to store various audio files and games not currently being used and invoked in a configuration or reconfiguration). Any other suitable magnetic, optical, and/or semiconductor memory may operate in conjunction with the EGM of the present disclosure. In certain embodiments, the at least one memory device 1016 resides within the housing of the EGM (described below), while in other embodiments at least one component of the at least one memory device 1016 resides outside of the housing of the EGM. In these embodiments, any combination of one or more computer readable media may be utilized. The computer readable media may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(60) A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electro-magnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

(61) The at least one memory device 1016 is configured to store, for example: (1) configuration software 1014, such as all the parameters and settings for a game playable on the EGM; (2) associations 1018 between configuration indicia read from an EGM with one or more parameters and settings; (3) communication protocols configured to enable the at least one processor 1010 to communicate with the peripheral devices 1022; and/or (4) communication transport protocols (such as TCP/IP, USB, Firewire, IEEE1394, Bluetooth, IEEE 802.11x (IEEE 802.11 standards), hiperlan/2, HomeRF, etc.) configured to enable the EGM to communicate with local and non-local devices using such protocols. In one implementation, the master gaming controller 1012 communicates with other devices using a serial communication protocol. A few non-limiting examples of serial communication protocols that other devices, such as peripherals (e.g., a bill validator or a ticket printer), may use to communicate with the master game controller 1012 include USB, RS-232, and Netplex (a proprietary protocol developed by IGT).

(62) As will be appreciated by one skilled in the art, aspects of the present disclosure may be illustrated and described herein in any of a number of patentable classes or context including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, aspects of the present

disclosure may be implemented entirely hardware, entirely software (including firmware, resident software, micro-code, etc.) or combining software and hardware implementation that may all generally be referred to herein as a "circuit," "module," "component," or "system." Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable media having computer readable program code embodied thereon.

(63) Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C#, VB.NET, Python or the like, conventional procedural programming languages, such as the "C" programming language, Visual Basic, Fortran 2003, Perl, COBOL 2002, PHP, ABAP, dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the player's computer, partly on the player's computer, as a stand-alone software package, partly on the player's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the player's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (SaaS).

(64) Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable instruction execution apparatus, create a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

(65) These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other programmable instruction execution apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

(66) In certain embodiments, the at least one memory device 1016 is configured to store program code and instructions executable by the at least one processor of the EGM to control the EGM. The at least one memory device 1016 of the EGM also stores other operating data, such as image data, event data, input data, random number generators (RNGs) or pseudo-RNGs, payable data or information, and/or applicable game rules that relate to the play of one or more games on the EGM. In various embodiments, part or all of the program code and/or the operating data described above is stored in at least one detachable or removable memory device including, but not limited to, a cartridge, a disk, a CD ROM, a DVD, a USB memory device, or any other suitable non-transitory computer readable medium. In certain such embodiments, an operator (such as a gaming establishment operator) and/or a player uses such a removable memory device in an EGM to implement at least part of the present disclosure. In other embodiments, part or all of the program code and/or the operating data is downloaded to the at least one memory device of the EGM through any suitable data network described above (such as an Internet or intranet).

(67) The at least one memory device 1016 also stores a plurality of device drivers 1042. Examples of different types of

device drivers include device drivers for EGM components and device drivers for the peripheral components 1022. Typically, the device drivers 1042 utilize various communication protocols that enable communication with a particular physical device. The device driver abstracts the hardware implementation of that device. For example, a device driver may be written for each type of card reader that could potentially be connected to the EGM. Non-limiting examples of communication protocols used to implement the device drivers include Netplex, USB, Serial, Ethernet 175, Firewire, I/O debouncer, direct memory map, serial, PCI, parallel, RF, Bluetooth™ near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), etc. In one embodiment, when one type of a particular device is exchanged for another type of the particular device, the at least one processor of the EGM loads the new device driver from the at least one memory device to enable communication with the new device. For instance, one type of card reader in the EGM can be replaced with a second different type of card reader when device drivers for both card readers are stored in the at least one memory device.

(68) In certain embodiments, the software units stored in the at least one memory device 1016 can be upgraded as needed. For instance, when the at least one memory device 1016 is a hard drive, new games, new game options, new parameters, new settings for existing parameters, new settings for new parameters, new device drivers, and new communication protocols can be uploaded to the at least one memory device 1016 from the master game controller 1012 or from some other external device. As another example, when the at least one memory device 1016 includes a CD/DVD drive including a CD/DVD configured to store game options, parameters, and settings, the software stored in the at least one memory device 1016 can be upgraded by replacing a first CD/DVD with a second CD/DVD. In yet another example, when the at least one memory device 1016 uses flash memory 1019 or EPROM 1008 units configured to store games, game options, parameters, and settings, the software stored in the flash and/or EPROM memory units can be upgraded by replacing one or more memory units with new memory units that include the upgraded software. In another embodiment, one or more of the memory devices, such as the hard drive, may be employed in a game software download process from a remote software server.

(69) In some embodiments, the at least one memory device 1016 also stores authentication and/or validation components 1044 configured to authenticate/validate specified EGM components and/or information, such as hardware components, software components, firmware components, peripheral device components, player input device components, information received from one or more player input devices, information stored in the at least one memory device 1016, etc. Examples of various authentication and/or validation components are described in U.S. Pat. No. 6,620,047, entitled "Electronic Gaming Apparatus Having Authentication Data Sets".

(70) In certain embodiments, the peripheral devices 1022 include several device interfaces, such as: (1) at least one output device 1020 including at least one display device 1035; (2) at least one input device 1030 (which may include contact and/or non-contact interfaces); (3) at least one transponder 1054; (4) at least one wireless communication component 1056; (5) at least one wired/wireless power distribution component 1058; (6) at least one sensor 1060; (7) at least one data preservation component 1062; (8) at least one motion/gesture analysis and interpretation component 1064; (9) at least one motion detection component 1066; (10) at least one portable power source 1068; (11) at least one geolocation module 1076; (12) at least one player identification module 1077; (13) at least one player/device tracking module 1078; and (14) at least one information filtering module 1079.

(71) The at least one output device 1020 includes at least one display device 1035 configured to display any game(s) displayed by the EGM and any suitable information associated with such game(s). In certain embodiments, the display devices are connected to or mounted on a housing of the EGM (described below). In various embodiments, the display devices serve as digital glass configured to advertise certain games or other aspects of the gaming establishment in which the EGM is located. In various embodiments, the EGM includes one or more of the following display devices: (a) a central display device; (b) a player tracking display configured to display various information regarding a player's player tracking status (as described below); (c) a secondary or upper display device in addition to the central display device and the player tracking display; (d) a credit display configured to display a current quantity of credits, amount of cash,

account balance, or the equivalent; and (e) a bet display configured to display an amount wagered for one or more plays of one or more games. The example EGM 2000a illustrated in FIG. 4A includes a central display device 2116, a player tracking display 2140, a credit display 2120, and a bet display 2122. The example EGM 2000b illustrated in FIG. 4B includes a central display device 2116, an upper display device 2118, a player tracking display 2140, a credit display 2120, and a bet display 2122.

(72) In various embodiments, the display devices include, without limitation: a monitor, a television display, a plasma display, a liquid crystal display (LCD), a display based on light emitting diodes (LEDs), a display based on a plurality of organic light-emitting diodes (OLEDs), a display based on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display device includes a touch-screen with an associated touch-screen controller. The display devices may be of any suitable sizes, shapes, and configurations.

(73) The display devices of the EGM are configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices of the EGM are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices of the EGM are configured to display one or more video reels, one or more video wheels, and/or one or more video dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

(74) In various embodiments, the at least one output device 1020 includes a payout device. In these embodiments, after the EGM receives an actuation of a cashout device (described below), the EGM causes the payout device to provide a payment to the player. In one embodiment, the payout device is one or more of: (a) a ticket printer and dispenser configured to print and dispense a ticket or credit slip associated with a monetary value, wherein the ticket or credit slip may be redeemed for its monetary value via a cashier, a kiosk, or other suitable redemption system; (b) a bill dispenser configured to dispense paper currency; (c) a coin dispenser configured to dispense coins or tokens (such as into a coin payout tray); and (d) any suitable combination thereof. The example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B each include a ticket printer and dispenser 2136. Examples of ticket-in ticket-out (TITO) technology are described in U.S. Pat. No. 5,429,361, entitled "Gaming Machine Information, Communication and Display System"; U.S. Pat. No. 5,470,079, entitled "Gaming Machine Accounting and Monitoring System"; U.S. Pat. No. 5,265,874, entitled "Cashless Gaming Apparatus and Method"; U.S. Pat. No. 6,729,957, entitled "Gaming Method and Host Computer with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 6,729,958, entitled "Gaming System with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 6,736,725, entitled "Gaming Method and Host Computer with Ticket-In/Ticket-Out Capability"; U.S. Pat. No. 7,275,991, entitled "Slot Machine with Ticket-In/Ticket-Out Capability"; and U.S. Pat. No. 6,048,269, entitled "Coinless Slot Machine System and Method".

(75) In certain embodiments, rather than dispensing bills, coins, or a physical ticket having a monetary value to the player following receipt of an actuation of the cashout device, the payout device is configured to cause a payment to be provided to the player in the form of an electronic funds transfer, such as via a direct deposit into a bank account, a casino account, or a prepaid account of the player; via a transfer of funds onto an electronically recordable identification card or smart card of the player; or via sending a virtual ticket having a monetary value to an electronic device of the player. Examples of providing payment using virtual tickets are described in U.S. Pat. No. 8,613,659, entitled "Virtual Ticket-In and Ticket-Out on a Gaming Machine".

(76) While any credit balances, any wagers, any values, and any awards are described herein as amounts of monetary credits or currency, one or more of such credit balances, such wagers, such values, and such awards may be for non-monetary credits, promotional credits, of player tracking points or credits.

(77) In certain embodiments, the at least one output device 1020 is a sound generating device controlled by one or more sound cards. In one such embodiment, the sound generating device includes one or more speakers or other sound generating hardware and/or software configured to generate sounds, such as by playing music for any games or by playing music for other modes of the EGM, such as an attract mode. The example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B each include a plurality of speakers 2150. In another such embodiment, the EGM provides dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the EGM. In certain embodiments, the EGM displays a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the EGM. The videos may be customized to provide any appropriate information.

(78) The at least one input device 1030 may include any suitable device that enables an input signal to be produced and received by the at least one processor 1010 of the EGM.

(79) In one embodiment, the at least one input device 1030 includes a payment device configured to communicate with the at least one processor of the EGM to fund the EGM. In certain embodiments, the payment device includes one or more of: (a) a bill acceptor into which paper money is inserted to fund the EGM; (b) a ticket acceptor into which a ticket or a voucher is inserted to fund the EGM; (c) a coin slot into which coins or tokens are inserted to fund the EGM; (d) a reader or a validator for credit cards, debit cards, or credit slips into which a credit card, debit card, or credit slip is inserted to fund the EGM; (e) a player identification card reader into which a player identification card is inserted to fund the EGM; or (f) any suitable combination thereof. The example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B each include a combined bill and ticket acceptor 2128 and a coin slot 2126.

(80) In one embodiment, the at least one input device 1030 includes a payment device configured to enable the EGM to be funded via an electronic funds transfer, such as a transfer of funds from a bank account. In another embodiment, the EGM includes a payment device configured to communicate with a mobile device of a player, such as a mobile phone, a radio frequency identification tag, or any other suitable wired or wireless device, to retrieve relevant information associated with that player to fund the EGM. Examples of funding an EGM via communication between the EGM and a mobile device (such as a mobile phone) of a player are described in U.S. Patent Application Publication No. 2013/0344942, entitled "Avatar as Security Measure for Mobile Device Use with Electronic Gaming Machine". When the EGM is funded, the at least one processor determines the amount of funds entered and displays the corresponding amount on a credit display or any other suitable display as described below.

(81) In certain embodiments, the at least one input device 1030 includes at least one wagering or betting device. In various embodiments, the one or more wagering or betting devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). One such wagering or betting device is as a maximum wager or bet device that, when actuated, causes the EGM to place a maximum wager on a play of a game. Another such wagering or betting device is a repeat bet device that, when actuated, causes the EGM to place a wager that is equal to the previously-placed wager on a play of a game. A further such wagering or betting device is a bet one device that, when actuated, causes the EGM to increase the wager by one credit. Generally, upon actuation of one of the wagering or betting devices, the quantity of credits displayed in a credit meter (described below) decreases by the amount of credits wagered, while the quantity of credits displayed in a bet display (described below) increases by the amount of credits wagered.

(82) In various embodiments, the at least one input device 1030 includes at least one game play activation device. In various embodiments, the one or more game play initiation devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). After a player appropriately funds the EGM and places a wager, the

EGM activates the game play activation device to enable the player to actuate the game play activation device to initiate a play of a game on the EGM (or another suitable sequence of events associated with the EGM). After the EGM receives an actuation of the game play activation device, the EGM initiates the play of the game. The example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B each include a game play activation device in the form of a game play initiation button 2132. In other embodiments, the EGM begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

(83) In other embodiments, the at least one input device 1030 includes a cashout device. In various embodiments, the cashout device is: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). When the EGM receives an actuation of the cashout device from a player and the player has a positive (i.e., greater-than-zero) credit balance, the EGM initiates a payout associated with the player's credit balance. The example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B each include a cashout device in the form of a cashout button 2134.

(84) In various embodiments, the at least one input device 1030 includes a plurality of buttons that are programmable by the EGM operator to, when actuated, cause the EGM to perform particular functions. For instance, such buttons may be hard keys, programmable soft keys, or icons icon displayed on a display device of the EGM (described below) that are actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). The example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B each include a plurality of such buttons 2130.

(85) In certain embodiments, the at least one input device 1030 includes a touch-screen coupled to a touch-screen controller or other touch-sensitive display overlay to enable interaction with any images displayed on a display device (as described below). One such input device is a conventional touch-screen button panel. The touch-screen and the touch-screen controller are connected to a video controller. In these embodiments, signals are input to the EGM by touching the touch screen at the appropriate locations.

(86) In embodiments including a player tracking system, as further described below, the at least one input device 1030 includes a card reader in communication with the at least one processor of the EGM. The example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B each include a card reader 2138. The card reader is configured to read a player identification card inserted into the card reader.

(87) The at least one wireless communication component 1056 includes one or more communication interfaces having different architectures and utilizing a variety of protocols, such as (but not limited to) 802.11 (WiFi); 802.15 (including Bluetooth™); 802.16 (WiMax); 802.22; cellular standards such as CDMA, CDMA2000, and WCDMA; Radio Frequency (e.g., RFID); infrared; and Near Field Magnetic communication protocols. The at least one wireless communication component 1056 transmits electrical, electromagnetic, or optical signals that carry digital data streams or analog signals representing various types of information.

(88) The at least one wired/wireless power distribution component 1058 includes components or devices that are configured to provide power to other devices. For example, in one embodiment, the at least one power distribution component 1058 includes a magnetic induction system that is configured to provide wireless power to one or more player input devices near the EGM. In one embodiment, a player input device docking region is provided, and includes a power distribution component that is configured to recharge a player input device without requiring metal-to-metal contact. In one embodiment, the at least one power distribution component 1058 is configured to distribute power to one or more internal components of the EGM, such as one or more rechargeable power sources (e.g., rechargeable batteries) located at the EGM.

(89) In certain embodiments, the at least one sensor 1060 includes at least one of: optical sensors, pressure sensors, RF sensors, infrared sensors, image sensors, thermal sensors, and biometric sensors. The at least one sensor 1060 may be used for a variety of functions, such as: detecting movements and/or gestures of various objects within a

predetermined proximity to the EGM; detecting the presence and/or identity of various persons (e.g., players, casino employees, etc.), devices (e.g., player input devices), and/or systems within a predetermined proximity to the EGM.

(90) The at least one data preservation component 1062 is configured to detect or sense one or more events and/or conditions that, for example, may result in damage to the EGM and/or that may result in loss of information associated with the EGM. Additionally, the data preservation system 1062 may be operable to initiate one or more appropriate action(s) in response to the detection of such events/conditions.

(91) The at least one motion/gesture analysis and interpretation component 1064 is configured to analyze and/or interpret information relating to detected player movements and/or gestures to determine appropriate player input information relating to the detected player movements and/or gestures. For example, in one embodiment, the at least one motion/gesture analysis and interpretation component 1064 is configured to perform one or more of the following functions: analyze the detected gross motion or gestures of a player; interpret the player's motion or gestures (e.g., in the context of a casino game being played) to identify instructions or input from the player; utilize the interpreted instructions/input to advance the game state; etc. In other embodiments, at least a portion of these additional functions may be implemented at a remote system or device.

(92) The at least one portable power source 1068 enables the EGM to operate in a mobile environment. For example, in one embodiment, the EGM 300 includes one or more rechargeable batteries.

(93) The at least one geolocation module 1076 is configured to acquire geolocation information from one or more remote sources and use the acquired geolocation information to determine information relating to a relative and/or absolute position of the EGM. For example, in one implementation, the at least one geolocation module 1076 is configured to receive GPS signal information for use in determining the position or location of the EGM. In another implementation, the at least one geolocation module 1076 is configured to receive multiple wireless signals from multiple remote devices (e.g., EGMs, servers, wireless access points, etc.) and use the signal information to compute position/location information relating to the position or location of the EGM.

(94) The at least one player identification module 1077 is configured to determine the identity of the current player or current owner of the EGM. For example, in one embodiment, the current player is required to perform a login process at the EGM in order to access one or more features. Alternatively, the EGM is configured to automatically determine the identity of the current player based on one or more external signals, such as an RFID tag or badge worn by the current player and that provides a wireless signal to the EGM that is used to determine the identity of the current player. In at least one embodiment, various security features are incorporated into the EGM to prevent unauthorized players from accessing confidential or sensitive information.

(95) The at least one information filtering module 1079 is configured to perform filtering (e.g., based on specified criteria) of selected information to be displayed at one or more displays 1035 of the EGM.

(96) In various embodiments, the EGM includes a plurality of communication ports configured to enable the at least one processor of the EGM to communicate with and to operate with external peripherals, such as: accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, SCSI ports, solenoids, speakers, thumbsticks, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices. U.S. Pat. No. 7,290,072 describes a variety of EGMs including one or more communication ports that enable the EGMs to communicate and operate with one or more external peripherals.

(97) As generally described above, in certain embodiments, such as the example EGMs 2000a and 2000b illustrated in FIGS. 5A and 5B, the EGM has a support structure, housing, or cabinet that provides support for a plurality of the input devices and the output devices of the EGM. Further, the EGM is configured such that a player may operate it while standing or sitting. In various embodiments, the EGM is positioned on a base or stand, or is configured as a pub-style tabletop game (not shown) that a player may operate typically while sitting. As illustrated by the different example EGMs

2000a and 2000b shown in FIGS. 5A and 5B, EGMs may have varying housing and display configurations.

(98) In certain embodiments, the EGM is a device that has obtained approval from a regulatory gaming commission, and in other embodiments, the EGM is a device that has not obtained approval from a regulatory gaming commission.

(99) The EGMs described above are merely three examples of different types of EGMs. Certain of these example EGMs may include one or more elements that may not be included in all gaming systems, and these example EGMs may not include one or more elements that are included in other gaming systems. For example, certain EGMs include a coin acceptor while others do not.

Operation of Primary or Base Games and/or Secondary or Bonus Games

(100) In various embodiments, an EGM may be implemented in one of a variety of different configurations. In various embodiments, the EGM may be implemented as one of: (a) a dedicated EGM in which computerized game programs executable by the EGM for controlling any primary or base games (referred to herein as “primary games”) and/or any secondary or bonus games or other functions (referred to herein as “secondary games”) displayed by the EGM are provided with the EGM before delivery to a gaming establishment or before being provided to a player; and (b) a changeable EGM in which computerized game programs executable by the EGM for controlling any primary games and/or secondary games displayed by the EGM are downloadable or otherwise transferred to the EGM through a data network or remote communication link; from a USB drive, flash memory card, or other suitable memory device; or in any other suitable manner after the EGM is physically located in a gaming establishment or after the EGM is provided to a player.

(101) As generally explained above, in various embodiments in which the gaming system includes a central server, central controller, or remote host and a changeable EGM, the at least one memory device of the central server, central controller, or remote host stores different game programs and instructions executable by the at least one processor of the changeable EGM to control one or more primary games and/or secondary games displayed by the changeable EGM. More specifically, each such executable game program represents a different game or a different type of game that the at least one changeable EGM is configured to operate. In one example, certain of the game programs are executable by the changeable EGM to operate games having the same or substantially the same game play but different paytables. In different embodiments, each executable game program is associated with a primary game, a secondary game, or both. In certain embodiments, an executable game program is executable by the at least one processor of the at least one changeable EGM as a secondary game to be played simultaneously with a play of a primary game (which may be downloaded to or otherwise stored on the at least one changeable EGM), or vice versa.

(102) In operation of such embodiments, the central server, central controller, or remote host is configured to communicate one or more of the stored executable game programs to the at least one processor of the changeable EGM. In different embodiments, a stored executable game program is communicated or delivered to the at least one processor of the changeable EGM by: (a) embedding the executable game program in a device or a component (such as a microchip to be inserted into the changeable EGM); (b) writing the executable game program onto a disc or other media; or (c) uploading or streaming the executable game program over a data network (such as a dedicated data network). After the executable game program is communicated from the central server, central controller, or remote host to the changeable EGM, the at least one processor of the changeable EGM executes the executable game program to enable the primary game and/or the secondary game associated with that executable game program to be played using the display device(s) and/or the input device(s) of the changeable EGM. That is, when an executable game program is communicated to the at least one processor of the changeable EGM, the at least one processor of the changeable EGM changes the game or the type of game that may be played using the changeable EGM.

(103) In certain embodiments, the gaming system randomly determines any game outcome(s) (such as a win outcome) and/or award(s) (such as a quantity of credits to award for the win outcome) for a play of a primary game and/or a play of a secondary game based on probability data. In certain such embodiments, this random determination is provided through utilization of an RNG, such as a true RNG or a pseudo RNG, or any other suitable randomization process. In

one such embodiment, each game outcome or award is associated with a probability, and the gaming system generates the game outcome(s) and/or the award(s) to be provided based on the associated probabilities. In these embodiments, since the gaming system generates game outcomes and/or awards randomly or based on one or more probability calculations, there is no certainty that the gaming system will ever provide any specific game outcome and/or award.

(104) In certain embodiments, the gaming system maintains one or more predetermined pools or sets of predetermined game outcomes and/or awards. In certain such embodiments, upon generation or receipt of a game outcome and/or award request, the gaming system independently selects one of the predetermined game outcomes and/or awards from the one or more pools or sets. The gaming system flags or marks the selected game outcome and/or award as used. Once a game outcome or an award is flagged as used, it is prevented from further selection from its respective pool or set; that is, the gaming system does not select that game outcome or award upon another game outcome and/or award request. The gaming system provides the selected game outcome and/or award. Examples of this type of award evaluation are described in U.S. Pat. No. 7,470,183, entitled "Finite Pool Gaming Method and Apparatus"; U.S. Pat. No. 7,563,163, entitled "Gaming Device Including Outcome Pools for Providing Game Outcomes"; U.S. Pat. No. 7,833,092, entitled "Method and System for Compensating for Player Choice in a Game of Chance"; U.S. Pat. No. 8,070,579, entitled "Bingo System with Downloadable Common Patterns"; and U.S. Pat. No. 8,398,472, entitled "Central Determination Poker Game".

(105) In certain embodiments, the gaming system determines a predetermined game outcome and/or award based on the results of a bingo, keno, or lottery game. In certain such embodiments, the gaming system utilizes one or more bingo, keno, or lottery games to determine the predetermined game outcome and/or award provided for a primary game and/or a secondary game. The gaming system is provided or associated with a bingo card. Each bingo card consists of a matrix or array of elements, wherein each element is designated with separate indicia. After a bingo card is provided, the gaming system randomly selects or draws a plurality of the elements. As each element is selected, a determination is made as to whether the selected element is present on the bingo card. If the selected element is present on the bingo card, that selected element on the provided bingo card is marked or flagged. This process of selecting elements and marking any selected elements on the provided bingo cards continues until one or more predetermined patterns are marked on one or more of the provided bingo cards. After one or more predetermined patterns are marked on one or more of the provided bingo cards, game outcome and/or award is determined based, at least in part, on the selected elements on the provided bingo cards. Examples of this type of award determination are described in U.S. Pat. No. 7,753,774, entitled "Using Multiple Bingo Cards to Represent Multiple Slot Paylines and Other Class III Game Options"; U.S. Pat. No. 7,731,581, entitled "Multi-Player Bingo Game with Multiple Alternative Outcome Displays"; U.S. Pat. No. 7,955,170, entitled "Providing Non-Bingo Outcomes for a Bingo Game"; U.S. Pat. No. 8,070,579, entitled "Bingo System with Downloadable Common Patterns"; and U.S. Pat. No. 8,500,538, entitled "Bingo Gaming System and Method for Providing Multiple Outcomes from Single Bingo Pattern".

(106) In certain embodiments in which the gaming system includes a central server, central controller, or remote host and an EGM, the EGM is configured to communicate with the central server, central controller, or remote host for monitoring purposes only. In such embodiments, the EGM determines the game outcome(s) and/or award(s) to be provided in any of the manners described above, and the central server, central controller, or remote host monitors the activities and events occurring on the EGM. In one such embodiment, the gaming system includes a real-time or online accounting and gaming information system configured to communicate with the central server, central controller, or remote host. In this embodiment, the accounting and gaming information system includes: (a) a player database configured to store player profiles, (b) a player tracking module configured to track players (as described below), and (c) a credit system configured to provide automated transactions. Examples of such accounting systems are described in U.S. Pat. No. 6,913,534, entitled "Gaming Machine Having a Lottery Game and Capability for Integration with Gaming Device Accounting System and Player Tracking System," and U.S. Pat. No. 8,597,116, entitled "Virtual Player Tracking and Related Services".

(107) As noted above, in various embodiments, the gaming system includes one or more executable game programs executable by at least one processor of the gaming system to provide one or more primary games and one or more secondary games. The primary game(s) and the secondary game(s) may comprise any suitable games and/or wagering games, such as, but not limited to: electro-mechanical or video slot or spinning reel type games; video card games such as video draw poker, multi-hand video draw poker, other video poker games, video blackjack games, and video baccarat games; video keno games; video bingo games; and video selection games.

(108) In certain embodiments in which the primary game is a slot or spinning reel type game, the gaming system includes one or more reels in either an electromechanical form with mechanical rotating reels or in a video form with simulated reels and movement thereof. Each reel displays a plurality of indicia or symbols, such as bells, hearts, fruits, numbers, letters, bars, or other images that typically correspond to a theme associated with the gaming system. In certain such embodiments, the gaming system includes one or more paylines associated with the reels. The example EGM 2000b shown in FIG. 5B includes a payline 1152 and a plurality of reels 1154. In certain embodiments, one or more of the reels are independent reels or unisymbol reels. In such embodiments, each independent reel generates and displays one symbol.

(109) In various embodiments, one or more of the paylines is horizontal, vertical, circular, diagonal, angled, or any suitable combination thereof. In other embodiments, each of one or more of the paylines is associated with a plurality of adjacent symbol display areas on a requisite number of adjacent reels. In one such embodiment, one or more paylines are formed between at least two symbol display areas that are adjacent to each other by either sharing a common side or sharing a common corner (i.e., such paylines are connected paylines). The gaming system enables a wager to be placed on one or more of such paylines to activate such paylines. In other embodiments in which one or more paylines are formed between at least two adjacent symbol display areas, the gaming system enables a wager to be placed on a plurality of symbol display areas, which activates those symbol display areas.

(110) In various embodiments, the gaming system provides one or more awards after a spin of the reels when specified types and/or configurations of the indicia or symbols on the reels occur on an active payline or otherwise occur in a winning pattern, occur on the requisite number of adjacent reels, and/or occur in a scatter pay arrangement.

(111) In certain embodiments, the gaming system employs a way to win award determination. In these embodiments, any outcome to be provided is determined based on a number of associated symbols that are generated in active symbol display areas on the requisite number of adjacent reels (i.e., not on paylines passing through any displayed winning symbol combinations). If a winning symbol combination is generated on the reels, one award for that occurrence of the generated winning symbol combination is provided. Examples of ways to win award determinations are described in U.S. Pat. No. 8,012,011, entitled "Gaming Device and Method Having Independent Reels and Multiple Ways of Winning"; U.S. Pat. No. 8,241,104, entitled "Gaming Device and Method Having Designated Rules for Determining Ways To Win"; and U.S. Pat. No. 8,430,739, entitled "Gaming System and Method Having Wager Dependent Different Symbol Evaluations".

(112) In various embodiments, the gaming system includes a progressive award. Typically, a progressive award includes an initial amount and an additional amount funded through a portion of each wager placed to initiate a play of a primary game. When one or more triggering events occurs, the gaming system provides at least a portion of the progressive award. After the gaming system provides the progressive award, an amount of the progressive award is reset to the initial amount and a portion of each subsequent wager is allocated to the next progressive award. Examples of progressive gaming systems are described in U.S. Pat. No. 7,585,223, entitled "Server Based Gaming System Having Multiple Progressive Awards"; U.S. Pat. No. 7,651,392, entitled "Gaming Device System Having Partial Progressive Payout"; U.S. Pat. No. 7,666,093, entitled "Gaming Method and Device Involving Progressive Wagers"; U.S. Pat. No. 7,780,523, entitled "Server Based Gaming System Having Multiple Progressive Awards"; and U.S. Pat. No. 8,337,298, entitled "Gaming Device Having Multiple Different Types of Progressive Awards".

(113) As generally noted above, in addition to providing winning credits or other awards for one or more plays of the

primary game(s), in various embodiments the gaming system provides credits or other awards for one or more plays of one or more secondary games. The secondary game typically enables an award to be obtained in addition to any award obtained through play of the primary game(s). The secondary game(s) typically produces a higher level of player excitement than the primary game(s) because the secondary game(s) provides a greater expectation of winning than the primary game(s) and is accompanied with more attractive or unusual features than the primary game(s). The secondary game(s) may be any type of suitable game, either similar to or completely different from the primary game.

(114) In various embodiments, the gaming system automatically provides or initiates the secondary game upon the occurrence of a triggering event or the satisfaction of a qualifying condition. In other embodiments, the gaming system initiates the secondary game upon the occurrence of the triggering event or the satisfaction of the qualifying condition and upon receipt of an initiation input. In certain embodiments, the triggering event or qualifying condition is a selected outcome in the primary game(s) or a particular arrangement of one or more indicia on a display device for a play of the primary game(s), such as a "BONUS" symbol appearing on three adjacent reels along a payline following a spin of the reels for a play of the primary game. In other embodiments, the triggering event or qualifying condition occurs based on a certain amount of game play (such as number of games, number of credits, amount of time) being exceeded, or based on a specified number of points being earned during game play. Any suitable triggering event or qualifying condition or any suitable combination of a plurality of different triggering events or qualifying conditions may be employed.

(115) In other embodiments, at least one processor of the gaming system randomly determines when to provide one or more plays of one or more secondary games. In one such embodiment, no apparent reason is provided for providing the secondary game. In this embodiment, qualifying for a secondary game is not triggered by the occurrence of an event in any primary game or based specifically on any of the plays of any primary game. That is, qualification is provided without any explanation or, alternatively, with a simple explanation. In another such embodiment, the gaming system determines qualification for a secondary game at least partially based on a game triggered or symbol triggered event, such as at least partially based on play of a primary game.

(116) In various embodiments, after qualification for a secondary game has been determined, the secondary game participation may be enhanced through continued play on the primary game. Thus, in certain embodiments, for each secondary game qualifying event, such as a secondary game symbol, that is obtained, a given number of secondary game wagering points or credits is accumulated in a "secondary game meter" configured to accrue the secondary game wagering credits or entries toward eventual participation in the secondary game. In one such embodiment, the occurrence of multiple such secondary game qualifying events in the primary game results in an arithmetic or exponential increase in the number of secondary game wagering credits awarded. In another such embodiment, any extra secondary game wagering credits may be redeemed during the secondary game to extend play of the secondary game.

(117) In certain embodiments, no separate entry fee or buy-in for the secondary game is required. That is, entry into the secondary game cannot be purchased; rather, in these embodiments entry must be won or earned through play of the primary game, thereby encouraging play of the primary game. In other embodiments, qualification for the secondary game is accomplished through a simple "buy-in." For example, qualification through other specified activities is unsuccessful, payment of a fee or placement of an additional wager "buys-in" to the secondary game. In certain embodiments, a separate side wager must be placed on the secondary game or a wager of a designated amount must be placed on the primary game to enable qualification for the secondary game. In these embodiments, the secondary game triggering event must occur and the side wager (or designated primary game wager amount) must have been placed for the secondary game to trigger.

(118) In various embodiments in which the gaming system includes a plurality of EGMs, the EGMs are configured to communicate with one another to provide a group gaming environment. In certain such embodiments, the EGMs enable players of those EGMs to work in conjunction with one another, such as by enabling the players to play together as a team or group, to win one or more awards. In other such embodiments, the EGMs enable players of those EGMs to

compete against one another for one or more awards. In one such embodiment, the EGMs enable the players of those EGMs to participate in one or more gaming tournaments for one or more awards. Examples of group gaming systems are described in U.S. Pat. No. 8,070,583, entitled "Server Based Gaming System and Method for Selectively Providing One or More Different Tournaments"; U.S. Pat. No. 8,500,548, entitled "Gaming System and Method for Providing Team Progressive Awards"; and U.S. Pat. No. 8,562,423, entitled "Method and Apparatus for Rewarding Multiple Game Players for a Single Win".

(119) In various embodiments, the gaming system includes one or more player tracking systems. Such player tracking systems enable operators of the gaming system (such as casinos or other gaming establishments) to recognize the value of customer loyalty by identifying frequent customers and rewarding them for their patronage. Such a player tracking system is configured to track a player's gaming activity. In one such embodiment, the player tracking system does so through the use of player tracking cards. In this embodiment, a player is issued a player identification card that has an encoded player identification number that uniquely identifies the player. When the player's playing tracking card is inserted into a card reader of the gaming system to begin a gaming session, the card reader reads the player identification number off the player tracking card to identify the player. The gaming system timely tracks any suitable information or data relating to the identified player's gaming session. The gaming system also timely tracks when the player tracking card is removed to conclude play for that gaming session. In another embodiment, rather than requiring insertion of a player tracking card into the card reader, the gaming system utilizes one or more portable devices, such as a mobile phone, a radio frequency identification tag, or any other suitable wireless device, to track when a gaming session begins and ends. In another embodiment, the gaming system utilizes any suitable biometric technology or ticket technology to track when a gaming session begins and ends.

(120) In such embodiments, during one or more gaming sessions, the gaming system tracks any suitable information or data, such as any amounts wagered, average wager amounts, and/or the time at which these wagers are placed. In different embodiments, for one or more players, the player tracking system includes the player's account number, the player's card number, the player's first name, the player's surname, the player's preferred name, the player's player tracking ranking, any promotion status associated with the player's player tracking card, the player's address, the player's birthday, the player's anniversary, the player's recent gaming sessions, or any other suitable data. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed on a player tracking display. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed via one or more service windows that are displayed on the central display device and/or the upper display device. Examples of player tracking systems are described in U.S. Pat. No. 6,722,985, entitled "Universal Player Tracking System"; U.S. Pat. No. 6,908,387, entitled "Player Tracking Communication Mechanisms in a Gaming Machine"; U.S. Pat. No. 7,311,605, entitled "Player Tracking Assembly for Complete Patron Tracking for Both Gaming and Non-Gaming Casino Activity"; U.S. Pat. No. 7,611,411, entitled "Player Tracking Instruments Having Multiple Communication Modes"; U.S. Pat. No. 7,617,151, entitled "Alternative Player Tracking Techniques"; and U.S. Pat. No. 8,057,298, entitled "Virtual Player Tracking and Related Services".

Web-Based Gaming

(121) In various embodiments, the gaming system includes one or more servers configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable web-based game play using the personal gaming device. In various embodiments, the player must first access a gaming website via an Internet browser of the personal gaming device or execute an application (commonly called an "app") installed on the personal gaming device before the player can use the personal gaming device to participate in web-based game play. In certain embodiments, the one or more servers and the personal gaming device operate in a thin-client environment. In these embodiments, the personal gaming device receives inputs via one or more input devices (such as a touch screen and/or physical buttons), the personal gaming device sends the received inputs to the one or more servers, the one or more servers make various determinations based on the inputs and determine content

to be displayed (such as a randomly determined game outcome and corresponding award), the one or more servers send the content to the personal gaming device, and the personal gaming device displays the content.

(122) In certain such embodiments, the one or more servers must identify the player before enabling game play on the personal gaming device (or, in some embodiments, before enabling monetary wager-based game play on the personal gaming device). In these embodiments, the player must identify herself to the one or more servers, such as by inputting the player's unique playername and password combination, providing an input to a biometric sensor (e.g., a fingerprint sensor, a retinal sensor, a voice sensor, or a facial-recognition sensor), or providing any other suitable information.

(123) Once identified, the one or more servers enable the player to establish an account balance from which the player can draw credits usable to wager on plays of a game. In certain embodiments, the one or more servers enable the player to initiate an electronic funds transfer to transfer funds from a bank account to the player's account balance. In other embodiments, the one or more servers enable the player to make a payment using the player's credit card, debit card, or other suitable device to add money to the player's account balance. In other embodiments, the one or more servers enable the player to add money to the player's account balance via a peer-to-peer type application, such as PayPal or Venmo. The one or more servers also enable the player to cash out the player's account balance (or part of it) in any suitable manner, such as via an electronic funds transfer, by initiating creation of a paper check that is mailed to the player, or by initiating printing of a voucher at a kiosk in a gaming establishment.

(124) In certain embodiments, the one or more servers include a payment server that handles establishing and cashing out players' account balances and a separate game server configured to determine the outcome and any associated award for a play of a game. In these embodiments, the game server is configured to communicate with the personal gaming device and the payment device, and the personal gaming device and the payment device are not configured to directly communicate with one another. In these embodiments, when the game server receives data representing a request to start a play of a game at a desired wager, the game server sends data representing the desired wager to the payment server. The payment server determines whether the player's account balance can cover the desired wager (i.e., includes a monetary balance at least equal to the desired wager).

(125) If the payment server determines that the player's account balance cannot cover the desired wager, the payment server notifies the game server, which then instructs the personal gaming device to display a suitable notification to the player that the player's account balance is too low to place the desired wager. If the payment server determines that the player's account balance can cover the desired wager, the payment server deducts the desired wager from the account balance and notifies the game server. The game server then determines an outcome and any associated award for the play of the game. The game server notifies the payment server of any nonzero award, and the payment server increases the player's account balance by the nonzero award. The game server sends data representing the outcome and any award to the personal gaming device, which displays the outcome and any award.

(126) In certain embodiments, the one or more servers enable web-based game play using a personal gaming device only if the personal gaming device satisfies one or more jurisdictional requirements. In one embodiment, the one or more servers enable web-based game play using the personal gaming device only if the personal gaming device is located within a designated geographic area (such as within certain state or county lines or within the boundaries of a gaming establishment). In this embodiment, the geolocation module of the personal gaming device determines the location of the personal gaming device and sends the location to the one or more servers, which determine whether the personal gaming device is located within the designated geographic area. In various embodiments, the one or more servers enable non-monetary wager-based game play if the personal gaming device is located outside of the designated geographic area.

(127) In various embodiments, the gaming system includes an EGM configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable tethered mobile game play using the personal gaming device. Generally, in these embodiments, the EGM establishes communication with the personal gaming device and enables the player to play games on the EGM remotely via the

personal gaming device. In certain embodiments, the gaming system includes a geo-fence system that enables tethered game play within a particular geographic area but not outside of that geographic area. Examples of tethering an EGM to a personal gaming device and geo-fencing are described in U.S. Patent Appl. Pub. No. 2013/0267324, entitled "Remote Gaming Method Allowing Temporary Inactivation Without Terminating Playing Session Due to Game Inactivity".

Social Network Integration

(128) In certain embodiments, the gaming system is configured to communicate with a social network server that hosts or partially hosts a social networking website via a data network (such as the Internet) to integrate a player's gaming experience with the player's social networking account. This enables the gaming system to send certain information to the social network server that the social network server can use to create content (such as text, an image, and/or a video) and post it to the player's wall, newsfeed, or similar area of the social networking website accessible by the player's connections (and in certain cases the public) such that the player's connections can view that information. This also enables the gaming system to receive certain information from the social network server, such as the player's likes or dislikes or the player's list of connections. In certain embodiments, the gaming system enables the player to link the player's player account to the player's social networking account(s). This enables the gaming system to, once it identifies the player and initiates a gaming session (such as via the player logging in to a website (or an application) on the player's personal gaming device or via the player inserting the player's player tracking card into an EGM), link that gaming session to the player's social networking account(s). In other embodiments, the gaming system enables the player to link the player's social networking account(s) to individual gaming sessions when desired by providing the required login information.

(129) For instance, in one embodiment, if a player wins a particular award (e.g., a progressive award or a jackpot award) or an award that exceeds a certain threshold (e.g., an award exceeding \$1,000), the gaming system sends information about the award to the social network server to enable the server to create associated content (such as a screenshot of the outcome and associated award) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to play). In another embodiment, if a player joins a multiplayer game and there is another seat available, the gaming system sends that information to the social network sever to enable the server to create associated content (such as text indicating a vacancy for that particular game) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to fill the vacancy). In another embodiment, if the player consents, the gaming system sends advertisement information or offer information to the social network server to enable the social network server to create associated content (such as text or an image reflecting an advertisement and/or an offer) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see. In another embodiment, the gaming system enables the player to recommend a game to the player's connections by posting a recommendation to the player's wall (or other suitable area) of the social networking website.

Differentiating Certain Gaming Systems from General Purpose Computing Devices

(130) Certain of the gaming systems described herein, such as EGMs located in a casino or another gaming establishment, include certain components and/or are configured to operate in certain manners that differentiate these systems from general purpose computing devices, i.e., certain personal gaming devices such as desktop computers and laptop computers.

(131) For instance, EGMs are highly regulated to ensure fairness and, in many cases, EGMs are configured to award monetary awards up to multiple millions of dollars. To satisfy security and regulatory requirements in a gaming environment, hardware and/or software architectures are implemented in EGMs that differ significantly from those of general purpose computing devices. For purposes of illustration, a description of EGMs relative to general purpose computing devices and some examples of these additional (or different) hardware and/or software architectures found in EGMs are described below.

(132) At first glance, one might think that adapting general purpose computing device technologies to the gaming industry and EGMs would be a simple proposition because both general purpose computing devices and EGMs employ processors that control a variety of devices. However, due to at least: (1) the regulatory requirements placed on EGMs, (2) the harsh environment in which EGMs operate, (3) security requirements, and (4) fault tolerance requirements, adapting general purpose computing device technologies to EGMs can be quite difficult. Further, techniques and methods for solving a problem in the general purpose computing device industry, such as device compatibility and connectivity issues, might not be adequate in the gaming industry. For instance, a fault or a weakness tolerated in a general purpose computing device, such as security holes in software or frequent crashes, is not tolerated in an EGM because in an EGM these faults can lead to a direct loss of funds from the EGM, such as stolen cash or loss of revenue when the EGM is not operating properly or when the random outcome determination is manipulated.

(133) Certain differences between general purpose computing devices and EGMs are described below. A first difference between EGMs and general purpose computing devices is that EGMs are state-based systems. A state-based system stores and maintains its current state in a non-volatile memory such that, in the event of a power failure or other malfunction, the state-based system can return to that state when the power is restored or the malfunction is remedied. For instance, for a state-based EGM, if the EGM displays an award for a game of chance but the power to the EGM fails before the EGM provides the award to the player, the EGM stores the pre-power failure state in a non-volatile memory, returns to that state upon restoration of power, and provides the award to the player. This requirement affects the software and hardware design on EGMs. General purpose computing devices are not state-based machines, and a majority of data is usually lost when a malfunction occurs on a general purpose computing device.

(134) A second difference between EGMs and general purpose computing devices is that, for regulatory purposes, the software on the EGM utilized to operate the EGM has been designed to be static and monolithic to prevent cheating by the operator of the EGM. For instance, one solution that has been employed in the gaming industry to prevent cheating and to satisfy regulatory requirements has been to manufacture an EGM that can use a proprietary processor running instructions to provide the game of chance from an EPROM or other form of non-volatile memory. The coding instructions on the EPROM are static (non-changeable) and must be approved by a gaming regulators in a particular jurisdiction and installed in the presence of a person representing the gaming jurisdiction. Any changes to any part of the software required to generate the game of chance, such as adding a new device driver used to operate a device during generation of the game of chance, can require burning a new EPROM approved by the gaming jurisdiction and reinstalling the new EPROM on the EGM in the presence of a gaming regulator. Regardless of whether the EPROM solution is used, to gain approval in most gaming jurisdictions, an EGM must demonstrate sufficient safeguards that prevent an operator or a player of an EGM from manipulating the EGM's hardware and software in a manner that gives him an unfair, and in some cases illegal, advantage.

(135) A third difference between EGMs and general purpose computing devices is authentication—EGMs storing code are configured to authenticate the code to determine if the code is unaltered before executing the code. If the code has been altered, the EGM prevents the code from being executed. The code authentication requirements in the gaming industry affect both hardware and software designs on EGMs. Certain EGMs use hash functions to authenticate code. For instance, one EGM stores game program code, a hash function, and an authentication hash (which may be encrypted). Before executing the game program code, the EGM hashes the game program code using the hash function to obtain a result hash and compares the result hash to the authentication hash. If the result hash matches the authentication hash, the EGM determines that the game program code is valid and executes the game program code. If the result hash does not match the authentication hash, the EGM determines that the game program code has been altered (i.e., may have been tampered with) and prevents execution of the game program code. Examples of EGM code authentication are described in U.S. Pat. No. 6,962,530, entitled "Authentication in a Secure Computerized Gaming System"; U.S. Pat. No. 7,043,641, entitled "Encryption in a Secure Computerized Gaming System"; U.S. Pat. No. 7,201,662, entitled "Method and Apparatus for Software Authentication"; and U.S. Pat. No. 8,627,097, entitled "System

and Method Enabling Parallel Processing of Hash Functions Using Authentication Checkpoint Hashes”.

(136) A fourth difference between EGMs and general purpose computing devices is that EGMs have unique peripheral device requirements that differ from those of a general purpose computing device, such as peripheral device security requirements not usually addressed by general purpose computing devices. For instance, monetary devices, such as coin dispensers, bill validators, and ticket printers and computing devices that are used to govern the input and output of cash or other items having monetary value (such as tickets) to and from an EGM have security requirements that are not typically addressed in general purpose computing devices. Therefore, many general purpose computing device techniques and methods developed to facilitate device connectivity and device compatibility do not address the emphasis placed on security in the gaming industry.

(137) To address some of the issues described above, a number of hardware/software components and architectures are utilized in EGMs that are not typically found in general purpose computing devices. These hardware/software components and architectures, as described below in more detail, include but are not limited to watchdog timers, voltage monitoring systems, state-based software architecture and supporting hardware, specialized communication interfaces, security monitoring, and trusted memory.

(138) Certain EGMs use a watchdog timer to provide a software failure detection mechanism. In a normally-operating EGM, the operating software periodically accesses control registers in the watchdog timer subsystem to “re-trigger” the watchdog. Should the operating software fail to access the control registers within a preset timeframe, the watchdog timer will timeout and generate a system reset. Typical watchdog timer circuits include a loadable timeout counter register to enable the operating software to set the timeout interval within a certain range of time. A differentiating feature of some circuits is that the operating software cannot completely disable the function of the watchdog timer. In other words, the watchdog timer always functions from the time power is applied to the board.

(139) Certain EGMs use several power supply voltages to operate portions of the computer circuitry. These can be generated in a central power supply or locally on the computer board. If any of these voltages falls out of the tolerance limits of the circuitry they power, unpredictable operation of the EGM may result. Though most modern general purpose computing devices include voltage monitoring circuitry, these types of circuits only report voltage status to the operating software. Out of tolerance voltages can cause software malfunction, creating a potential uncontrolled condition in the general purpose computing device. Certain EGMs have power supplies with relatively tighter voltage margins than that required by the operating circuitry. In addition, the voltage monitoring circuitry implemented in certain EGMs typically has two thresholds of control. The first threshold generates a software event that can be detected by the operating software and an error condition then generated. This threshold is triggered when a power supply voltage falls out of the tolerance range of the power supply, but is still within the operating range of the circuitry. The second threshold is set when a power supply voltage falls out of the operating tolerance of the circuitry. In this case, the circuitry generates a reset, halting operation of the EGM.

(140) As described above, certain EGMs are state-based machines. Different functions of the game provided by the EGM (e.g., bet, play, result, points in the graphical presentation, etc.) may be defined as a state. When the EGM moves a game from one state to another, the EGM stores critical data regarding the game software in a custom non-volatile memory subsystem. This ensures that the player's wager and credits are preserved and to minimize potential disputes in the event of a malfunction on the EGM. In general, the EGM does not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been stored. This feature enables the EGM to recover operation to the current state of play in the event of a malfunction, loss of power, etc. that occurred just before the malfunction. In at least one embodiment, the EGM is configured to store such critical information using atomic transactions.

(141) Generally, an atomic operation in computer science refers to a set of operations that can be combined so that they appear to the rest of the system to be a single operation with only two possible outcomes: success or failure. As related to data storage, an atomic transaction may be characterized as series of database operations which either all occur, or

all do not occur. A guarantee of atomicity prevents updates to the database occurring only partially, which can result in data corruption.

(142) To ensure the success of atomic transactions relating to critical information to be stored in the EGM memory before a failure event (e.g., malfunction, loss of power, etc.), memory that includes one or more of the following criteria be used: direct memory access capability; data read/write capability which meets or exceeds minimum read/write access characteristics (such as at least 5.08 Mbytes/sec (Read) and/or at least 38.0 Mbytes/sec (Write)). Memory devices that meet or exceed the above criteria may be referred to as “fault-tolerant” memory devices.

(143) Typically, battery-backed RAM devices may be configured to function as fault-tolerant devices according to the above criteria, whereas flash RAM and/or disk drive memory are typically not configurable to function as fault-tolerant devices according to the above criteria. Accordingly, battery-backed RAM devices are typically used to preserve EGM critical data, although other types of non-volatile memory devices may be employed. These memory devices are typically not used in typical general purpose computing devices.

(144) Thus, in at least one embodiment, the EGM is configured to store critical information in fault-tolerant memory (e.g., battery-backed RAM devices) using atomic transactions. Further, in at least one embodiment, the fault-tolerant memory is able to successfully complete all desired atomic transactions (e.g., relating to the storage of EGM critical information) within a time period of 200 milliseconds or less. In at least one embodiment, the time period of 200 milliseconds represents a maximum amount of time for which sufficient power may be available to the various EGM components after a power outage event has occurred at the EGM.

(145) As described previously, the EGM may not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been atomically stored. After the state of the EGM is restored during the play of a game of chance, game play may resume and the game may be completed in a manner that is no different than if the malfunction had not occurred. Thus, for example, when a malfunction occurs during a game of chance, the EGM may be restored to a state in the game of chance just before when the malfunction occurred. The restored state may include metering information and graphical information that was displayed on the EGM in the state before the malfunction. For example, when the malfunction occurs during the play of a card game after the cards have been dealt, the EGM may be restored with the cards that were previously displayed as part of the card game. As another example, a bonus game may be triggered during the play of a game of chance in which a player is required to make a number of selections on a video display screen. When a malfunction has occurred after the player has made one or more selections, the EGM may be restored to a state that shows the graphical presentation just before the malfunction including an indication of selections that have already been made by the player. In general, the EGM may be restored to any state in a plurality of states that occur in the game of chance that occurs while the game of chance is played or to states that occur between the play of a game of chance.

(146) Game history information regarding previous games played such as an amount wagered, the outcome of the game, and the like may also be stored in a non-volatile memory device. The information stored in the non-volatile memory may be detailed enough to reconstruct a portion of the graphical presentation that was previously presented on the EGM and the state of the EGM (e.g., credits) at the time the game of chance was played. The game history information may be utilized in the event of a dispute. For example, a player may decide that in a previous game of chance that they did not receive credit for an award that they believed they won. The game history information may be used to reconstruct the state of the EGM before, during, and/or after the disputed game to demonstrate whether the player was correct or not in the player's assertion. Examples of a state-based EGM, recovery from malfunctions, and game history are described in U.S. Pat. No. 6,804,763, entitled “High Performance Battery Backed RAM Interface”; U.S. Pat. No. 6,863,608, entitled “Frame Capture of Actual Game Play”; U.S. Pat. No. 7,111,141, entitled “Dynamic NV-RAM”; and U.S. Pat. No. 7,384,339, entitled, “Frame Capture of Actual Game Play”.

(147) Another feature of EGMs is that they often include unique interfaces, including serial interfaces, to connect to specific subsystems internal and external to the EGM. The serial devices may have electrical interface requirements that

differ from the “standard” EIA serial interfaces provided by general purpose computing devices. These interfaces may include, for example, Fiber Optic Serial, optically coupled serial interfaces, current loop style serial interfaces, etc. In addition, to conserve serial interfaces internally in the EGM, serial devices may be connected in a shared, daisy-chain fashion in which multiple peripheral devices are connected to a single serial channel.

(148) The serial interfaces may be used to transmit information using communication protocols that are unique to the gaming industry. For example, IGT's Netplex is a proprietary communication protocol used for serial communication between EGMs. As another example, SAS is a communication protocol used to transmit information, such as metering information, from an EGM to a remote device. Often SAS is used in conjunction with a player tracking system.

(149) Certain EGMs may alternatively be treated as peripheral devices to a casino communication controller and connected in a shared daisy chain fashion to a single serial interface. In both cases, the peripheral devices are assigned device addresses. If so, the serial controller circuitry must implement a method to generate or detect unique device addresses. General purpose computing device serial ports are not able to do this.

(150) Security monitoring circuits detect intrusion into an EGM by monitoring security switches attached to access doors in the EGM cabinet. Access violations result in suspension of game play and can trigger additional security operations to preserve the current state of game play. These circuits also function when power is off by use of a battery backup. In power-off operation, these circuits continue to monitor the access doors of the EGM. When power is restored, the EGM can determine whether any security violations occurred while power was off, e.g., via software for reading status registers. This can trigger event log entries and further data authentication operations by the EGM software.

(151) Trusted memory devices and/or trusted memory sources are included in an EGM to ensure the authenticity of the software that may be stored on less secure memory subsystems, such as mass storage devices. Trusted memory devices and controlling circuitry are typically designed to not enable modification of the code and data stored in the memory device while the memory device is installed in the EGM. The code and data stored in these devices may include authentication algorithms, random number generators, authentication keys, operating system kernels, etc. The purpose of these trusted memory devices is to provide gaming regulatory authorities a root trusted authority within the computing environment of the EGM that can be tracked and verified as original. This may be accomplished via removal of the trusted memory device from the EGM computer and verification of the secure memory device contents is a separate third party verification device. Once the trusted memory device is verified as authentic, and based on the approval of the verification algorithms included in the trusted device, the EGM is enabled to verify the authenticity of additional code and data that may be located in the gaming computer assembly, such as code and data stored on hard disk drives. Examples of trusted memory devices are described in U.S. Pat. No. 6,685,567, entitled “Process Verification”.

(152) In at least one embodiment, at least a portion of the trusted memory devices/sources may correspond to memory that cannot easily be altered (e.g., “unalterable memory”) such as EPROMS, PROMS, Bios, Extended Bios, and/or other memory sources that are able to be configured, verified, and/or authenticated (e.g., for authenticity) in a secure and controlled manner.

(153) According to one embodiment, when a trusted information source is in communication with a remote device via a network, the remote device may employ a verification scheme to verify the identity of the trusted information source. For example, the trusted information source and the remote device may exchange information using public and private encryption keys to verify each other's identities. In another embodiment, the remote device and the trusted information source may engage in methods using zero knowledge proofs to authenticate each of their respective identities.

(154) EGMs storing trusted information may utilize apparatuses or methods to detect and prevent tampering. For instance, trusted information stored in a trusted memory device may be encrypted to prevent its misuse. In addition, the trusted memory device may be secured behind a locked door. Further, one or more sensors may be coupled to the memory device to detect tampering with the memory device and provide some record of the tampering. In yet another example, the memory device storing trusted information might be designed to detect tampering attempts and clear or

erase itself when an attempt at tampering has been detected. Examples of trusted memory devices/sources are described in U.S. Pat. No. 7,515,718, entitled "Secured Virtual Network in a Gaming Environment".

(155) Mass storage devices used in a general purpose computing devices typically enable code and data to be read from and written to the mass storage device. In a gaming environment, modification of the gaming code stored on a mass storage device is strictly controlled and would only be enabled under specific maintenance type events with electronic and physical enablers required. Though this level of security could be provided by software, EGMs that include mass storage devices include hardware level mass storage data protection circuitry that operates at the circuit level to monitor attempts to modify data on the mass storage device and will generate both software and hardware error triggers should a data modification be attempted without the proper electronic and physical enablers being present. Examples of using a mass storage device are described in U.S. Pat. No. 6,149,522, entitled "Method of Authenticating Game Data Sets in an Electronic Casino Gaming System".

(156) It should be appreciated that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. For example, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. In another example, the terms "including" and "comprising" and variations thereof, when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. Additionally, a listing of items does not imply that any or all of the items are mutually exclusive nor does a listing of items imply that any or all of the items are collectively exhaustive of anything or in a particular order, unless expressly specified otherwise. Moreover, as used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. It should be further appreciated that headings of sections provided in this document and the titles are for convenience only, and are not to be taken as limiting the disclosure in any way. Furthermore, unless expressly specified otherwise, devices that are in communication with each other need not be in continuous communication with each other and may communicate directly or indirectly through one or more intermediaries.

(157) Various changes and modifications to the present embodiments described herein will be apparent to those skilled in the art. For example, a description of an embodiment with several components in communication with each other does not imply that all such components are required, or that each of the disclosed components must communicate with every other component. On the contrary a variety of optional components are described to illustrate the wide variety of possible embodiments of the present disclosure. As such, these changes and modifications can be made without departing from the spirit and scope of the present subject matter and without diminishing its intended technical scope. It is therefore intended that such changes and modifications be covered by the appended claims.

claimsHtml: 1. A method of operating a gaming system to provide a play of a multi-player game, said method comprising: causing a display, by a display device, of a wheel and a plurality of selectable sections of the wheel, wherein the sections are respectively associated with a plurality of different awards for the play of the multi-player game, and wherein the sections are displayed without displaying any of the awards in association with the sections; causing a display, by the display device, of a quantity of the sections that can be selected by each of a plurality of qualified players; after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, causing a display, by the display device, for each qualified player, of a player indication associated with the qualified player and with the section selected by the qualified player; after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, causing a display, by the display device, for each of the sections selected by the qualified players, of an award symbol associated with that section; causing a display, by the display device, of the wheel spinning; causing a display, by the display device, of the wheel stopped after spinning and an indicator indicating one of the sections; and causing a display, by the display device, of an indication associated with the qualified player that selected the indicated section.

2. The method of claim 1, which comprises, for the play of the multi-player game, causing the player indications

associated with the qualified players to be displayed on respective sections selected by the qualified players.

3. The method of claim 1, which comprises, for the play of the multi-player game, causing a display, by the display device, of two player indications associated with two of the qualified players associated with a same one of the sections selected by the two qualified players.

4. The method of claim 1, which comprises, for the play of the multi-player game, for each section selected by one of the qualified players, causing a display, by the display device, of the award symbol associated with that section on the section adjacent to the player indication displayed on the section.

5. The method of claim 1, which comprises, for the play of the multi-player game, causing a display, by the display device, of the quantity of the sections that can be selected by each of the plurality of qualified players as only one section.

6. The method of claim 1, which comprises, for the play of the multi-player game, causing a display, by the display device, of the quantity of the sections that can be selected by each of two of the qualified players as different quantities.

7. The method of claim 1, which comprises, for the play of the multi-player game, after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, causing a display, by the display device, for each of the sections not selected by any of the qualified players, of an award symbol associated with that section.

8. A method of operating a gaming system to provide a play of a multi-player game, the method comprising: causing a display, by a display device, of a wheel, a plurality of selectable sections of the wheel, and a plurality of different awards on the sections; causing a display, by the display device, of a quantity of the sections that can be selected by each of a plurality of qualified players, wherein each qualified player is based on the qualified player having accumulated qualifications at least of a minimum qualification threshold; after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, causing a display, by the display device, for each qualified player, of a player indication associated with the qualified player on the section selected by the qualified player; causing a display, by the display device, of the wheel spinning; causing a display, by the display device, of the wheel stopped after spinning and an indicator indicating one of the sections; and causing a display, by the display device, of an indication of the qualified player that selected the indicated section.

9. The method of claim 8, which comprises, for the play of the multi-player game, causing a display, by the display device, of two player indications associated with two of the qualified players associated with a same one of the sections selected by the two qualified players.

10. The method of claim 8, which comprises, for the play of the multi-player game, for each section selected by one of the qualified players, causing a display, by the display device, of the award symbol associated with that section on the section adjacent to the player indication displayed on the section.

11. The method of claim 8, which comprises, for the play of the multi-player game, causing a display, by the display device, of the quantity of the sections that can be selected by each of the plurality of qualified players as only one section.

12. The method of claim 8, which comprises, for the play of the multi-player game, causing a display, by the display device, of the quantity of the sections that can be selected by each of two of the plurality of qualified players as different quantities.

13. A method of operating a gaming system to provide a play of a multi-player game, the method comprising: causing a display, by a display device, of a symbol display object and a plurality of selectable sections of the symbol display object, wherein the sections are respectively associated with a plurality of different awards for the play of the multi-player game, and wherein the sections are displayed without displaying any of the awards in association with the sections; causing a display, by the display device, of a quantity of the sections that can be selected by each of a plurality of qualified players; after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, causing a display, by the display device, for each qualified player, of a player indication associated

with the qualified player on the section selected by the qualified player; after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, causing a display, by the display device, for each of the sections selected by the qualified players, of an award symbol on that section; causing a display, by the display device, of an indication of one of the sections of the symbol display object; and causing a display, by the display device, of an indication of each of the qualified players that selected the indicated section of the symbol display object.

14. The method of claim 13, which comprises, for the play of the multi-player game, causing a display, by the display device, of two player indications associated with two of the qualified players associated with a same one of the sections selected by the two qualified players.

15. The method of claim 13, which comprises, for the play of the multi-player game, for each section selected by one of the qualified players, causing a display, by the display device, of the award symbol associated with that section on the section adjacent to the player indication displayed on the section.

16. The method of claim 13, which comprises, for the play of the multi-player game, causing a display, by the display device, of the quantity of the sections that can be selected by each of the plurality of qualified players as only one section.

17. The method of claim 13, which comprises, for the play of the multi-player game, causing a display, by the display device, of the quantity of the sections that can be selected by each of two of the plurality of qualified players as different quantities.

18. The method of claim 13, which comprises, for the play of the multi-player game, after receipt of player selections for each of the sections that can be selected by the qualified players, causing a display, by the display device, for each of the sections not selected by any of the qualified players, of an award symbol associated with that section.

19. The method of claim 13, which comprises, for multiple different plays of the multi-player game, causing a display, by the display device, of different award symbols on the same sections of the symbol award display.

20. The method of claim 13, wherein the symbol display object comprises a wheel.

briefHtml: BACKGROUND
(1) The present disclosure relates to a banked bonus player selection feature in a gaming environment.
(2) Gaming machines may provide players awards in primary games. Gaming machines generally require the player to place a wager to activate the primary game. The award may be based on the player obtaining a winning symbol or symbol combination and on the amount of the wager. Gaming machines may also provide players awards in secondary games. The secondary games may be wheel based games.
BRIEF SUMMARY

(3) In various embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions, that when executed by the processor, cause the processor to, for a play of a multi-player game: cause a display, by a display device, of a wheel and a plurality of selectable sections of the wheel, wherein the sections are respectively associated with a plurality of different awards for the play of the multi-player game, and wherein the sections are displayed without displaying any of the awards in association with the sections; and cause a display, by the display device, of a quantity of the sections that can be selected by each of a plurality of qualified players. The plurality of instructions, when executed by the processor, further cause the processor to: after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, cause a display, by the display device, for each qualified player, a player indication associated with the qualified player and with the section selected by the qualified player; after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, cause a display, by the display device, for each of the sections selected by the qualified players, an award symbol associated with that section; cause a display, by the display device, of the wheel spinning; cause a display, by the display device, of the wheel stopped after spinning and an indicator indicating one of the sections; and cause a display, by the display device, of an indication associated with the qualified player that selected the indicated section.
(4) In various other embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions, that when executed by the processor, cause the processor to, for a play of a multi-player game: cause a display, by a display device, of a wheel, a plurality of

selectable sections of the wheel, and a plurality of different awards on the sections; and cause a display, by the display device, of a quantity of the sections that can be selected by each of a plurality of qualified players, wherein each qualified player is based on the qualified player having accumulated qualifications at least of a minimum qualification threshold. The plurality of instructions, when executed by the processor, further cause the processor to: after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, cause a display, by the display device, for each qualified player, a player indication associated with the qualified player on the section selected by the qualified player; cause a display, by the display device, of the wheel spinning; cause a display, by the display device, of the wheel stopped after spinning and an indicator indicating one of the sections; and cause a display, by the display device, of an indication of the qualified player that selected the indicated section.

(5) In various other embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions, that when executed by the processor, cause the processor to, for a play of a multi-player game: cause a display, by a display device, of a symbol display object and a plurality of selectable sections of the symbol display object, wherein the sections are respectively associated with a plurality of different awards for the play of the multi-player game, and wherein the sections are displayed without displaying any of the awards in association with the sections; and cause a display, by the display device, of a quantity of the sections that can be selected by each of a plurality of qualified players. The plurality of instructions, when executed by the processor, further cause the processor to: after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, cause a display, by the display device, for each qualified player, a player indication associated with the qualified player on the section selected by the qualified player; after receipt of player selections for each of the quantity of the sections that can be selected by the qualified players, cause a display, by the display device, for each of the sections selected by the qualified players, an award symbol on that section; cause a display, by the display device, of an indication of one of the sections of the symbol display object; and cause a display, by the display device, of an indication of each of the qualified players that selected the indicated section of the symbol display object.

(6) Additional features are described herein, and will be apparent from the following Detailed Description and the figures.

backgroundTextHtml: PRIORITY

(1) This application is a continuation of, claims priority to and the benefit of U.S. patent application Ser. No. 18/536,699, filed on Dec. 12, 2023, which is a continuation of, claims priority to and the benefit of U.S. patent application Ser. No. 17/804,158, filed on May 26, 2022, the entire contents of both of which are incorporated herein by reference.

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usRefGroup: US 6364314 B1 Canterbury 20020400 cited by applicant, US 6663489 B2 Baerlocher 20031200 cited by applicant, US 7425177 B2 Rodgers et al. 20080900 cited by applicant, US 7431649 B2 Webb et al. 20081000 cited by applicant, US 7591723 B2 Cregan et al. 20090900 cited by applicant, US 7625278 B2 Paulsen et al. 20091200 cited by applicant, US 7708628 B2 Baerlocher 20100500 cited by applicant, US 7806760 B2 Baerlocher 20101000 cited by applicant, US 8100754 B2 Bigelow et al. 20120100 cited by applicant, US 8262458 B2 Rodgers et al. 20120900 cited by applicant, US 8308547 B2 Walker et al. 20121100 cited by applicant, US 8353762 B2 Rodgers et al. 20130100 cited by applicant, US 8425303 B2 Kennedy et al. 20130400 cited by applicant, US 8597103 B2 Bigelow et al. 20131200 cited by applicant, US 8628403 B2 Cole et al. 20140100 cited by applicant, US 8932125 B1 Harris 20150100 cited by applicant, US 10115273 B2 Uss et al. 20181000 cited by applicant, US 10297108 B2 De Waal et al. 20190500 cited by applicant, US 10540851 B2 Nicely et al. 20200100 cited by applicant, US 10643435 B1 Harris 20200500 cited by applicant, US 11302139 B2 Namaziyan 20220400 cited by applicant, US 2005/0059471 A1 Cannon 20050300 cited by applicant, US 2006/0009283 A1 Englman et al. 20060100 cited by applicant, US 2008/0113768 A1 Baerlocher 20080500 cited by applicant, US 2009/0111577 A1 Mead 20090400 cited by examiner G07F 17/3211 463/31, US 2009/0117993 A1 Bigelow et al. 20090500 cited by applicant, US 2010/0016065 A1 Nicely et al. 20100100 cited by

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abstractHtml: Various aspects of the present disclosure generally relate to wireless communication. In some aspects, a user equipment (UE) may receive a first physical broadcast channel (PBCH) communication on a channel. The UE may decode the first PBCH communication as a decoded PBCH payload. The UE may reencode the decoded PBCH payload as a first pilot signal. The UE may receive a first data communication with first data. The UE may correct the channel for the first data based on the first pilot signal. The UE may decode the first data. Numerous other aspects are described.

descriptionHtml: BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects. The same reference numbers in different drawings may identify the same or similar elements.

(2) FIG. 1 is a diagram illustrating an example of a wireless network, in accordance with the present disclosure.

(3) FIG. 2 is a diagram illustrating an example of a network node in communication with a user equipment (UE) in a wireless network, in accordance with the present disclosure.

(4) FIG. 3 is a diagram illustrating an example disaggregated base station architecture, in accordance with the present disclosure.

(5) FIG. 4 is a diagram illustrating an example of physical channels and reference signals in a wireless network, in accordance with the present disclosure.

(6) FIG. 5 is a diagram illustrating an example of training data for channel estimation, in accordance with the present disclosure.

(7) FIG. 6 is a diagram illustrating an example of encoding, in accordance with the present disclosure.

(8) FIG. 7 is a diagram illustrating an example of a polar encoding butterfly lattice, in accordance with the present disclosure.

(9) FIG. 8 is a diagram illustrating an example associated with reencoding a physical broadcast channel payload as a pilot signal, in accordance with the present disclosure.

(10) FIG. 9 is a diagram illustrating an example process performed, for example, by a UE, in accordance with the present disclosure.

(11) FIG. 10 is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.

DETAILED DESCRIPTION

(12) Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. One skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

(13) Several aspects of telecommunication systems will now be presented with reference to various apparatuses and techniques. These apparatuses and techniques will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes, algorithms, or the like (collectively referred to as “elements”). These elements may be implemented using hardware, software, or combinations thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

(14) While aspects may be described herein using terminology commonly associated with a 5G or New Radio (NR) radio access technology (RAT), aspects of the present disclosure can be applied to other RATs, such as a 3G RAT, a 4G RAT, and/or a RAT subsequent to 5G (e.g., 6G).

(15) FIG. 1 is a diagram illustrating an example of a wireless network 100, in accordance with the present disclosure.

The wireless network 100 may be or may include elements of a 5G (e.g., NR) network and/or a 4G (e.g., Long Term Evolution (LTE)) network, among other examples. The wireless network 100 may include one or more network nodes 110 (shown as a network node 110a, a network node 110b, a network node 110c, and a network node 110d), a user equipment (UE) 120 or multiple UEs 120 (shown as a UE 120a, a UE 120b, a UE 120c, a UE 120d, and a UE 120e), and/or other entities. A network node 110 is a network node that communicates with UEs 120. As shown, a network node 110 may include one or more network nodes. For example, a network node 110 may be an aggregated network node, meaning that the aggregated network node is configured to utilize a radio protocol stack that is physically or logically integrated within a single radio access network (RAN) node (e.g., within a single device or unit). As another example, a network node 110 may be a disaggregated network node (sometimes referred to as a disaggregated base station), meaning that the network node 110 is configured to utilize a protocol stack that is physically or logically distributed among two or more nodes (such as one or more central units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)).

(16) In some examples, a network node 110 is or includes a network node that communicates with UEs 120 via a radio access link, such as an RU. In some examples, a network node 110 is or includes a network node that communicates with other network nodes 110 via a fronthaul link or a midhaul link, such as a DU. In some examples, a network node 110 is or includes a network node that communicates with other network nodes 110 via a midhaul link or a core network via a backhaul link, such as a CU. In some examples, a network node 110 (such as an aggregated network node 110 or a disaggregated network node 110) may include multiple network nodes, such as one or more RUs, one or more CUs, and/or one or more DUs. A network node 110 may include, for example, an NR base station, an LTE base station, a Node B, an eNB (e.g., in 4G), a gNB (e.g., in 5G), an access point, a transmission reception point (TRP), a DU, an RU, a CU, a mobility element of a network, a core network node, a network element, a network equipment, a RAN node, or a combination thereof. In some examples, the network nodes 110 may be interconnected to one another or to one or more other network nodes 110 in the wireless network 100 through various types of fronthaul, midhaul, and/or backhaul interfaces, such as a direct physical connection, an air interface, or a virtual network, using any suitable transport network.

(17) In some examples, a network node 110 may provide communication coverage for a particular geographic area. In the Third Generation Partnership Project (3GPP), the term “cell” can refer to a coverage area of a network node 110 and/or a network node subsystem serving this coverage area, depending on the context in which the term is used. A network node 110 may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or another type of cell. A macro cell may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs 120 with service subscriptions. A pico cell may cover a relatively small geographic area and may allow unrestricted access by UEs 120 with service subscriptions. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs 120 having association with the femto cell (e.g., UEs 120 in a closed subscriber group (CSG)). A network node 110 for a macro cell may be referred to as a macro network node. A network node 110 for a pico cell may be referred to as a pico network node. A network node 110 for a femto cell may be referred to as a femto network node or an in-home network node. In the example shown in FIG. 1, the network node 110a may be a macro network node for a macro cell 102a, the network node 110b may be a pico network node for a pico cell 102b, and the network node 110c may be a femto network node for a femto cell 102c. A network node may support one or multiple (e.g., three) cells. In some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a network node 110 that is mobile (e.g., a mobile network node).

(18) In some aspects, the term “base station” or “network node” may refer to an aggregated base station, a disaggregated base station, an integrated access and backhaul (IAB) node, a relay node, or one or more components thereof. For example, in some aspects, “base station” or “network node” may refer to a CU, a DU, an RU, a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, or a combination thereof. In some

aspects, the term “base station” or “network node” may refer to one device configured to perform one or more functions, such as those described herein in connection with the network node 110. In some aspects, the term “base station” or “network node” may refer to a plurality of devices configured to perform the one or more functions. For example, in some distributed systems, each of a quantity of different devices (which may be located in the same geographic location or in different geographic locations) may be configured to perform at least a portion of a function, or to duplicate performance of at least a portion of the function, and the term “base station” or “network node” may refer to any one or more of those different devices. In some aspects, the term “base station,” “network entity,” or “network node” may refer to one or more virtual base stations or one or more virtual base station functions. For example, in some aspects, two or more base station functions may be instantiated on a single device. In some aspects, the term “base station,” “network entity,” or “network node” may refer to one of the base station functions and not another. In this way, a single device may include more than one base station.

(19) The wireless network 100 may include one or more relay stations. A relay station is a network node that can receive a transmission of data from an upstream node (e.g., a network node 110 or a UE 120) and send a transmission of the data to a downstream node (e.g., a UE 120 or a network node 110). A relay station may be a UE 120 that can relay transmissions for other UEs 120. In the example shown in FIG. 1, the network node 110d (e.g., a relay network node) may communicate with the network node 110a (e.g., a macro network node) and the UE 120d in order to facilitate communication between the network node 110a and the UE 120d. A network node 110 that relays communications may be referred to as a relay station, a relay base station, a relay network node, a relay node, a relay, or the like.

(20) The wireless network 100 may be a heterogeneous network that includes network nodes 110 of different types, such as macro network nodes, pico network nodes, femto network nodes, relay network nodes, or the like. These different types of network nodes 110 may have different transmit power levels, different coverage areas, and/or different impacts on interference in the wireless network 100. For example, macro network nodes may have a high transmit power level (e.g., 5 to 40 watts) whereas pico network nodes, femto network nodes, and relay network nodes may have lower transmit power levels (e.g., 0.1 to 2 watts).

(21) A network controller 130 may couple to or communicate with a set of network nodes 110 and may provide coordination and control for these network nodes 110. The network controller 130 may communicate with the network nodes 110 via a backhaul communication link or a midhaul communication link. The network nodes 110 may communicate with one another directly or indirectly via a wireless or wireline backhaul communication link. In some aspects, the network controller 130 may be a CU or a core network device, or may include a CU or a core network device.

(22) The UEs 120 may be dispersed throughout the wireless network 100, and each UE 120 may be stationary or mobile. A UE 120 may include, for example, an access terminal, a terminal, a mobile station, and/or a subscriber unit. A UE 120 may be a cellular phone (e.g., a smart phone), a personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a laptop computer, a cordless phone, a wireless local loop (WLL) station, a tablet, a camera, a gaming device, a netbook, a smartbook, an ultrabook, a medical device, a biometric device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, a smart wristband, smart jewelry (e.g., a smart ring or a smart bracelet)), an entertainment device (e.g., a music device, a video device, and/or a satellite radio), a vehicular component or sensor, a smart meter/sensor, industrial manufacturing equipment, a global positioning system device, a UE function of a network node, and/or any other suitable device that is configured to communicate via a wireless or wired medium.

(23) Some UEs 120 may be considered machine-type communication (MTC) or evolved or enhanced machine-type communication (eMTC) UEs. An MTC UE and/or an eMTC UE may include, for example, a robot, a drone, a remote device, a sensor, a meter, a monitor, and/or a location tag, that may communicate with a network node, another device (e.g., a remote device), or some other entity. Some UEs 120 may be considered Internet-of-Things (IoT) devices, and/or may be implemented as NB-IoT (narrowband IoT) devices. Some UEs 120 may be considered a Customer Premises

Equipment. A UE 120 may be included inside a housing that houses components of the UE 120, such as processor components and/or memory components. In some examples, the processor components and the memory components may be coupled together. For example, the processor components (e.g., one or more processors) and the memory components (e.g., a memory) may be operatively coupled, communicatively coupled, electronically coupled, and/or electrically coupled.

(24) In general, any number of wireless networks 100 may be deployed in a given geographic area. Each wireless network 100 may support a particular RAT and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic area in order to avoid interference between wireless networks of different RATs. In some cases, NR or 5G RAT networks may be deployed.

(25) In some examples, two or more UEs 120 (e.g., shown as UE 120a and UE 120e) may communicate directly using one or more sidelink channels (e.g., without using a network node 110 as an intermediary to communicate with one another). For example, the UEs 120 may communicate using peer-to-peer (P2P) communications, device-to-device (D2D) communications, a vehicle-to-everything (V2X) protocol (e.g., which may include a vehicle-to-vehicle (V2V) protocol, a vehicle-to-infrastructure (V2I) protocol, or a vehicle-to-pedestrian (V2P) protocol), and/or a mesh network. In such examples, a UE 120 may perform scheduling operations, resource selection operations, and/or other operations described elsewhere herein as being performed by the network node 110.

(26) Devices of the wireless network 100 may communicate using the electromagnetic spectrum, which may be subdivided by frequency or wavelength into various classes, bands, channels, or the like. For example, devices of the wireless network 100 may communicate using one or more operating bands. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

(27) The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

(28) With the above examples in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like, if used herein, may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like, if used herein, may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band. It is contemplated that the frequencies included in these operating bands (e.g., FR1, FR2, FR3, FR4, FR4-a, FR4-1, and/or FR5) may be modified, and techniques described herein are applicable to those modified frequency ranges.

(29) In some aspects, the UE 120 may include a communication manager 140. As described in more detail elsewhere herein, the communication manager 140 may receive a first physical broadcast channel (PBCH) communication on a channel. The communication manager 140 may decode the first PBCH communication as a decoded PBCH payload. The communication manager 140 may reencode the decoded PBCH payload as a first pilot signal. The communication manager 140 may receive a first data communication with first data. The communication manager 140 may correct the

channel for the first data based on the first pilot signal. The communication manager 140 may decode the first data. Additionally, or alternatively, the communication manager 140 may perform one or more other operations described herein.

(30) As indicated above, FIG. 1 is provided as an example. Other examples may differ from what is described with regard to FIG. 1.

(31) FIG. 2 is a diagram illustrating an example 200 of a network node 110 in communication with a UE 120 in a wireless network 100, in accordance with the present disclosure. The network node 110 may be equipped with a set of antennas 234a through 234t, such as T antennas ($T \geq 1$). The UE 120 may be equipped with a set of antennas 252a through 252r, such as R antennas ($R \geq 1$). The network node 110 of example 200 includes one or more radio frequency components, such as antennas 234 and a modem 254. In some examples, a network node 110 may include an interface, a communication component, or another component that facilitates communication with the UE 120 or another network node. Some network nodes 110 may not include radio frequency components that facilitate direct communication with the UE 120, such as one or more CUs, or one or more DUs.

(32) At the network node 110, a transmit processor 220 may receive data, from a data source 212, intended for the UE 120 (or a set of UEs 120). The transmit processor 220 may select one or more modulation and coding schemes (MCSs) for the UE 120 based at least in part on one or more channel quality indicators (CQIs) received from that UE 120. The network node 110 may process (e.g., encode and modulate) the data for the UE 120 based at least in part on the MCS(s) selected for the UE 120 and may provide data symbols for the UE 120. The transmit processor 220 may process system information (e.g., for semi-static resource partitioning information (SRPI)) and control information (e.g., CQI requests, grants, and/or upper layer signaling) and provide overhead symbols and control symbols. The transmit processor 220 may generate reference symbols for reference signals (e.g., a cell-specific reference signal (CRS) or a demodulation reference signal (DMRS)) and synchronization signals (e.g., a primary synchronization signal (PSS) or a secondary synchronization signal (SSS)). A transmit (TX) multiple-input multiple-output (MIMO) processor 230 may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, the overhead symbols, and/or the reference symbols, if applicable, and may provide a set of output symbol streams (e.g., T output symbol streams) to a corresponding set of modems 232 (e.g., T modems), shown as modems 232a through 232t. For example, each output symbol stream may be provided to a modulator component (shown as MOD) of a modem 232. Each modem 232 may use a respective modulator component to process a respective output symbol stream (e.g., for OFDM) to obtain an output sample stream. Each modem 232 may further use a respective modulator component to process (e.g., convert to analog, amplify, filter, and/or upconvert) the output sample stream to obtain a downlink signal. The modems 232a through 232t may transmit a set of downlink signals (e.g., T downlink signals) via a corresponding set of antennas 234 (e.g., T antennas), shown as antennas 234a through 234t.

(33) At the UE 120, a set of antennas 252 (shown as antennas 252a through 252r) may receive the downlink signals from the network node 110 and/or other network nodes 110 and may provide a set of received signals (e.g., R received signals) to a set of modems 254 (e.g., R modems), shown as modems 254a through 254r. For example, each received signal may be provided to a demodulator component (shown as DEMOD) of a modem 254. Each modem 254 may use a respective demodulator component to condition (e.g., filter, amplify, downconvert, and/or digitize) a received signal to obtain input samples. Each modem 254 may use a demodulator component to further process the input samples (e.g., for OFDM) to obtain received symbols. A MIMO detector 256 may obtain received symbols from the modems 254, may perform MIMO detection on the received symbols if applicable, and may provide detected symbols. A receive processor 258 may process (e.g., demodulate and decode) the detected symbols, may provide decoded data for the UE 120 to a data sink 260, and may provide decoded control information and system information to a controller/processor 280. The term “controller/processor” may refer to one or more controllers, one or more processors, or a combination thereof. A channel processor may determine a reference signal received power (RSRP) parameter, a received signal strength indicator (RSSI) parameter, a reference signal received quality (RSRQ) parameter, and/or a CQI parameter, among

other examples. In some examples, one or more components of the UE 120 may be included in a housing 284.

(34) The network controller 130 may include a communication unit 294, a controller/processor 290, and a memory 292. The network controller 130 may include, for example, one or more devices in a core network. The network controller 130 may communicate with the network node 110 via the communication unit 294.

(35) One or more antennas (e.g., antennas 234a through 234t and/or antennas 252a through 252r) may include, or may be included within, one or more antenna panels, one or more antenna groups, one or more sets of antenna elements, and/or one or more antenna arrays, among other examples. An antenna panel, an antenna group, a set of antenna elements, and/or an antenna array may include one or more antenna elements (within a single housing or multiple housings), a set of coplanar antenna elements, a set of non-coplanar antenna elements, and/or one or more antenna elements coupled to one or more transmission and/or reception components, such as one or more components of FIG. 2.

(36) On the uplink, at the UE 120, a transmit processor 264 may receive and process data from a data source 262 and control information (e.g., for reports that include RSRP, RSSI, RSRQ, and/or CQI) from the controller/processor 280. The transmit processor 264 may generate reference symbols for one or more reference signals. The symbols from the transmit processor 264 may be precoded by a TX MIMO processor 266 if applicable, further processed by the modems 254 (e.g., for DFT-s-OFDM or CP-OFDM), and transmitted to the network node 110. In some examples, the modem 254 of the UE 120 may include a modulator and a demodulator. In some examples, the UE 120 includes a transceiver. The transceiver may include any combination of the antenna(s) 252, the modem(s) 254, the MIMO detector 256, the receive processor 258, the transmit processor 264, and/or the TX MIMO processor 266. The transceiver may be used by a processor (e.g., the controller/processor 280) and the memory 282 to perform aspects of any of the methods described herein (e.g., with reference to FIGS. 4-10).

(37) At the network node 110, the uplink signals from UE 120 and/or other UEs may be received by the antennas 234, processed by the modem 232 (e.g., a demodulator component, shown as DEMOD, of the modem 232), detected by a MIMO detector 236 if applicable, and further processed by a receive processor 238 to obtain decoded data and control information sent by the UE 120. The receive processor 238 may provide the decoded data to a data sink 239 and provide the decoded control information to the controller/processor 240. The network node 110 may include a communication unit 244 and may communicate with the network controller 130 via the communication unit 244. The network node 110 may include a scheduler 246 to schedule one or more UEs 120 for downlink and/or uplink communications. In some examples, the modem 232 of the network node 110 may include a modulator and a demodulator. In some examples, the network node 110 includes a transceiver. The transceiver may include any combination of the antenna(s) 234, the modem(s) 232, the MIMO detector 236, the receive processor 238, the transmit processor 220, and/or the TX MIMO processor 230. The transceiver may be used by a processor (e.g., the controller/processor 240) and the memory 242 to perform aspects of any of the methods described herein (e.g., with reference to FIGS. 4-10).

(38) The controller/processor 240 of the network node 110, the controller/processor 280 of the UE 120, and/or any other component(s) of FIG. 2 may perform one or more techniques associated with using a PBCH for channel correction, as described in more detail elsewhere herein. For example, the controller/processor 240 of the network node 110, the controller/processor 280 of the UE 120, and/or any other component(s) of FIG. 2 may perform or direct operations of, for example, process 900 of FIG. 9, and/or other processes as described herein. The memory 242 and the memory 282 may store data and program codes for the network node 110 and the UE 120, respectively. In some examples, the memory 242 and/or the memory 282 may include a non-transitory computer-readable medium storing one or more instructions (e.g., code and/or program code) for wireless communication. For example, the one or more instructions, when executed (e.g., directly, or after compiling, converting, and/or interpreting) by one or more processors of the network node 110 and/or the UE 120, may cause the one or more processors, the UE 120, and/or the network node 110 to perform or direct operations of, for example, process 900 of FIG. 9, and/or other processes as described herein. In

some examples, executing instructions may include running the instructions, converting the instructions, compiling the instructions, and/or interpreting the instructions, among other examples.

(39) In some aspects, the UE 120 includes means for receiving a first PBCH communication on a channel; means for decoding the first PBCH communication as a decoded PBCH payload; means for reencoding the decoded PBCH payload as a first pilot signal; means for receiving a first data communication with first data; means for correcting the channel for the first data based on the first pilot signal; and/or means for decoding the first data. The means for the UE 120 to perform operations described herein may include, for example, one or more of communication manager 140, antenna 252, modem 254, MIMO detector 256, receive processor 258, transmit processor 264, TX MIMO processor 266, controller/processor 280, or memory 282.

(40) While blocks in FIG. 2 are illustrated as distinct components, the functions described above with respect to the blocks may be implemented in a single hardware, software, or combination component or in various combinations of components. For example, the functions described with respect to the transmit processor 264, the receive processor 258, and/or the TX MIMO processor 266 may be performed by or under the control of the controller/processor 280.

(41) As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

(42) Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a RAN node, a core network node, a network element, a base station, or a network equipment may be implemented in an aggregated or disaggregated architecture. For example, a base station (such as a Node B (NB), an evolved NB (eNB), an NR BS, a 5G NB, an access point (AP), a TRP, or a cell, among other examples), or one or more units (or one or more components) performing base station functionality, may be implemented as an aggregated base station (also known as a standalone base station or a monolithic base station) or a disaggregated base station. "Network entity" or "network node" may refer to a disaggregated base station, or to one or more units of a disaggregated base station (such as one or more CUs, one or more DUs, one or more RUs, or a combination thereof).

(43) An aggregated base station (e.g., an aggregated network node) may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node (e.g., within a single device or unit). A disaggregated base station (e.g., a disaggregated network node) may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more CUs, one or more DUs, or one or more RUs). In some examples, a CU may be implemented within a network node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other network nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, such as a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU), among other examples.

(44) Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an IAB network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)) to facilitate scaling of communication systems by separating base station functionality into one or more units that can be individually deployed. A disaggregated base station may include functionality implemented across two or more units at various physical locations, as well as functionality implemented for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station can be configured for wired or wireless communication with at least one other unit of the disaggregated base station.

(45) FIG. 3 is a diagram illustrating an example disaggregated base station architecture 300, in accordance with the present disclosure. The disaggregated base station architecture 300 may include a CU 310 that can communicate directly with a core network 320 via a backhaul link, or indirectly with the core network 320 through one or more

disaggregated control units (such as a Near-RT RIC 325 via an E2 link, or a Non-RT RIC 315 associated with a Service Management and Orchestration (SMO) Framework 305, or both). A CU 310 may communicate with one or more DUs 330 via respective midhaul links, such as through F1 interfaces. Each of the DUs 330 may communicate with one or more RUs 340 via respective fronthaul links. Each of the RUs 340 may communicate with one or more UEs 120 via respective radio frequency (RF) access links. In some implementations, a UE 120 may be simultaneously served by multiple RUs 340.

(46) Each of the units, including the CUs 310, the DUs 330, the RUs 340, as well as the Near-RT RICs 325, the Non-RT RICs 315, and the SMO Framework 305, may include one or more interfaces or be coupled with one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to one or multiple communication interfaces of the respective unit, can be configured to communicate with one or more of the other units via the transmission medium. In some examples, each of the units can include a wired interface, configured to receive or transmit signals over a wired transmission medium to one or more of the other units, and a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

(47) In some aspects, the CU 310 may host one or more higher layer control functions. Such control functions can include radio resource control (RRC) functions, packet data convergence protocol (PDCP) functions, or service data adaptation protocol (SDAP) functions, among other examples. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 310. The CU 310 may be configured to handle user plane functionality (for example, Central Unit-User Plane (CU-UP) functionality), control plane functionality (for example, Central Unit-Control Plane (CU-CP) functionality), or a combination thereof. In some implementations, the CU 310 can be logically split into one or more CU-UP units and one or more CU-CP units. A CU-UP unit can communicate bidirectionally with a CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU 310 can be implemented to communicate with a DU 330, as necessary, for network control and signaling.

(48) Each DU 330 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 340. In some aspects, the DU 330 may host one or more of a radio link control (RLC) layer, a MAC layer, and one or more high physical (PHY) layers depending, at least in part, on a functional split, such as a functional split defined by the 3GPP. In some aspects, the one or more high PHY layers may be implemented by one or more modules for forward error correction (FEC) encoding and decoding, scrambling, and modulation and demodulation, among other examples. In some aspects, the DU 330 may further host one or more low PHY layers, such as implemented by one or more modules for a fast Fourier transform (FFT), an inverse FFT (iFFT), digital beamforming, or physical random access channel (PRACH) extraction and filtering, among other examples. Each layer (which also may be referred to as a module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 330, or with the control functions hosted by the CU 310.

(49) Each RU 340 may implement lower-layer functionality. In some deployments, an RU 340, controlled by a DU 330, may correspond to a logical node that hosts RF processing functions or low-PHY layer functions, such as performing an FFT, performing an iFFT, digital beamforming, or PRACH extraction and filtering, among other examples, based on a functional split (for example, a functional split defined by the 3GPP), such as a lower layer functional split. In such an architecture, each RU 340 can be operated to handle over the air (OTA) communication with one or more UEs 120. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) 340 can be controlled by the corresponding DU 330. In some scenarios, this configuration can enable each DU 330 and the CU 310 to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

(50) The SMO Framework 305 may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework 305 may be configured to

support the deployment of dedicated physical resources for RAN coverage requirements, which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework 305 may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) platform 390) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs 310, DUs 330, RUs 340, non-RT RICs 315, and Near-RT RICs 325. In some implementations, the SMO Framework 305 can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) 311, via an O1 interface. Additionally, in some implementations, the SMO Framework 305 can communicate directly with each of one or more RUs 340 via a respective O1 interface. The SMO Framework 305 also may include a Non-RT RIC 315 configured to support functionality of the SMO Framework 305.

(51) The Non-RT RIC 315 may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC 325. The Non-RT RIC 315 may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC 325. The Near-RT RIC 325 may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs 310, one or more DUs 330, or both, as well as an O-eNB, with the Near-RT RIC 325.

(52) In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC 325, the Non-RT MC 315 may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT MC 325 and may be received at the SMO Framework 305 or the Non-RT RIC 315 from non-network data sources or from network functions. In some examples, the Non-RT RIC 315 or the Near-RT RIC 325 may be configured to tune RAN behavior or performance. For example, the Non-RT RIC 315 may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework 305 (such as reconfiguration via an O1 interface) or via creation of RAN management policies (such as A1 interface policies).

(53) As indicated above, FIG. 3 is provided as an example. Other examples may differ from what is described with regard to FIG. 3.

(54) FIG. 4 is a diagram illustrating an example 400 of physical channels and reference signals in a wireless network, in accordance with the present disclosure. As shown in FIG. 4, downlink channels and downlink reference signals may carry information from a network node 110 to a UE 120, and uplink channels and uplink reference signals may carry information from a UE 120 to a network node 110.

(55) As shown, a downlink channel may include a physical downlink control channel (PDCCH) that carries downlink control information (DCI), a physical downlink shared channel (PDSCH) that carries downlink data, or a PBCH that carries system information, among other examples. In some aspects, PDSCH communications may be scheduled by PDCCH communications. As further shown, an uplink channel may include a physical uplink control channel (PUCCH) that carries uplink control information (UCI), a physical uplink shared channel (PUSCH) that carries uplink data, or a physical random access channel (PRACH) used for initial network access, among other examples. In some aspects, the UE 120 may transmit acknowledgement (ACK) or negative acknowledgement (NACK) feedback (e.g., ACK/NACK feedback or ACK/NACK information) in UCI on the PUCCH and/or the PUSCH.

(56) As further shown, a downlink reference signal may include a synchronization signal block (SSB), a channel state information (CSI) reference signal (CSI-RS), a DMRS, a positioning reference signal (PRS), or a phase tracking reference signal (PTRS), among other examples. As also shown, an uplink reference signal may include a sounding reference signal (SRS), a DMRS, or a PTRS, among other examples.

(57) An SSB may carry information used for initial network acquisition and synchronization, such as a PSS, an SSS, a PBCH, and a PBCH DMRS. An SSB is sometimes referred to as a synchronization signal/PBCH (SS/PBCH) block. In some aspects, the network node 110 may transmit multiple SSBs on multiple corresponding beams, and the SSBs may

be used for beam selection.

(58) A CSI-RS may carry information used for downlink channel estimation (e.g., downlink CSI acquisition), which may be used for scheduling, link adaptation, or beam management, among other examples. The network node 110 may configure a set of CSI-RSs for the UE 120, and the UE 120 may measure the configured set of CSI-RSs. Based at least in part on the measurements, the UE 120 may perform channel estimation and may report channel estimation parameters to the network node 110 (e.g., in a CSI report), such as a channel quality indicator (CQI), a precoding matrix indicator (PMI), a CSI-RS resource indicator (CRI), a layer indicator (LI), a rank indicator (RI), or a reference signal received power (RSRP), among other examples. The network node 110 may use the CSI report to select transmission parameters for downlink communications to the UE 120, such as a number of transmission layers (e.g., a rank), a precoding matrix (e.g., a precoder), a modulation and coding scheme (MCS), or a refined downlink beam (e.g., using a beam refinement procedure or a beam management procedure), among other examples.

(59) A DMRS may carry information used to estimate a radio channel for demodulation of an associated physical channel (e.g., PDCCH, PDSCH, PBCH, PUCCH, or PUSCH). The design and mapping of a DMRS may be specific to a physical channel for which the DMRS is used for estimation. DMRSs are UE-specific, can be beamformed, can be confined in a scheduled resource (e.g., rather than transmitted on a wideband), and can be transmitted only when necessary. As shown, DMRSs are used for both downlink communications and uplink communications.

(60) A PTRS may carry information used to compensate for oscillator phase noise. Typically, the phase noise increases as the oscillator carrier frequency increases. Thus, PTRS can be utilized at high carrier frequencies, such as millimeter wave frequencies, to mitigate phase noise. The PTRS may be used to track the phase of the local oscillator and to enable suppression of phase noise and common phase error (CPE). As shown, PTRSs are used for both downlink communications (e.g., on the PDSCH) and uplink communications (e.g., on the PUSCH).

(61) A PRS may carry information used to enable timing or ranging measurements of the UE 120 based on signals transmitted by the network node 110 to improve observed time difference of arrival (OTDOA) positioning performance. For example, a PRS may be a pseudo-random Quadrature Phase Shift Keying (QPSK) sequence mapped in diagonal patterns with shifts in frequency and time to avoid collision with cell-specific reference signals and control channels (e.g., a PDCCH). In general, a PRS may be designed to improve detectability by the UE 120, which may need to detect downlink signals from multiple neighboring network nodes in order to perform OTDOA-based positioning. Accordingly, the UE 120 may receive a PRS from multiple cells (e.g., a reference cell and one or more neighbor cells), and may report a reference signal time difference (RSTD) based on OTDOA measurements associated with the PRSs received from the multiple cells. In some aspects, the network node 110 may then calculate a position of the UE 120 based on the RSTD measurements reported by the UE 120.

(62) An SRS may carry information used for uplink channel estimation, which may be used for scheduling, link adaptation, precoder selection, or beam management, among other examples. The network node 110 may configure one or more SRS resource sets for the UE 120, and the UE 120 may transmit SRSs on the configured SRS resource sets. An SRS resource set may have a configured usage, such as uplink CSI acquisition, downlink CSI acquisition for reciprocity-based operations, uplink beam management, among other examples. The network node 110 may measure the SRSs, may perform channel estimation based at least in part on the measurements, and may use the SRS measurements to configure communications with the UE 120.

(63) Previous network designs (e.g., 3G, LTE) have used frequent common or pilot signals to help with channel correction. Channel correction involves estimating a channel, or changes to the channel, for received communications and correcting the channel based on the channel estimate. Channel correction helps communications to be decoded properly in case there is a shift in time or a frequency phase while traveling on the channel. In an example, LTE uses a CRS or a common reference signal as a pilot signal. The CRS is a frequent (e.g., every 1 millisecond (ms)), full-bandwidth signal that consumes a significant amount of power. 5G NR networks have replaced the CRS with a less frequent tracking reference signal (TRS). The TRS may be every 20-80 ms. However, the time period between TRSs

may be too long such that some communications are affected by changes in time or frequency on the channel. There is also less training data for estimating tracking loops and for channel estimation. Without appropriate channel correction, communications may degrade, which increases latency and wastes power, processing resources, and signaling resources.

(64) As indicated above, FIG. 4 is provided as an example. Other examples may differ from what is described with regard to FIG. 4.

(65) FIG. 5 is a diagram illustrating an example 500 of training data for channel estimation, in accordance with the present disclosure.

(66) According to various aspects described herein, a UE may decode a PBCH communication to obtain a decoded PBCH payload. The UE may then reencode the decoded PBCH payload to serve as a pilot signal. The pilot signal may be used for channel estimation and correction. Because PBCH communications arrive every 5-20 ms (as compared to 20-80 ms for a TRS), pilot signals from PBCH communications may provide more frequent opportunities for channel estimation. The PBCH may also provide more training data for channel estimation. For example, instead of using only the DMRS in the PBCH, additional PBCH data may be added to the training data. Example 500 shows resource elements (REs) in one symbol and 12 tones for an SSS signal currently used for loop or channel estimation. However, example 500 also shows a greater quantity of REs for a PBCH that may be used as additional training data for channel estimation. More training data (e.g., multiple of 4 or 5) results in a more accurate channel estimation and a greater processing gain (e.g., 6 dB).

(67) As indicated above, FIG. 5 is provided as an example. Other examples may differ from what is described with regard to FIG. 5.

(68) FIG. 6 is a diagram illustrating an example 600 of encoding, in accordance with the present disclosure.

(69) Example 600 shows an existing process of encoding as described in 3GPP Technical Specification (TS) 38.212. Such a process may be used to encode a PBCH payload to obtain a PBCH communication or to encode data to obtain a data communication. Once a payload is generated, the payload may be scrambled and a cyclic redundancy check (CRC) may be attached for reliability checking. The process may continue with payload interleaving to rearrange the data according to a defined pattern, before bit mapping. The process may continue with polar encoding, which transforms the data through reproduction and combination into a larger amount of data for redundancy and reliability. The process further continues with subblock interleaving and bit selection.

(70) Another process may be used to decode the PBCH communication to obtain a PBCH payload. This may involve a decoding process that reverses the encoding process.

(71) As indicated above, FIG. 6 is provided as an example. Other examples may differ from what is described with regard to FIG. 6.

(72) FIG. 7 is a diagram illustrating an example 700 of a polar encoding butterfly lattice, in accordance with the present disclosure.

(73) Polar encoding may use a defined butterfly lattice, which is a defined pattern for increasing the redundancy and reliability of data. In example 700, a letter represents a portion of data. Portions of data may be duplicated and/or combined with other portions of data for redundancy (e.g., using XOR or deal operations). For example, in a first stage (first column), data portion "a" may be combined with data portion "b." Combined data portion "ab" may be combined with combined data portion "cd." Combined data portion "abcd" may be combined with combined data portion "efgh" to form combined data portion "abcdefgh." In some aspects, the UE may encode a decoded PBCH payload based at least in part on per-stage instructions for a butterfly lattice that is used for polar encoding.

(74) As indicated above, FIG. 7 is provided as an example. Other examples may differ from what is described with regard to FIG. 7.

(75) FIG. 8 is a diagram illustrating an example 800 associated with reencoding a PBCH payload as a pilot signal, in accordance with the present disclosure. As shown in FIG. 8, a network entity 810 (e.g., network node 110) and a UE

820 (e.g., UE 120) may communicate with one another via a wireless network (e.g., wireless network 100).

(76) As shown by reference number 825, the network entity 810 may transmit a first PBCH communication on a channel. As shown by reference number 830, the UE 820 may decode the first PBCH communication to obtain a decoded PBCH payload. As shown by reference number 835, the UE 820 may reencode the decoded PBCH payload to obtain a first pilot signal. The reencoding may use an encoding process, such as the encoding process described in connection with FIGS. 6 and 7. In some aspects, reencoding may occur if the first PBCH communication is successfully decoded. If the decoding of the first PBCH communication is not successful, the UE 820 may not reencode the decoded PBCH payload.

(77) As shown by reference number 840, the network entity 810 may transmit a first data communication with first data on the channel. As shown by reference number 845, the UE 820 may correct the channel for the first data based on the first pilot signal. Correcting the channel may involve obtaining a channel estimate based on comparing a received antenna point signal of the first PBCH communication and the first pilot signal (an expected antenna point signal), as shown by reference number 850. The received antenna point signal may be a signal as received at the antenna, before decoding. The expected antenna point signal may be the signal at the antenna after reencoding. The comparing may result in a difference between the received antenna point signal and the expected antenna point signal. The difference may be in time (e.g., 1 or 2 symbols) and/or frequency (e.g., phase shift). As shown by reference number 855, the UE 820 may correct the channel for the first data based on this difference, which may include a channel estimate that is enhanced with respect to previous channel estimation techniques. For example, the difference in time or frequency (or channel estimate) may be added to or subtracted from the received antenna point signal of the first data communication before decoding the first data communication, as shown by reference number 860, to obtain decoded first data.

(78) In some aspects, the UE 820 may perform other actions with the channel estimate that is associated with reencoding the decoded PBCH payload. For example, the UE 820 may perform beam selection, RSRP estimation, delay bin estimation, Doppler estimation, and/or time to live (TTL) estimation.

(79) In some aspects, the same decoded PBCH payload from the first PBCH communication may be used for subsequent PBCH communication reencoding. For example, as shown by reference number 865, the network entity 810 may transmit a second PBCH communication. Rather than consuming time and processing resources decoding the second PBCH communication, the UE 820 may reencode the decoded PBCH payload (from the first PBCH communication) to obtain a second pilot signal, as shown by reference number 870, to be used for channel correction of subsequent data communications. Note that the reencoding of the decoded PBCH payload for the second pilot signal may involve a change in frame number or scrambling from when the first PBCH communication. The antenna point signal of the first pilot signal and the antenna point signal of the second pilot signal are expected to be different. The UE 820 may receive a second data communication and correct the channel for the second data communication based on the second pilot signal, using a similar process as shown by reference numbers 850 and 855.

(80) The PBCH payload is not expected to change frequently. However, the PBCH payload may change occasionally (e.g., every 100 PBCH communications). In some aspects, the second PBCH communication may be decoded to obtain a new decoded PBCH payload if there is a sufficient change in the PBCH communication signal. For example, the UE 820 may decode the second PBCH communication to obtain a new decoded PBCH payload based on a difference between a DMRS signal-to-noise ratio (SNR) and a PBCH SNR. For example, if the difference (e.g., in dBm) satisfies a threshold (e.g., greater than 5 dB), the UE 820 may reencode the decoded PBCH payload. If the difference does not satisfy the threshold (e.g., less than 5 dB), the UE 820 may not reencode the decoded PBCH payload.

(81) A PBCH payload may include two parts, dynamic bits (e.g., system frame number (SFN), half frame number, SSB identifier (ID)) and semi-static bits (rest of the bits). The dynamic bits can change from one SSB occasion to another. However, the UE 820 can predict these bits at any desired SSB occasion. Any other change in these bits can occur at a known modification boundary and may be indicated by the network entity 810 through a short message. There is not a reason for frequent PBCH decoding. That is, the UE 820 may be able to determine if a change in the PBCH communication is expected to result in decoding a new PBCH payload.

(82) By reencoding a PBCH payload as a pilot signal, channel estimation may be improved without the need for a power consuming, frequent, wide-band pilot signal. As a result, the network entity 810 and the UE 820 conserve power and signaling resources.

(83) In some aspects, the UE 820 may use other communications to increase a frequency of a pilot signal or to add more training data for channel estimation. For example, the UE 820 may decode a data communication to obtain decoded first data and the UE 820 may reencode the data communication as a pilot signal for channel correction of subsequent data communications. This may involve data that is more consistent for use as pilot signals. In some aspects, the UE 820 may decode control information or control channel communications and use decoded control information as a pilot signal for channel correction of subsequent communications. This may involve control information that is more consistent for use as pilot signals.

(84) As indicated above, FIG. 8 is provided as an example. Other examples may differ from what is described with respect to FIG. 8.

(85) FIG. 9 is a diagram illustrating an example process 900 performed, for example, by a UE, in accordance with the present disclosure. Example process 900 is an example where the UE (e.g., UE 120, UE 820) performs operations associated with using a PBCH for channel correction.

(86) As shown in FIG. 9, in some aspects, process 900 may include receiving a first PBCH communication on a channel (block 910). For example, the UE (e.g., using communication manager 1008 and/or reception component 1002 depicted in FIG. 10) may receive a first PBCH communication on a channel, as described above.

(87) As further shown in FIG. 9, in some aspects, process 900 may include decoding the first PBCH communication as a decoded PBCH payload (block 920). For example, the UE (e.g., using communication manager 1008 and/or decoding component 1010 depicted in FIG. 10) may decode the first PBCH communication as a decoded PBCH payload, as described above.

(88) As further shown in FIG. 9, in some aspects, process 900 may include reencoding the decoded PBCH payload as a first pilot signal (block 930). For example, the UE (e.g., using communication manager 1008 and/or encoding component 1012 depicted in FIG. 10) may reencode the decoded PBCH payload as a first pilot signal, as described above.

(89) As further shown in FIG. 9, in some aspects, process 900 may include receiving a first data communication with first data (block 940). For example, the UE (e.g., using communication manager 1008 and/or reception component 1002 depicted in FIG. 10) may receive a first data communication with first data, as described above.

(90) As further shown in FIG. 9, in some aspects, process 900 may include correcting the channel for the first data based on the first pilot signal (block 950). For example, the UE (e.g., using communication manager 140 and/or channel correction component 1014 depicted in FIG. 10) may correct the channel for the first data based on the first pilot signal, as described above.

(91) As further shown in FIG. 9, in some aspects, process 900 may include decoding the first data (block 960). For example, the UE (e.g., using communication manager 1008 and/or decoding component 1010 depicted in FIG. 10) may decode the first data, as described above.

(92) Process 900 may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

(93) In a first aspect, the first pilot signal is an expected antenna point signal, and correcting the channel for the first data includes comparing the expected antenna point signal and a received antenna point signal of the first PBCH communication, obtaining an estimate of the channel based on a result of the comparing, and correcting the channel for the first data based on the estimate of the channel.

(94) In a second aspect, alone or in combination with the first aspect, process 900 includes performing beam selection based at least in part on the estimate of the channel.

(95) In a third aspect, alone or in combination with one or more of the first and second aspects, the reencoding the decoded PBCH payload includes reencoding the decoded PBCH payload based on the decoding of the first PBCH

communication being successful.

(96) In a fourth aspect, alone or in combination with one or more of the first through third aspects, process 900 includes receiving a second PBCH communication on the channel, reencoding the decoded PBCH payload as a second pilot signal, receiving a second data communication with second data on the channel, correcting the channel for the second data based on the second pilot signal, and decoding the second data.

(97) In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, the second pilot signal is an expected antenna point signal, and correcting the channel for the second data includes comparing the expected antenna point signal and a received antenna point signal of the second PBCH communication, obtaining an estimate of the channel based on a result of the comparing, and correcting the channel for the second data based on the estimate of the channel.

(98) In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, process 900 includes decoding a second PBCH communication to obtain a new decoded PBCH payload based on a difference between a DMRS SNR of the second PBCH communication and a PBCH SNR of the second PBCH communication satisfying a difference threshold.

(99) In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, the reencoding the decoded PBCH payload includes polar encoding the decoded PBCH payload based at least in part on per-stage instructions for a butterfly lattice used for the polar encoding.

(100) In an eighth aspect, alone or in combination with one or more of the first through seventh aspects, process 900 includes receiving a second data communication with second data on the channel, correcting the channel for the second data based on the first pilot signal, and decoding the second data.

(101) In a ninth aspect, alone or in combination with one or more of the first through eighth aspects, the decoding the first data includes decoding the first data to obtain decoded first data, and process 900 includes reencoding the decoded first data as a second pilot signal, receiving a second data communication with second data on the channel, correcting the channel for the second data based on the second pilot signal, and decoding the second data.

(102) In a tenth aspect, alone or in combination with one or more of the first through ninth aspects, process 900 includes receiving a first control communication with control information on the channel, decoding the control information to obtain decoded control information, reencoding the decoded control information as a second pilot signal, receiving a second data communication with second data on the channel, correcting the channel for the second data based on the second pilot signal, and decoding the second data.

(103) Although FIG. 9 shows example blocks of process 900, in some aspects, process 900 may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 9. Additionally, or alternatively, two or more of the blocks of process 900 may be performed in parallel.

(104) FIG. 10 is a diagram of an example apparatus 1000 for wireless communication, in accordance with the present disclosure. The apparatus 1000 may be a UE (e.g., UE 120, UE 820), or a UE may include the apparatus 1000. In some aspects, the apparatus 1000 includes a reception component 1002 and a transmission component 1004, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus 1000 may communicate with another apparatus 1006 (such as a UE, a base station, or another wireless communication device) using the reception component 1002 and the transmission component 1004. As further shown, the apparatus 1000 may include the communication manager 1008. The communication manager 1008 may control and/or otherwise manage one or more operations of the reception component 1002 and/or the transmission component 1004. In some aspects, the communication manager 1008 may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. The communication manager 1008 may be, or be similar to, the communication manager 140 depicted in FIGS. 1 and 2. For example, in some aspects, the communication manager 1008 may be configured to perform one or more of the functions described as being performed by the communication manager 140. In some aspects, the communication

manager 1008 may include the reception component 1002 and/or the transmission component 1004. The communication manager 1008 may include one or more of a decoding component 1010, an encoding component 1012, and/or a channel correction component 1014, among other examples.

(105) In some aspects, the apparatus 1000 may be configured to perform one or more operations described herein in connection with FIGS. 1-8. Additionally, or alternatively, the apparatus 1000 may be configured to perform one or more processes described herein, such as process 900 of FIG. 9. In some aspects, the apparatus 1000 and/or one or more components shown in FIG. 10 may include one or more components of the UE described in connection with FIG. 2. Additionally, or alternatively, one or more components shown in FIG. 10 may be implemented within one or more components described in connection with FIG. 2. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

(106) The reception component 1002 may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus 1006. The reception component 1002 may provide received communications to one or more other components of the apparatus 1000. In some aspects, the reception component 1002 may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus 1000. In some aspects, the reception component 1002 may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2.

(107) The transmission component 1004 may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus 1006. In some aspects, one or more other components of the apparatus 1000 may generate communications and may provide the generated communications to the transmission component 1004 for transmission to the apparatus 1006. In some aspects, the transmission component 1004 may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus 1006. In some aspects, the transmission component 1004 may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. 2. In some aspects, the transmission component 1004 may be co-located with the reception component 1002 in a transceiver.

(108) The reception component 1002 may receive a first PBCH communication on a channel. The decoding component 1010 may decode the first PBCH communication as a decoded PBCH payload. The encoding component 1012 may reencode the decoded PBCH payload as a first pilot signal. The reception component 1002 may receive a first data communication with first data. The channel correction component 1014 may correct the channel for the first data based on the first pilot signal. The decoding component 1010 may decode the first data.

(109) The channel correction component 1014 may perform beam selection or RSRP estimation based at least in part on the estimate of the channel. The reception component 1002 may receive a second PBCH communication on the channel. The encoding component 1012 may reencode the decoded PBCH payload as a second pilot signal.

(110) The reception component 1002 may receive a second data communication with second data on the channel. The channel correction component 1014 may correct the channel for the second data based on the second pilot signal. The decoding component 1010 may decode the second data.

(111) The decoding component 1010 may decode a second PBCH communication to obtain a new decoded PBCH payload based on a difference between a DMRS SNR of the second PBCH communication and a PBCH SNR of the second PBCH communication satisfying a difference threshold.

(112) The reception component 1002 may receive a second data communication with second data on the channel. The channel correction component 1014 may correct the channel for the second data based on the first pilot signal. The decoding component 1010 may decode the second data.

(113) The reception component 1002 may receive a first control communication with control information on the channel. The decoding component 1010 may decode the control information to obtain decoded control information. The reception component 1002 may receive a second data communication with second data on the channel. The channel correction component 1014 may correct the channel for the second data based on the second pilot signal. The decoding component 1010 may decode the second data.

(114) The number and arrangement of components shown in FIG. 10 are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. 10. Furthermore, two or more components shown in FIG. 10 may be implemented within a single component, or a single component shown in FIG. 10 may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. 10 may perform one or more functions described as being performed by another set of components shown in FIG. 10.

(115) The following provides an overview of some Aspects of the present disclosure: Aspect 1: A method of wireless communication performed by an apparatus of a user equipment (UE), comprising: receiving a first physical broadcast channel (PBCH) communication on a channel; decoding the first PBCH communication as a decoded PBCH payload; reencoding the decoded PBCH payload as a first pilot signal; receiving a first data communication with first data; correcting the channel for the first data based on the first pilot signal; and decoding the first data. Aspect 2: The method of Aspect 1, wherein the first pilot signal is an expected antenna point signal, and wherein correcting the channel for the first data includes: comparing the expected antenna point signal and a received antenna point signal of the first PBCH communication; obtaining an estimate of the channel based on a result of the comparing; and correcting the channel for the first data based on the estimate of the channel. Aspect 3: The method of Aspect 2, further comprising performing beam selection based at least in part on the estimate of the channel. Aspect 4: The method of any of Aspects 1-3, wherein the reencoding the decoded PBCH payload includes reencoding the decoded PBCH payload based on the decoding of the first PBCH communication being successful. Aspect 5: The method of any of Aspects 1-4, further comprising: receiving a second PBCH communication on the channel; reencoding the decoded PBCH payload as a second pilot signal; receiving a second data communication with second data on the channel; correcting the channel for the second data based on the second pilot signal; and decoding the second data. Aspect 6: The method of Aspect 5, wherein the second pilot signal is an expected antenna point signal, and wherein correcting the channel for the second data includes: comparing the expected antenna point signal and a received antenna point signal of the second PBCH communication; obtaining an estimate of the channel based on a result of the comparing; and correcting the channel for the second data based on the estimate of the channel. Aspect 7: The method of any of Aspects 1-6, further comprising decoding a second PBCH communication to obtain a new decoded PBCH payload based on a difference between a demodulation reference signal signal-to-noise ratio (SNR) of the second PBCH communication and a PBCH SNR of the second PBCH communication satisfying a difference threshold. Aspect 8: The method of any of Aspects 1-7, wherein the reencoding the decoded PBCH payload includes polar encoding the decoded PBCH payload based at least in part on per-stage instructions for a butterfly lattice used for the polar encoding. Aspect 9: The method of any of Aspects 1-8, further comprising: receiving a second data communication with second data on the channel; correcting the channel for the second data based on the first pilot signal; and decoding the second data. Aspect 10: The method of any of Aspects 1-9, wherein the decoding the first data includes decoding the first data to obtain decoded first data, and wherein the method further comprises: reencoding the decoded first data as a second pilot signal; receiving a second data communication with second data on the channel; correcting the channel for the second data based on the second pilot signal; and decoding the second data. Aspect 11: The method of any of Aspects 1-10, further comprising: receiving a first control communication with control information on the channel; decoding the control information to obtain decoded

control information; reencoding the decoded control information as a second pilot signal; receiving a second data communication with second data on the channel; correcting the channel for the second data based on the second pilot signal; and decoding the second data. Aspect 12: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 1-11. Aspect 13: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 1-11. Aspect 14: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 1-11. Aspect 15: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 1-11. Aspect 16: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 1-11.

(116) The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects.

(117) As used herein, the term “component” is intended to be broadly construed as hardware and/or a combination of hardware and software. “Software” shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, and/or functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. As used herein, a “processor” is implemented in hardware and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the aspects. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code, since those skilled in the art will understand that software and hardware can be designed to implement the systems and/or methods based, at least in part, on the description herein.

(118) As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

(119) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various aspects. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. The disclosure of various aspects includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

(120) No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the terms “set” and “group” are intended to include one or more items and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used

herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

claimsHtml: 1. A method of wireless communication performed by an apparatus of a user equipment (UE), comprising: receiving a first physical broadcast channel (PBCH) communication, carrying system information, on a channel; decoding the first PBCH communication as a decoded PBCH payload; reencoding the decoded PBCH payload as a first pilot signal used for channel estimation; receiving, on the channel, a first data communication with first data; correcting the channel for the first data based on the first pilot signal; and decoding the first data.

2. The method of claim 1, wherein the first pilot signal is an expected antenna point signal, and wherein correcting the channel for the first data includes: comparing the expected antenna point signal and a received antenna point signal of the first PBCH communication; obtaining an estimate of the channel based on a result of the comparing; and correcting the channel for the first data based on the estimate of the channel.

3. The method of claim 2, further comprising performing beam selection based at least in part on the estimate of the channel.

4. The method of claim 1, wherein the reencoding the decoded PBCH payload includes reencoding the decoded PBCH payload based on the decoding of the first PBCH communication being successful.

5. The method of claim 1, further comprising: receiving a second PBCH communication on the channel; reencoding the decoded PBCH payload as a second pilot signal; receiving a second data communication with second data on the channel; correcting the channel for the second data based on the second pilot signal; and decoding the second data.

6. The method of claim 5, wherein the second pilot signal is an expected antenna point signal, and wherein correcting the channel for the second data includes: comparing the expected antenna point signal and a received antenna point signal of the second PBCH communication; obtaining an estimate of the channel based on a result of the comparing; and correcting the channel for the second data based on the estimate of the channel.

7. The method of claim 1, further comprising decoding a second PBCH communication to obtain a new decoded PBCH payload based on a difference between a demodulation reference signal signal-to-noise ratio (SNR) of the second PBCH communication and a PBCH SNR of the second PBCH communication satisfying a difference threshold.

8. The method of claim 1, wherein the reencoding the decoded PBCH payload includes polar encoding the decoded PBCH payload based at least in part on per-stage instructions for a butterfly lattice used for the polar encoding.

9. The method of claim 1, further comprising: receiving a second data communication with second data on the channel; correcting the channel for the second data based on the first pilot signal; and decoding the second data.

10. The method of claim 1, wherein the decoding the first data includes decoding the first data to obtain decoded first data, and wherein the method further comprises: reencoding the decoded first data as a second pilot signal; receiving a second data communication with second data on the channel; correcting the channel for the second data based on the second pilot signal; and decoding the second data.

11. The method of claim 1, further comprising: receiving a first control communication with control information on the channel; decoding the control information to obtain decoded control information; reencoding the decoded control information as a second pilot signal; receiving a second data communication with second data on the channel; correcting the channel for the second data based on the second pilot signal; and decoding the second data.

12. An apparatus of a user equipment (UE) for wireless communication, comprising: memory; and one or more processors coupled to the memory, the memory comprising instructions executable by the one or more processors to cause the UE to: receive a first physical broadcast channel (PBCH) communication, carrying system information, on a channel; decode the first PBCH communication as a decoded PBCH payload; reencode the decoded PBCH payload as a first pilot signal used for channel estimation; receive, on the channel, a first data communication with first data; correct

the channel for the first data based on the first pilot signal; and decode the first data.

13. The apparatus of claim 12, wherein the first pilot signal is an expected antenna point signal, and wherein the instructions, when correcting the channel for the first data, are further executable by the one or more processors to cause the UE to: compare the expected antenna point signal and a received antenna point signal of the first PBCH communication; obtain an estimate of the channel based on a result of the comparing; and correct the channel for the first data based on the estimate of the channel.

14. The apparatus of claim 13, wherein the memory further comprises instructions executable by the one or more processors to cause the UE to perform reference signal received power estimation based at least in part on the estimate of the channel.

15. The apparatus of claim 12, wherein the instructions, when reencoding the decoded PBCH payload, are further executable by the one or more processors to cause the UE to reencode the decoded PBCH payload based on the decoding of the first PBCH communication being successful.

16. The apparatus of claim 12, wherein the memory further comprises instructions executable by the one or more processors to cause the UE to: receive a second PBCH communication on the channel; reencode the decoded PBCH payload as a second pilot signal; receive a second data communication with second data on the channel; correct the channel for the second data based on the second pilot signal; and decode the second data.

17. The apparatus of claim 16, wherein the second pilot signal is an expected antenna point signal, and wherein the instructions, when correcting the channel for the second data, are further executable by the one or more processors to cause the UE to: compare the expected antenna point signal and a received antenna point signal of the second PBCH communication; obtain an estimate of the channel based on a result of the comparing; and correct the channel for the second data based on the estimate of the channel.

18. The apparatus of claim 12, the memory further comprises instructions executable by the one or more processors to cause the UE to decode a second PBCH communication to obtain a new decoded PBCH payload based on a difference between a demodulation reference signal signal-to-noise ratio (SNR) of the second PBCH communication and a PBCH SNR of the second PBCH communication satisfying a difference threshold.

19. The apparatus of claim 12, wherein the instructions, when reencoding the first PBCH communication, are further executable by the one or more processors to polar encode the decoded PBCH payload based at least in part on per-stage instructions for a butterfly lattice used for the polar encoding.

20. The apparatus of claim 12, wherein the memory further comprises instructions executable by the one or more processors to cause the UE to: receive a second data communication with second data on the channel; correct the channel for the second data based on the first pilot signal; and decode the second data.

21. The apparatus of claim 12, wherein the instructions, when decoding the first data, are further executable by the one or more processors to decode the first data to obtain decoded first data, and wherein the memory further comprises instructions executable by the one or more processors to cause the UE to: reencode the decoded first data as a second pilot signal; receive a second data communication with second data on the channel; correct the channel for the second data based on the second pilot signal; and decode the second data.

22. The apparatus of claim 12, wherein the memory further comprises instructions executable by the one or more processors to cause the UE to: receive a first control communication with control information on the channel; decode the control information to obtain decoded control information; reencode the decoded control information as a second pilot signal; receive a second data communication with second data on the channel; correct the channel for the second data based on the second pilot signal; and decode the second data.

23. A non-transitory computer-readable medium storing one or more instructions for wireless communication, the one or more instructions, when executed by one or more processors of a user equipment (UE), cause the UE to: receive a first physical broadcast channel (PBCH) communication, carrying system information, on a channel; decode the first PBCH communication as a decoded PBCH payload; reencode the decoded PBCH payload as a first pilot signal used for

channel estimation; receive, on the channel, a first data communication with first data; correct the channel for the first data based on the first pilot signal; and decode the first data.

24. The non-transitory computer-readable medium of claim 23, wherein the first pilot signal is an expected antenna point signal, and wherein the one or more instructions, when correcting the channel for the first data, further cause the UE to: compare the expected antenna point signal and a received antenna point signal of the first PBCH communication; obtain an estimate of the channel based on a result of the comparing; and correct the channel for the first data based on the estimate of the channel.

25. The non-transitory computer-readable medium of claim 23, wherein the one or more instructions further cause the UE to: receive a second PBCH communication on the channel; reencode the decoded PBCH payload as a second pilot signal; receive a second data communication with second data on the channel; correct the channel for the second data based on the second pilot signal; and decode the second data.

26. The non-transitory computer-readable medium of claim 23, wherein the one or more instructions further cause the UE to: receive a second data communication with second data on the channel; correct the channel for the second data based on the first pilot signal; and decode the second data.

27. An apparatus for wireless communication, comprising: means for receiving a first physical broadcast channel (PBCH) communication, carrying system information, on a channel; means for decoding the first PBCH communication as a decoded PBCH payload; means for reencoding the decoded PBCH payload as a first pilot signal used for channel estimation; means for receiving, on the channel, a first data communication with first data; means for correcting the channel for the first data based on the first pilot signal; and means for decoding the first data.

28. The apparatus of claim 27, wherein the first pilot signal is an expected antenna point signal, and wherein the means for correcting the channel for the first data includes: means for comparing the expected antenna point signal and a received antenna point signal of the first PBCH communication; means for obtaining an estimate of the channel based on a result of the comparing; and means for correcting the channel for the first data based on the estimate of the channel.

29. The apparatus of claim 27, further comprising: means for receiving a second PBCH communication on the channel; means for reencoding the decoded PBCH payload as a second pilot signal; means for receiving a second data communication with second data on the channel; means for correcting the channel for the second data based on the second pilot signal; and means for decoding the second data.

30. The apparatus of claim 27, further comprising: means for receiving a second data communication with second data on the channel; means for correcting the channel for the second data based on the first pilot signal; and means for decoding the second data.

FIELD OF THE DISCLOSURE (1) Aspects of the present disclosure generally relate to wireless communication and to techniques and apparatuses for using a physical broadcast channel for channel correction.

BACKGROUND (2) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources (e.g., bandwidth, transmit power, or the like). Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, time division synchronous code division multiple access (TD-SCDMA) systems, and Long Term Evolution (LTE). LTE/LTE-Advanced is a set of enhancements to the Universal Mobile Telecommunications System (UMTS) mobile standard promulgated by the Third Generation Partnership Project (3GPP).

(3) A wireless network may include one or more network nodes that support communication for wireless communication devices, such as a user equipment (UE) or multiple UEs. A UE may communicate with a network node via downlink communications and uplink communications. “Downlink” (or “DL”) refers to a communication link from the

network node to the UE, and “uplink” (or “UL”) refers to a communication link from the UE to the network node. Some wireless networks may support device-to-device communication, such as via a local link (e.g., a sidelink (SL), a wireless local area network (WLAN) link, and/or a wireless personal area network (WPAN) link, among other examples).
(4) The above multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different UEs to communicate on a municipal, national, regional, and/or global level. New Radio (NR), which may be referred to as 5G, is a set of enhancements to the LTE mobile standard promulgated by the 3GPP. NR is designed to better support mobile broadband internet access by improving spectral efficiency, lowering costs, improving services, making use of new spectrum, and better integrating with other open standards using orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) (CP-OFDM) on the downlink, using CP-OFDM and/or single-carrier frequency division multiplexing (SC-FDM) (also known as discrete Fourier transform spread OFDM (DFT-s-OFDM)) on the uplink, as well as supporting beamforming, multiple-input multiple-output (MIMO) antenna technology, and carrier aggregation. As the demand for mobile broadband access continues to increase, further improvements in LTE, NR, and other radio access technologies remain useful.
SUMMARY
(5) Some aspects described herein relate to a method of wireless communication performed by a user equipment (UE) or an apparatus of the UE. The method may include receiving a first physical broadcast channel (PBCH) communication on a channel. The method may include decoding the first PBCH communication as a decoded PBCH payload. The method may include reencoding the decoded PBCH payload as a first pilot signal. The method may include receiving a first data communication with first data. The method may include correcting the channel for the first data based on the first pilot signal. The method may include decoding the first data.
(6) Some aspects described herein relate to an apparatus of a UE for wireless communication. The apparatus may include memory, one or more processors coupled to the memory, and instructions stored in the memory and executable by the one or more processors. The instructions may be executable by the one or more processors to cause the UE to receive a first PBCH communication on a channel. The instructions may be executable by the one or more processors to cause the UE to decode the first PBCH communication as a decoded PBCH payload. The instructions may be executable by the one or more processors to cause the UE to reencode the decoded PBCH payload as a first pilot signal. The instructions may be executable by the one or more processors to cause the UE to receive a first data communication with first data. The instructions may be executable by the one or more processors to cause the UE to correct the channel for the first data based on the first pilot signal. The instructions may be executable by the one or more processors to cause the user equipment to decode the first data.
(7) Some aspects described herein relate to a non-transitory computer-readable medium that stores one or more instructions for wireless communication by a UE. The one or more instructions, when executed by one or more processors of the UE, may cause the UE to receive a first PBCH communication on a channel. The one or more instructions, when executed by one or more processors of the UE, may cause the UE to decode the first PBCH communication as a decoded PBCH payload. The one or more instructions, when executed by one or more processors of the UE, may cause the UE to reencode the decoded PBCH payload as a first pilot signal. The one or more instructions, when executed by one or more processors of the UE, may cause the UE to receive a first data communication with first data. The one or more instructions, when executed by one or more processors of the UE, may cause the UE to correct the channel for the first data based on the first pilot signal. The one or more instructions, when executed by one or more processors of the UE, may cause the UE to decode the first data.
(8) Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving a first PBCH communication on a channel. The apparatus may include means for decoding the first PBCH communication as a decoded PBCH payload. The apparatus may include means for reencoding the decoded PBCH payload as a first pilot signal. The apparatus may include means for receiving a first data communication with first data. The apparatus may include means for correcting the channel for the first data based on the first pilot signal. The apparatus may include means for decoding the first data.
(9) Aspects generally include a method, apparatus, system, computer program product, non-transitory computer-readable medium, UE, base station, network entity, network node, wireless

communication device, and/or processing system as substantially described herein with reference to and as illustrated by the drawings and specification.

(10) The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter. The conception and specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages, will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purposes of illustration and description, and not as a definition of the limits of the claims.

(11) While aspects are described in the present disclosure by illustration to some examples, those skilled in the art will understand that such aspects may be implemented in many different arrangements and scenarios. Techniques described herein may be implemented using different platform types, devices, systems, shapes, sizes, and/or packaging arrangements. For example, some aspects may be implemented via integrated chip embodiments or other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, and/or artificial intelligence devices). Aspects may be implemented in chip-level components, modular components, non-modular components, non-chip-level components, device-level components, and/or system-level components. Devices incorporating described aspects and features may include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals may include one or more components for analog and digital purposes (e.g., hardware components including antennas, radio frequency (RF) chains, power amplifiers, modulators, buffers, processors, interleavers, adders, and/or summers). It is intended that aspects described herein may be practiced in a wide variety of devices, components, systems, distributed arrangements, and/or end-user devices of varying size, shape, and constitution.

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